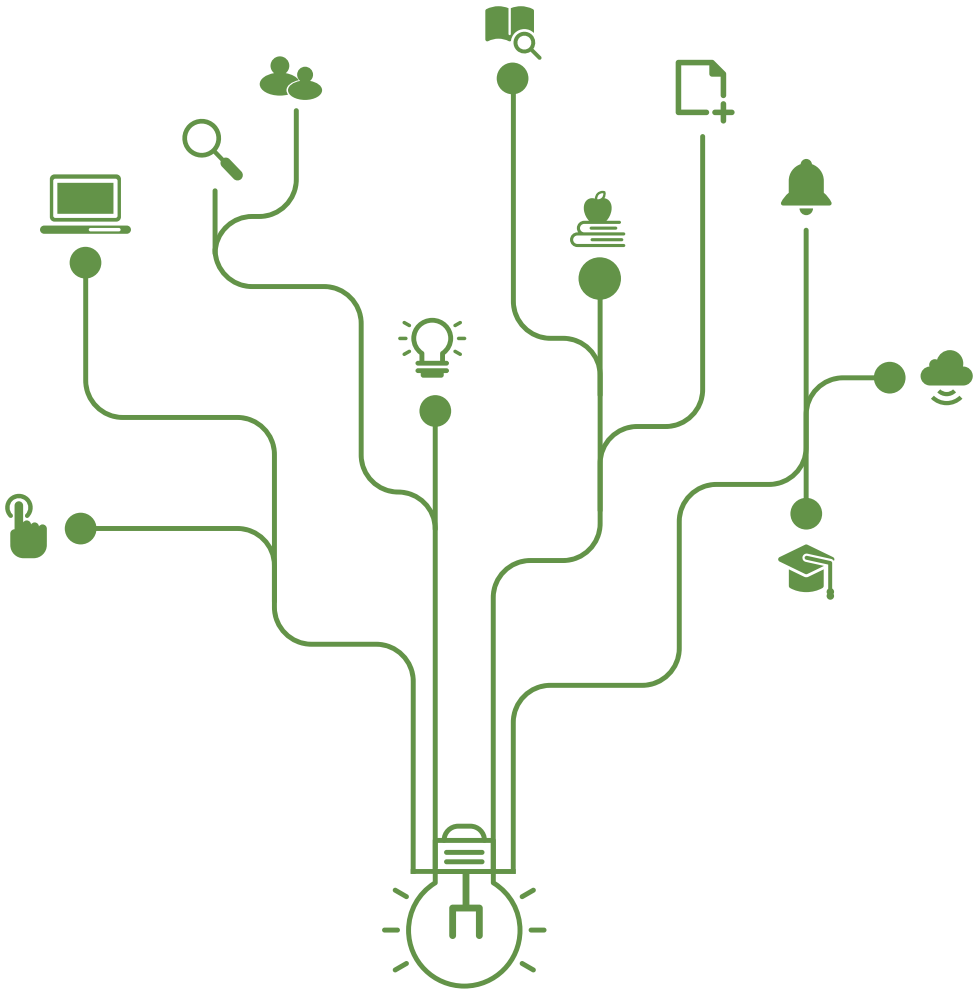
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The Energy Impacts of Green Remodeling: Evidence from the Building Sector

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We examine the impact of South Korea's Green Remodeling Program on the energy consumption of public daycare centers and public health centers. Combining monthly building-level utility records from 2018 to 2025 with administrative project data, propensity-score weighting, and staggered difference-in-differences estimators, we find that retrofits reduce total delivered energy by about 10% in the preferred Callaway and Sant'Anna (2021) specification (11–14% across alternative estimators) and electricity use by about 14%. Gas reductions of 8–9% appear in two-way fixed-effects and Sun and Abraham (2021) estimates but are not robust to the heterogeneity-robust Callaway–Sant'Anna estimator. Savings are larger in health centers (about 15%) than in daycare centers (7–11%), and are concentrated in wall and roof insulation, renewable energy, and complete envelope packages. Realized savings fall well short of engineering projections: building-level realized total-energy savings average 11.7%, compared with 37.9% predicted by the program's ECO2 simulation, and a machine-learning counterfactual recovers only 53–55% of the engineering benchmark. As a result, the realized abatement cost is about \$2,376/tCO_2, roughly 3.3 times the engineering-implied cost. Placebo, free-riding, spatial-spillover, and behind-the-meter-generation checks support the main estimates. The results suggest that subsidies tied to engineering projections overstate program benefits and that targeting high-return envelope measures, recalibrating ex ante predictions with realized utility data, and strengthening post-retrofit verification would improve the cost-effectiveness of public-building decarbonization.

Keywords— Building, Energy Efficiency, Policy Evaluation, Green Remodelling

JEL— Q50, Q58, Q48, O13

1 Introduction

With population growth, increased built-up area, and higher building services and comfort levels, building energy consumption has increased on average by 1.2% per year (González-Torres et al., 2022). The building sector is the largest energy consumer, accounting for approximately 34% of global greenhouse gas (GHG) emissions and 32% of end-use energy consumption, followed by industry and transport (UNEP, 2025). Accordingly, the South Korean government's nationally determined contribution commits to reducing building-sector GHG emissions from 52.1 $MtCO_2e$ in 2018 to 35.0 $MtCO_2e$ by 2030, a reduction of approximately 32.8% (MOLIT, 2021b).

However, as of 2022 the building sector had reduced emissions by only 8.8% relative to 2018, showing a lack of evident progress. Recognizing this gap, the Ministry of Land, Infrastructure, and Transport (MOLIT) announced the "National Land Transport 2050 Carbon Neutral Roadmap" in December 2022, which identifies zero-energy standards for new buildings and green remodeling of existing buildings as major initiatives. This includes increasingly stringent mandates on new public buildings through the Zero Energy Building (ZEB) policy, as well as subsidy programs for public building retrofits. Although newly constructed buildings warrant attention, the existing building stock is of greater structural significance: in 2022, buildings over 20 years old accounted for approximately 58.2% of the total stock in South Korea, and those over 30 years old accounted for 37% (Seo et al., 2022). Improving the energy performance of existing, aging buildings is therefore an essential and widely applicable mitigation measure.

Despite the importance of public support in achieving climate policy goals, significant gaps remain in our understanding of the realized benefits of public building energy-efficiency retrofit programs. A growing empirical literature shows that realized savings from energy-efficiency investments systematically fall short of engineering predictions (Fowlie et al., 2018; **?**), but most of this evidence comes from residential settings, and much of it from low-income single-family homes in the United States. Comparable evidence on publicly owned buildings outside the United States, where the government acts as both the subsidy provider and the building owner, remains limited.

Public buildings provide a useful setting for studying this engineering–realized gap because they hold ownership structure fixed while the other determinants of the gap vary. Ex ante, it is unclear whether the gap should be smaller or larger than in residential settings. On the one hand, the classic landlord–tenant split-incentive problem is largely absent because the owner is also the bill payer, which should narrow the gap. On the other hand, building occupants do not pay energy bills directly and therefore have a weak stake in conserving energy, and occupants may respond to efficiency gains by increasing usage (Aydin et al., 2017). In addition, public contracting and subsidy-eligibility rules may shape which measures are installed independently of a building's usage pattern, and information asymmetries between contractors and building owners can make installation quality difficult to verify (Giraudet, 2020). Both channels could widen the gap. Whether

realized savings in public buildings come close to engineering projections is therefore an empirical question.

Specifically, our analysis focuses on public daycare centers and public health centers retrofitted through the 2021 and 2022 Green Remodeling Program cohorts. We combine building-level administrative records on project timing, installed technologies, and project costs with eight years (2018–2025) of monthly electricity and gas consumption from the National Building Energy Database. To construct credible counterfactuals, we reweight non-participating public buildings using propensity-score inverse probability weights estimated separately by building type and outcome, restrict each sample to common support, and use later-cohort projects that had not yet started construction, accepted projects that were later cancelled, and never-treated public buildings as comparison groups. Our preferred estimator is the staggered-adoption difference-in-differences estimator of Callaway and Sant'Anna (2021), and we report conventional two-way fixed-effects (TWFE) and Sun and Abraham (2021) estimates for comparison. Because electricity and gas meters are not observed symmetrically across buildings, we construct a total-delivered-energy outcome that treats missing gas as zero only for buildings that show no gas use before 2021.

We find that participation in the Green Remodeling Program leads to meaningful and persistent reductions in energy use. In the preferred Callaway–Sant'Anna specification, retrofits reduce total delivered energy by 10.2% and electricity consumption by 13.9% over the first post-retrofit year. TWFE and Sun–Abraham estimates on the matched sample are similar for electricity (13.7–14.3%) and somewhat larger for total energy (13.7–14.4%). Event-study estimates show no systematic pre-trends once never-treated buildings are included in the comparison pool, and the post-retrofit reductions persist throughout the twelve-month window. For gas, TWFE and Sun–Abraham estimates indicate reductions of 8.6–9.4%, but the Callaway–Sant'Anna estimate is close to zero and statistically insignificant, so we do not regard the gas savings as robust.

Effects are heterogeneous. Savings are larger and more precisely estimated in health centers, where total energy and electricity fall by about 15%, than in daycare centers, where electricity falls by 10–12% and the Callaway–Sant'Anna estimate for total energy is smaller and imprecise. Calendar-month profiles suggest that health-center savings are concentrated in the cooling season. By retrofit measure, the largest and most precisely estimated reductions are associated with wall and roof insulation (about 11% for electricity), renewable energy (about 13% for electricity), and complete envelope retrofit packages (about 15% for total energy), whereas boilers, HVAC, ventilation, and lighting are not associated with measurable savings.

Realized savings fall well short of the program's engineering projections. At the building level, the ECO2 simulation used by the program predicts average total-energy savings of 37.9% for the matched sample, whereas realized savings average 11.7%, so engineering projections overstate realized savings by roughly a factor of three, consistent with Fowlie et al. (2018). A machine-learning counterfactual that follows **?** and does not rely on a comparison group recovers only 53–55% of the engineering benchmark. Translating the realized savings into emissions implies an abatement cost of about \$2,376/tCO_2 based on total project costs,

roughly 3.3 times the engineering-implied cost of \$711/tCO_2 and well above standard estimates of the social cost of carbon.

A series of robustness checks supports these conclusions. An early-cohort placebo test that assigns a fake treatment date to 2023 projects before any construction yields a near-zero effect. Approved buildings show larger post-project reductions than similar applicants whose projects were cancelled, which is inconsistent with free-riding. Excluding comparison buildings located near retrofitted buildings, or drawing comparison buildings from outside treated municipalities, does not change the estimates, which suggests that local spillovers do not bias them. Finally, excluding retrofits that installed on-site renewable energy reduces the electricity estimate but leaves a statistically significant reduction of about 10%, so the results are not driven solely by behind-the-meter generation.

This paper contributes to the existing literature in three ways. First, it contributes to the literature on the engineering–realized gap in energy-efficiency programs (Fowlie et al., 2018; **?**; Davis et al., 2014). Whereas most evidence comes from residential programs in the United States, we provide evidence from publicly owned buildings in Korea, where the split-incentive problem is largely absent. We find a wedge of similar magnitude to the residential evidence, which suggests that the gap between projected and realized savings is not specific to the residential sector or to the U.S. institutional context. Our building-level data on installed technologies also allow us to examine which retrofit technologies are associated with the largest engineering overprediction.

Second, our study contributes to the literature that combines matching or weighting with difference-in-differences designs to estimate the impact of building energy-efficiency investments on energy consumption. Our study is most similar to Boampong (2020), who combine a DID framework with coarsened exact matching to identify energy savings from the Gainesville Regional Utilities high-efficiency air-conditioner rebate program. We differ in that we focus on building-level outcomes from a whole-building retrofit program, and we combine propensity-score weighting with heterogeneity-robust staggered DID estimators (Callaway and Sant'Anna, 2021; Sun and Abraham, 2021; Goodman-Bacon, 2021) and a machine-learning counterfactual (**??**). We also follow Papineau et al. (2025) in estimating measure-specific savings.

Third, our paper provides insights into the broader literature on the effectiveness and cost-effectiveness of government subsidy programs aimed at improving building energy efficiency (Fowlie et al., 2018; Ito, 2015; Giandomenico et al., 2022).[1] Fowlie et al. (2018) evaluated the Weatherization Assistance Program, the largest residential energy-efficiency program in the United States, using a randomized encouragement design, and found that projected savings were roughly three times realized savings. We find a remarkably similar ratio and, using project-level cost data, show that this wedge translates directly into a cost-effectiveness gap.

[1] An exhaustive review of alternative methods for estimating the impact of energy-efficiency programs on energy consumption is beyond the scope of this paper, but it is well summarized in Gillingham et al. (2018).

A large share of the Korean green remodeling literature remains engineering- and case-study oriented, often combining (i) a specific retrofit package, (ii) simulation, and (iii) short-horizon pre- and post-consumption comparisons. Lee and Choi (2023) evaluated a single Korean public community center retrofitted over roughly four months. From an engineering perspective, Woo et al. (2024) evaluated South Korea's public building Green Remodeling Program by combining field surveys and design-consulting information for 1,065 public welfare facilities (daycare centers, health centers, and hospitals) with Ecological & Economical Combined Optimization (ECO2) simulations to estimate energy-saving potential. They found an average predicted energy-performance improvement of about 30%, with substantial variation by region and building use and the largest gains for public health centers. Many other studies have supported Korea's green remodeling program by showing that retrofits in aging public buildings can deliver sizable energy and emissions reductions and can be more cost-effective when coordinated with other renovation work (Cho et al., 2023; Jun et al., 2021; Nam et al., 2021). However, this evidence base relies heavily on simulations and single or small-sample case studies, so it often cannot separate retrofit effects from concurrent changes in weather, occupancy, and operations. Our large-scale, comparison-group-based evaluation complements this literature, and our finding that health centers deliver the largest realized savings is consistent with the engineering ranking in Woo et al. (2024), although the realized magnitudes are far smaller.

The rest of the paper is structured as follows. Section 2 describes the program, the conceptual framework, and the data. Section 3 presents the empirical design. Section 4 reports the main results, heterogeneity, and robustness checks. Section 5 compares realized savings with engineering projections and evaluates cost-effectiveness. Section 6 concludes with policy implications and limitations.

2 Background and Data

2.1 Program Description

South Korea's green building policy is implemented through the Green Building Promotion Act, which requires the establishment of a national basic plan every five years that defines targets and promotion directions for green buildings (MOLIT, 2021a). Although earlier efforts emphasized strengthening performance standards for new buildings and specific building components, structural changes in the building stock have made large-scale improvement of existing buildings increasingly central to national decarbonization. Currently, South Korea has approximately 7.28 million buildings, and more than 5.42 million are older than 15 years (MOLIT, 2014). This scale of aging stock implies that achieving national carbon neutrality in the building sector is difficult through a new-building–focused ZEB policy alone. Therefore, improving the energy performance of older buildings has become a critical strategy for reducing GHG emissions. Green remodeling specifically targets measurable improvements in energy performance and reductions in GHG emissions through technological advancement in both passive and active building design.

Public buildings have been positioned as the leading sector for green remodeling, both because of their large floor areas and because public-sector leadership can encourage adoption in private buildings. Program interventions typically include improvements to the building envelope, such as insulation upgrades and airtightness enhancement, replacement or modernization of mechanical systems with high-efficiency equipment, and integration of new and renewable energy technologies. Policy initiatives have also included public-building assessments and design consulting (2015–2019) to identify and upgrade buildings with low performance and high energy consumption, aiming to reduce emissions and scale green remodeling across sectors (MOLIT, 2014). More recently, national decarbonization strategies have further reinforced green remodeling as a major initiative implementing the program at large annual volumes nationwide starting from 2020.

Figure 1 summarizes the evolution of the program. The legal basis for green remodeling support was established by the Green Buildings Support Act in 2013, and the public-building support program was launched in 2020 with daycare centers, public health centers, and public medical facilities as the initial priority building types. The program selected 821 buildings in 2020, 895 in 2021, 575 in 2022, 617 in 2023, and 562 in 2024, for a cumulative total of 3,470 supported buildings by the end of 2024 (MOLIT, 2025). Figure 3 shows the number of retrofitted public buildings by year.

Our analysis focuses on public daycare centers and public health centers retrofitted in the 2021 and 2022 programs (see Appendix Figures D.1–D.3 for the locations of participating facilities). This focus reflects the program's initial policy of prioritizing aging public buildings predominantly used by vulnerable populations (children, older adults, and patients) to improve energy performance and indoor environmental conditions. Beginning with the 2023 cohort, the program scope expanded to include additional community facilities such as senior centers and libraries, and the 2023 and 2024 cohorts enter our analysis as not-yet-treated comparison buildings. Each program year follows the same administrative model. Figure 2 illustrates a typical project: a public daycare center before and after a retrofit that combined exterior insulation, window replacement, and rooftop solar panels.

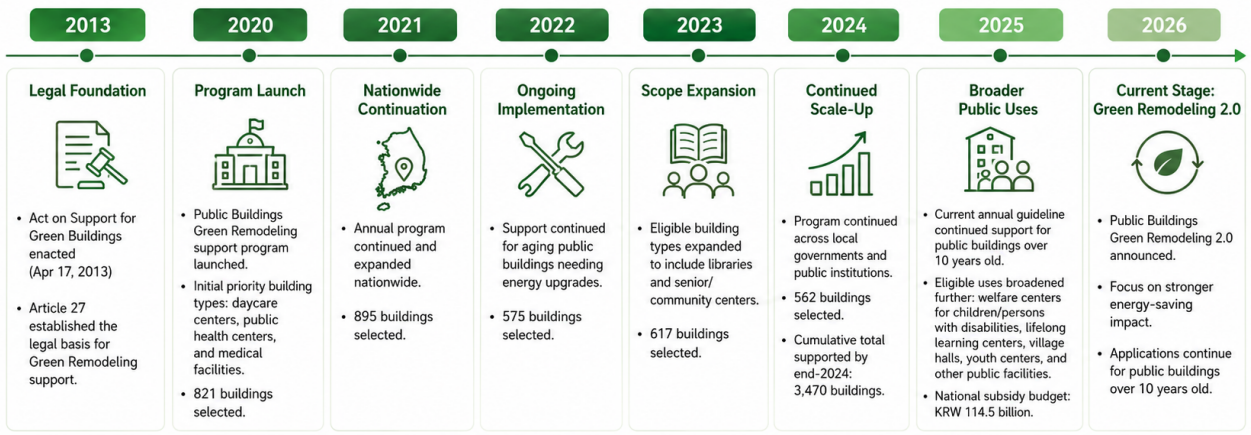


Figure 1: Green Remodeling Program Timeline

Note: Source: Ministry of Land, Infrastructure and Transport. Counts refer to all public buildings selected in each program year, including building types outside the analysis sample.



Figure 2: Example of a Green Remodeled Public Daycare Center: Before and After

Note: The left panel shows the building before the retrofit and the middle and right panels show the building after the retrofit. Source: Ministry of Land, Infrastructure and Transport.

The program provides financial support for renovation projects that upgrade the energy performance of aging public buildings, particularly those that have been in operation for at least ten years. Eligible applicants include central and local government bodies, as well as other public institutions that own or manage public facilities. Support is primarily used to cover retrofit measures, such as envelope improvements and equipment upgrades, including high-efficiency HVAC and ventilation, boilers, and lighting. Projects typically combine multiple measures rather than a single component. The subsidy structure also differs by applicant type and region, with higher support rates for many local governments and lower rates for some central institutions. In addition, the program follows a staged administrative process that includes application and screening, pre-construction support, project selection, and phased disbursement of funds during implementation.

Building owners are to first submit an application for the grant during the annual call, after which the MOLIT conducts site visits and evaluates candidate buildings based on urgency and expected effectiveness. The evaluation is guided by a quantitative scoring rubric that weights the planned application of energy-saving measures, such as insulation and window upgrades, high-efficiency equipment, and renewables, as well as the applicant institution's demonstrated commitment and capacity to execute the project; the final list is

confirmed through deliberation by the Green Remodeling Review Committee. Once a building is selected, it undergoes a pre-retrofit audit by government-designated architectural engineering firms. The audit consists of a detailed inventory of the building's characteristics, including the number of windows and doors, and the equipment installed, and assesses the overall condition of the building.

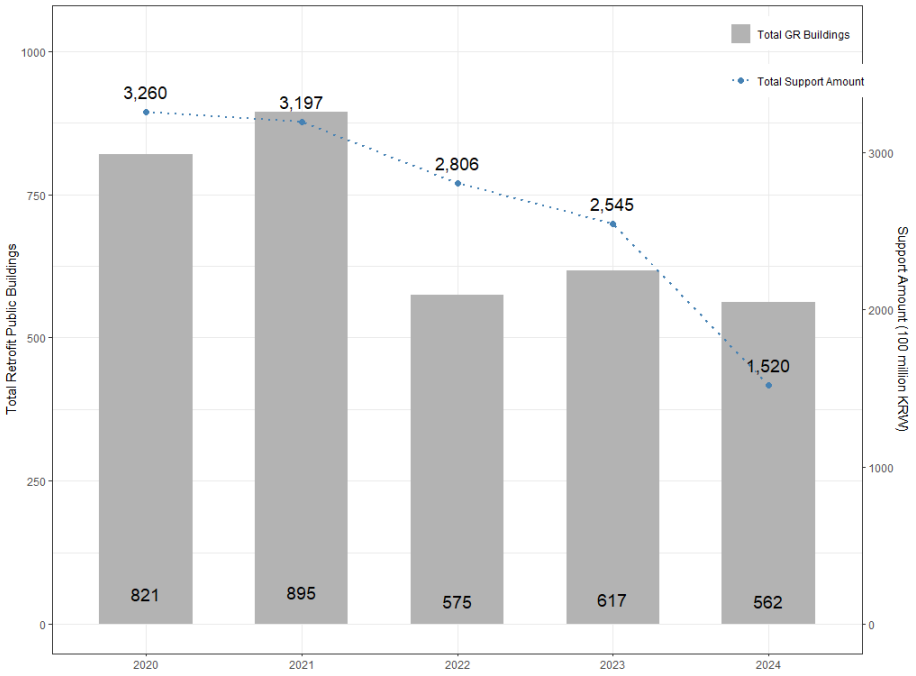


Figure 3: Green Remodeled Public Buildings by Year

The public-building green remodeling program specifies a defined set of retrofit measures eligible for government support, as shown in Table 1. Eligible measures are grouped into mandatory, selective, and additional support items. Mandatory items include high-efficiency envelope upgrades such as interior and exterior insulation, and systems-oriented measures, including heat-recovery and high-efficiency heating, ventilation, and air-conditioning (HVAC) systems, as well as building energy management systems and renewable energy. Selective items include measures such as cool roofs, awnings, electric water heaters, and smart air showers. In addition, the program can support certain ancillary works, including structural and safety upgrades (enabling works), as well as project-related costs such as design, supervision, relocation, and procurement.

Financial support is provided as a cost-share subsidy with a 50% subsidy rate for projects led by Seoul, the central government, and public-sector institutions, and a 70% subsidy rate for other local governments. The program also sets maximum support ceilings per unit floor area (Korean Won (KRW) per 3.3 m^2), differentiated by building size and applicant type: for general buildings with gross floor area of at least 300 m^2, the ceiling is KRW 1.90 million per 3.3 m^2 (alternative ceiling: KRW 2.28 million) for Seoul, central, and public applicants, and KRW 2.66 million per 3.3 m^2 (alternative ceiling: KRW 2.52 million) for other local governments. For small buildings with a gross floor area under 300 m^2, the ceiling is KRW 2.00 million

per 3.3 m^2 (alternative ceiling: KRW 2.40 million) for Seoul, central, and public applicants, and KRW 2.80 million per 3.3 m^2 (alternative ceiling: KRW 3.36 million) for other local governments. Because support is provided as a percentage cost share and is subject to per-area ceilings, participating institutions generally finance the remaining portion of retrofit costs.

Table 1: Green remodeling Mandatory and Optional Scope

Division	**Green Remodeling Criteria**
Mandatory item	High-efficiency item: Interior and exterior insulation Equipment: Heat recovery system and high-efficiency HVAC system, Building energy management system and renewable energy
Selective item	Cool roof, awning, electric water heater, and smart air shower
Additional support item	Structural/safety upgrades and enabling works Design, supervision, relocation, and procurement costs

Table 2 reports the funding composition of the 2021 and 2022 cohorts for the three initial priority building types. The 509 projects received total funding of KRW 147.8 billion, of which KRW 100.9 billion (68%) came from the national subsidy and KRW 47.0 billion from local governments. Public health centers account for the largest number of projects (315), whereas daycare centers (186 projects) account for a similar share of total spending at a lower cost per project. Only eight public hospitals were retrofitted in these cohorts.

Table 2: Funding Composition by Building Type, 2021–2022 Program Cohorts

		Funding amounts (KRW billion)		
Building type	Projects	National	Local	Total project cost
Public daycare center	186	44.79	22.45	67.25
Public health center	315	49.29	21.13	70.42
Public hospital	8	6.78	3.38	10.16
Total	509	100.86	46.96	147.82

Note: The table reports project funding for the 2021 and 2022 program cohorts. Funding amounts are expressed in billions of Korean won (KRW). Total project cost is the sum of national and local government funding. Source: Korea Authority of Land and Infrastructure Safety.

2.2 Conceptual Framework: Realized versus Projected Savings in Public Buildings

The Green Remodeling Program selects projects and sizes support on the basis of engineering assessments, so the central policy question is how closely realized savings track the savings projected ex ante. In residential settings, a large literature documents that realized savings fall short of engineering projections (Fowlie et al., 2018; **?**; Davis et al., 2014). Whether the same is true in public buildings is not obvious, because public ownership changes several of the mechanisms that are thought to generate the gap. Figure 4 summarizes three such mechanisms.

First, public buildings largely eliminate the landlord–tenant split-incentive problem: the public institution that owns the building and applies for the retrofit also pays the energy bill. This should push toward a smaller gap. Second, the staff and users who operate public buildings do not pay energy bills themselves and, depending on budget rules, may not benefit from energy-cost savings. This weak occupant stake means that efficiency gains may be partly offset by longer operating hours, expanded conditioned areas, or relaxed thermostat setpoints (Aydin et al., 2017; Gillingham et al., 2016), which pushes toward a larger gap. Third, public procurement and subsidy-eligibility rules shape which measures are selected and how they are installed. If measures are selected because they satisfy program requirements rather than because they suit the building's usage pattern, or if installation quality is difficult to verify ex post (Giraudet, 2020), realized savings may fall further below projections. Because these mechanisms work in opposite directions, the net size of the engineering–realized gap in public buildings is ambiguous ex ante and must be determined empirically.

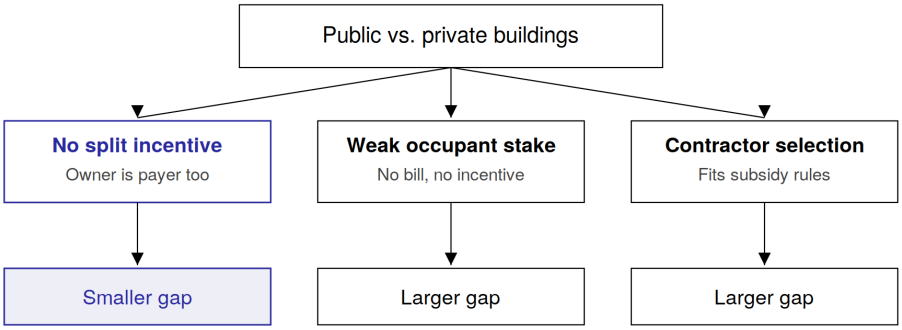


Figure 4: Mechanisms Shaping the Engineering–Realized Gap in Public Buildings

2.3 Data

This study draws on multiple administrative and public datasets to evaluate the impact of South Korea's green remodeling policy on energy consumption in the building sector, particularly for public daycare centers, health centers, and hospitals. We utilized energy consumption data for individual buildings from the National Building Energy Database (NBED), run by the MOLIT and the Korea Real Estate Board (KREB). The dataset contains monthly electricity and gas consumption data from 2018 onwards, being updated monthly. The dataset also includes comprehensive information such as floor area, building use classification, construction year, and location.

A list of green remodeled public buildings for the 2021–2024 programs was provided by the MOLIT and the Korea Authority of Land and Infrastructure Safety (KALIS), including buildings whose applications were accepted but whose projects were later cancelled. The list contains building location, type, project start and end dates, the technological components implemented, project costs and funding sources, and,

for the 2021–2022 cohorts, the engineering-predicted energy performance before and after the retrofit. We matched addresses from the green remodeling database with those in the NBED database to identify the energy accounts of these buildings, and we geocoded building addresses to latitude and longitude using the Kakao Developers geocoding API. The universe of public daycare centers, health centers, and hospitals that serve as potential comparison buildings is obtained from facility lists published by the Korea Childcare Promotion Institute and the Ministry of Health and Welfare. Weather is a key determinant of energy demand, particularly for heating and cooling. To account for these exogenous factors, regional weather data are obtained from the Korea Meteorological Administration (KMA). These variables are matched to buildings based on their regional location, ensuring that energy consumption estimates are not biased by local climate variability.

Incomplete records were identified and categorized, including cases with missing data in specific months or years. The energy consumption dataset spans 2018 to 2025 and includes multiple unique building observations. However, not all buildings have a complete set of monthly records over this period; some are missing data in particular months, and others have gaps that extend over longer periods.

To address this inconsistency and account for seasonal variation in energy use, we restrict the sample to buildings with at least one year of observations both before and after the policy implementation date. We further exclude observations affected by duplicated monthly entries and extreme outliers. A total of 1,895 public buildings (844 daycare centers, 961 health centers, and 90 hospitals) were selected in the four program years from 2021 to 2024. After these restrictions, the analysis sample contains 634 participating buildings and 5,225 non-participating public buildings observed monthly from January 2018 to December 2025. Treatment is defined by the 2021 and 2022 cohorts that completed their retrofits; buildings selected in 2023 and 2024 contribute to the comparison group only before their own construction begins.

Table 3 summarizes energy consumption outcomes and baseline building characteristics for public buildings that participated in the green remodeling program (634 buildings) and those that did not (5,225 buildings) after filtering. For each group, the table reports the mean and standard deviation (SD) for the variables used in the analysis. Overall, both energy outcomes show substantial dispersion, with SDs that are large relative to the means, suggesting considerable heterogeneity in energy use across buildings.

We use daily mean temperature data from the KMA for all regions to calculate cooling degree days (CDD) and heating degree days (HDD). CDD and HDD are calculated by comparing each day's average outdoor temperature to a chosen base temperature (24 °C for CDD and 18 °C for HDD). For CDD, if the average temperature for that day is above the base, the daily contribution is the average temperature minus the base. For HDD, if the average temperature is below the base, the daily contribution is the base minus the average temperature. The total CDD and HDD are the sums of these daily contributions over a month.

For electricity consumption, program participants have a higher average level than non-participants (8.5 kWh/m^2 versus 7.0 kWh/m^2). Within the participant group, mean electricity use is higher in the pre-retrofit period (8.9 kWh/m^2) and lower in the post-retrofit period (7.5 kWh/m^2), consistent with a reduction after the intervention (about a 16 % decline relative to the pre-retrofit mean). A similar pattern appears for gas consumption. Participants have a higher average gas use than non-participants (9.7 kWh/m^2 versus 8.8 kWh/m^2), and within participants the mean falls from 10.4 kWh/m^2 pre-retrofit to 8.1 kWh/m^2 post-retrofit (about a 22 % decline). These descriptive changes motivate the staggered DID analysis, while the large SDs indicate that effects may vary across buildings. Panel B of Table 3 previews the engineering–realized gap studied in Section 5. For the 2021–2022 cohorts, actual energy intensity falls from 10.5 to 9.3 kWh/m^2 (about 11%), whereas the engineering model predicts a decline from 9.1 to 5.5 kWh/m^2 (about 40%). The average project cost KRW 290.4 million, of which KRW 198.1 million was financed by the national subsidy.

Building characteristics also differ between groups. Participating buildings are smaller on average (total floor area 498 m^2) than non-participating buildings (965 m^2), and the variability in floor area is especially large among non-participants (SD 7,083 m^2), indicating a highly spread-out size distribution. Participants are, on average, slightly shorter and have fewer stories, whereas basement floors are similar across groups. Participating buildings received 6.6 retrofit measures on average, and the average construction duration is 280 days, providing context for the timing of the intervention and the definition of pre- and post-retrofit periods in the main analysis.

To complement the administrative panel data and directly assess behavioral responses to the efficiency improvements, we designed and fielded a survey of program participants. The survey targeted institutions from the 2021 and 2022 program cohorts, two of the four cohorts in our full sample, restricted to public daycare centers, health centers, and hospitals that had completed their retrofit. We focus on these two cohorts because engineering-predicted energy consumption estimates and program cost data are already available for them, allowing us to directly compare realized savings against engineering projections; extending the survey to the remaining cohorts was beyond the scope of the current budget. Eligible respondents were staff with direct knowledge of the institution's energy use and operations, and the instrument (approximately 40 items, 10–15 minutes) covered institution characteristics, the specific technological components adopted and their costs, and self-reported changes in energy-related behavior and outcomes following the retrofit.

The survey was designed specifically to help identify the behavioral rebound effect that the administrative consumption data alone cannot separate from mechanical or engineering-driven changes in energy use. Respondents were asked to report, separately for the pre- and post-retrofit periods, the conditioned floor area actually served by heating and cooling equipment, representative thermostat setpoints and daily operating hours by season, and any change in the primary heating source or fuel. Where respondents indicated that heating or cooling equipment usage increased or remained level despite the efficiency upgrade, follow-up items

asked them to attribute this to specific mechanisms, for example, longer operating hours, expanded conditioned area, a relaxed setpoint, or reduced sensitivity to energy costs following the retrofit, allowing us to distinguish take-back behavior from other confounding factors. The survey also captured perceived satisfaction with realized electricity and gas savings, and how energy cost savings are treated within the institution's budget (e.g., whether savings are retained, reallocated, or reduce subsequent-year appropriations), which speaks to the incentive structure institutions face when deciding whether to moderate consumption post-retrofit.

The survey is designed to be used in three complementary ways. First, self-reported pre/post changes in setpoints, operating hours, and conditioned area provide direct, building-level evidence on the behavioral margins through which rebound could occur, which can corroborate and help interpret the gap between engineering-predicted and realized savings documented in the administrative data. Second, the budget-attribution items let us test whether the rebound effect varies systematically with the institution's financial incentive to conserve—for instance, whether institutions that do not directly bear energy costs, or that lose unspent budget at year-end, exhibit larger increases in usage than those that retain savings. Third, because our administrative program records provide only aggregate retrofit costs at the building level, we use the survey to collect technology-specific information—which components were installed and their individual costs—for each institution. This allows us to decompose the engineering-realized savings gap by technology type rather than by building alone, and to construct a more precise cost-effectiveness analysis based on component-level, rather than aggregate, expenditure. Because the survey covers only the 2021–2022 cohorts, we treat these results as a targeted sub-sample analysis rather than as representative of the full set of treated buildings, and survey-based evidence should be interpreted with this scope in mind. The estimates reported in this paper rely on the administrative consumption and program records; we use the survey to interpret the mechanisms behind the engineering–realized gap in future work.

Table 3: Summary Statistics: Before and After Intervention

	Program participants		Non-program participants	
	Mean (1)	SD (2)	Mean (3)	SD (4)
Panel A: All Cohorts (2021–2024)				
Electricity Consumption Data				
Electricity Consumption (kWh/m^2)	8.5	9.5	7.0	11.2
Pre-retrofit Consumption (kWh/m^2)	8.9	10.1		
Post-retrofit Consumption (kWh/m^2)	7.5	8.2		
Gas Consumption Data				
Gas Consumption (kWh/m^2)	9.7	9.8	8.8	10.6
Pre-retrofit Consumption (kWh/m^2)	10.4	10.5		
Post-retrofit Consumption (kWh/m^2)	8.1	8.0		
Building Characteristics				
Total Floor Area (m^2)	498	1,218	965	7,083
Building Height (m)	7.4	4.3	8.8	4.7
Number of Stories (floors)	1.9	0.8	2.5	1.0
Number of Basement Floors (floors)	0.2	0.4	0.3	0.5
Retrofit Measures (count)	6.6	2.6		
Construction Days (days)	280	139		
Weather Characteristics				
Cooling Degree Days (#)	0.45	0.99	0.49	1.05
Heating Degree Days (#)	6.84	7.01	6.74	7.03
Average Temperature (°C)	13.5	9.3	13.6	9.4
Average Rainfall (mm)	117.2	113.3	116.3	117.6
Total observations (#)	634		5,225	
Panel B: 2021–2022 Cohorts Only				
Actual and Predicted Energy				
Actual Pre-retrofit Energy (kWh/m^2)	10.5	13.2		
Actual Post-retrofit Energy (kWh/m^2)	9.3	9.0		
Predicted Pre-retrofit Energy (kWh/m^2)	9.1	2.7		
Predicted Post-retrofit Energy (kWh/m^2)	5.5	2.0		
Project Costs				
Total Project Cost (KRW, mil.)	290.4	267.5		
National Funding (KRW, mil.)	198.1	184.8		
Local Funding (KRW, mil.)	92.3	86.6		
Total observations (#)	507			

Note: Panel A reports summary statistics for all program and non-program participants across the 2021–2024 cohorts. Panel B is restricted to the 2021–2022 cohorts for which predicted energy estimates and project cost data are available. Columns (1)–(2) present means and standard deviations for program participants; columns (3)–(4) present the same for non-program participants. Retrofit measures counts the number of the following interventions present: wall insulation, roof insulation, floor insulation, windows, doors, HVAC, ventilation, boiler, lighting, renewables, cool roof, monitoring, EPD finishing, water saving, instant heater, solar control, asbestos removal, demolition, general green works, landscaping, structural works, and electrical upgrade.

3 Empirical Design

The methodology used to estimate energy-efficiency gains from the Green Remodeling Program for public buildings is based on a staggered difference-in-differences (DID) framework. Our preferred estimator is the doubly robust staggered-adoption estimator of Callaway and Sant'Anna (2021), and we report conventional two-way fixed-effects (TWFE) and Sun and Abraham (2021) estimates as robustness checks. Using longitudinal monthly energy-consumption data at the building level, the empirical strategy compares changes in electricity and gas consumption for buildings participating in the Green Remodeling Program with changes for comparable non-participating public buildings.

To improve the comparability of participating and non-participating buildings, we use inverse probability weighting based on estimated propensity scores. The weighting procedure is implemented separately by building type—daycare centers, health centers, and hospitals—because the determinants and levels of building energy consumption differ substantially across these building categories. Electricity and gas are also weighted separately because the availability and pre-treatment patterns of the two energy sources differ across buildings.

Our primary unit of analysis is monthly electricity and gas consumption at the building level. After constructing the weighted comparison samples, we estimate fixed-effects panel models and staggered DID specifications to examine the effect of the Green Remodeling Program on building energy consumption.

3.1 Propensity-Score Weighting

To construct a comparison group that is more similar to buildings participating in the Green Remodeling Program, we use propensity-score inverse probability weighting rather than one-to-one nearest-neighbor matching. Specifically, we estimate inverse probability weights for the average treatment effect on the treated (ATT). Participating buildings receive a weight of one, while non-participating buildings are reweighted so that their distribution of observed pre-treatment characteristics more closely resembles that of the treated buildings.

The propensity-score models are estimated separately by building type and outcome. For the component outcomes, electricity and gas are weighted separately because meter availability and pre-treatment energy-use histories differ across buildings. Estimating the models within building type avoids using facilities with substantially different energy-use profiles as counterfactuals for one another. The total-energy outcomes introduced in Section 3.2 are also assigned their own common-support samples and ATT weights because the structural-adjusted and strict dual-fuel definitions imply different sets of eligible building-months.

The propensity-score specification uses only pre-treatment information. The primary continuous covariates are the logarithm of total floor area, the logarithm of average pre-treatment energy consumption, the logarithm of energy consumption in the most recent pre-treatment year, and the pre-treatment trend in energy

consumption. The pre-treatment trend is estimated from annual pre-intervention energy consumption and captures whether a building's energy use was increasing or decreasing before participation in the Green Remodeling Program. Region is additionally included when there is sufficient treated–control overlap within regions.

The validity of propensity-score weighting depends on two related conditions: conditional independence and common support. Conditional independence, or selection on observables, requires that after conditioning on observed pre-treatment characteristics, program participation is independent of potential outcomes (Qiu et al., 2017, 2018). Although this assumption cannot be directly tested, combining weighting with DID reduces concerns arising from time-invariant unobserved differences between participating and non-participating buildings (Fowlie et al., 2012).

Common support requires treated and control buildings to have overlapping probabilities of participating in the program (Qiu et al., 2017, 2018). To avoid extrapolating from control buildings that are not comparable to treated buildings, we restrict each building-type and outcome-specific sample to the overlapping portion of the treated and control propensity-score distributions. We first estimate the propensity score, identify the intersection of the treated and control propensity-score ranges, remove observations outside this common-support region, and then re-estimate the propensity-score model on the restricted sample. This procedure is repeated until the common-support sample stabilizes.

We estimate propensity scores using bias-reduced logistic regression. This specification is useful in relatively small building-type subsamples, where conventional logistic propensity-score estimation can suffer from complete or quasi-complete separation. After restricting the sample to common support, we calculate ATT inverse probability weights. The primary analysis uses the resulting raw ATT weights. As a robustness check, we also calculate weights in which the upper tail of the control weights is winsorized at the 99th percentile.

Covariate balance is evaluated using absolute standardized mean differences before and after weighting. We use an absolute standardized mean difference of 0.10 as the principal diagnostic threshold and also inspect propensity-score overlap, the distribution of the control weights, and the effective sample size of the weighted comparison group. Figure 5 summarizes balance for the component-outcome samples that support the weighted analysis: daycare-center electricity, daycare-center gas, and health-center electricity. Before weighting, several covariates exceed the 0.10 threshold: for example, the standardized difference in log floor area is about 0.19 for daycare-center electricity and about 0.45 for daycare-center gas, and the standardized differences in pre-treatment consumption are between 0.13 and 0.20 in all three samples. After ATT-IPW weighting, all displayed standardized differences fall below 0.05, well within the 0.10 threshold.

The weighting procedure therefore yields usable component-outcome ATT-IPW samples for daycare-center electricity, daycare-center gas, and health-center electricity. Health-center gas does not provide a sufficiently

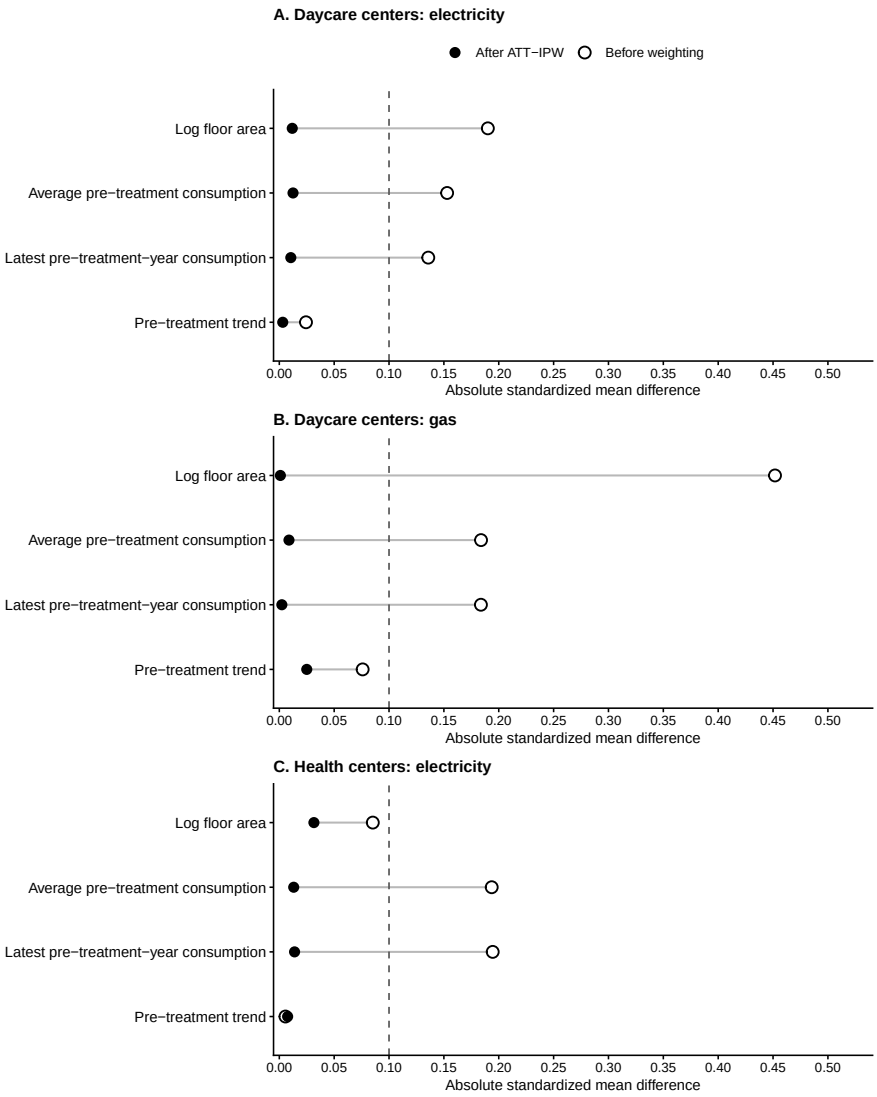


Figure 5: Covariate balance before and after ATT-IPW weighting: component outcomes

Note: Panels A and B report daycare-center electricity and gas; Panel C reports health-center electricity. Hollow circles indicate absolute standardized mean differences before weighting and filled circles indicate differences after ATT-IPW weighting. The dashed vertical line marks the 0.10 balance threshold. Regional indicators remain included in the propensity-score models where sufficient treated–control overlap exists but are omitted from the displayed panels for readability.

large treated sample with usable pre-treatment gas histories, while the hospital samples do not yield a final weighted comparison satisfying the common-support and balance requirements used here. We therefore do not treat those building-type–outcome combinations as primary ATT-IPW treatment-effect samples.

Detailed before- and after-weighting statistics are reported in Appendix Tables A.1 and A.2. Complementary plots of pre-treatment monthly energy-use profiles are reported in Appendix Figures A.1 and A.2. These profile figures hold the final common-support population fixed and compare participants with the same non-participant population before and after ATT-IPW reweighting. Additional propensity-score-overlap, regional-balance, weight-distribution, and effective-sample-size diagnostics are discussed in Appendix Section A.3.

3.2 Construction of Total Delivered Energy

Electricity and gas records are not observed symmetrically across public buildings. In particular, an absent gas record can indicate either a genuinely non-gas building or a missing meter/record for a building that uses gas. A single mechanical sum of electricity and gas would therefore either discard potentially valid non-gas buildings or risk treating missing gas observations as zero too broadly. We address this issue by constructing two total-delivered- energy outcomes. Both are measured in kWh/m^2 and are transformed to natural logs for the event-study analysis.

Our *structural-adjusted total* is the primary combined-energy outcome. We classify a building as having structurally no recorded gas using only information observed before 2021, so that the classification cannot be influenced by post-retrofit energy use. A building is classified as structural no-gas when (i) it has no positive gas-consumption month before 2021, (ii) it has at least 24 pre-2021 panel months, and (iii) electricity consumption is positive in at least 75 percent of those pre-2021 months. For buildings that meet all three conditions, a missing gas value is treated as zero when total delivered energy is constructed. For all other buildings, missing gas remains missing. Thus, the structural-adjusted outcome can be written schematically as

$$T_{it}^{\text{struct}} = E_{it} + G_{it}^{*}, \tag{1}$$

where E_{it} is electricity consumption and G_{it}^{*} equals observed gas consumption when available, equals zero only for buildings classified ex ante as structural no-gas, and remains missing otherwise. This construction retains buildings with persistent pre-period evidence of electricity use but no evidence of gas use, rather than dropping them solely because no gas record is present.

As a robustness outcome, we also construct a *strict dual-fuel total*. This measure is defined only in building-months in which both electricity and gas are observed, finite, and strictly positive:

$$T_{it}^{\text{strict}} = E_{it} + G_{it} \quad \text{if } E_{it} > 0 \text{ and } G_{it} > 0, \tag{2}$$

and is otherwise missing. The strict definition therefore avoids any zero-gas classification assumption, but it restricts the analysis to a smaller set of observed dual-fuel building-months and can exclude buildings that are genuinely non-gas.

Because the two constructions imply different eligible samples, we estimate common support and ATT inverse-probability weights separately for the structural-adjusted and strict total-energy outcomes. The structural-adjusted total is used as the primary combined-energy measure, while the strict dual-fuel total is reported as a robustness check. The total-energy analyses are restricted to public daycare centers and public health centers; hospitals are excluded from these combined-energy specifications. We apply the same two total-energy definitions when reporting the CSDID estimates and the TWFE and Sun–Abraham robustness estimates so that differences across estimators are not confounded by changes in outcome construction.

Because these two total-energy definitions generate different eligible samples, we also evaluate their weighting balance directly. Figure 6 reports the standardized-mean-difference diagnostics for daycare centers under both total-energy definitions and for health centers under the structural-adjusted definition. The corresponding post-weighting standardized differences are below the 0.10 threshold. A health-center strict dual-fuel model is not shown because a final common-support ATT-IPW sample could not be constructed for that building-type–outcome combination.

3.3 Staggered Difference-in-Differences and Event Study

Retrofit timing differs across participating buildings, so we estimate the effect of the program in a staggered-adoption framework using the longitudinal monthly energy-consumption data from 2018 to 2025. Treated buildings are non-cancelled 2021–2022 projects that satisfy the common-support restrictions in Section 3.1. Event time is measured in months relative to retrofit completion: event month 0 is the first full month after the end of construction, and months during construction are excluded from the estimation sample so that temporary closure or partial occupancy during works does not contaminate the estimates. We report dynamic effects for twelve months before and twelve months after treatment.

Preferred estimator. Conventional TWFE estimators can be biased when treatment timing is staggered and effects vary across cohorts or over time, because already-treated units implicitly serve as controls for newly treated units (Goodman-Bacon, 2021). Our preferred estimator is therefore the doubly robust estimator of Callaway and Sant'Anna (2021), which first estimates group-time average treatment effects

$$ATT(g,t) = \mathbb{E}\left[Y_{it}(1) - Y_{it}(0) \mid G_i = g\right], \tag{3}$$

where $G_i = g$ denotes the treatment cohort of building i and $Y_{it}(1)$ and $Y_{it}(0)$ are the potential log energy outcomes with and without the retrofit. Each $ATT(g,t)$ compares the change in outcomes of cohort g between a pre-treatment base period and period t with the corresponding change in a comparison group that is not

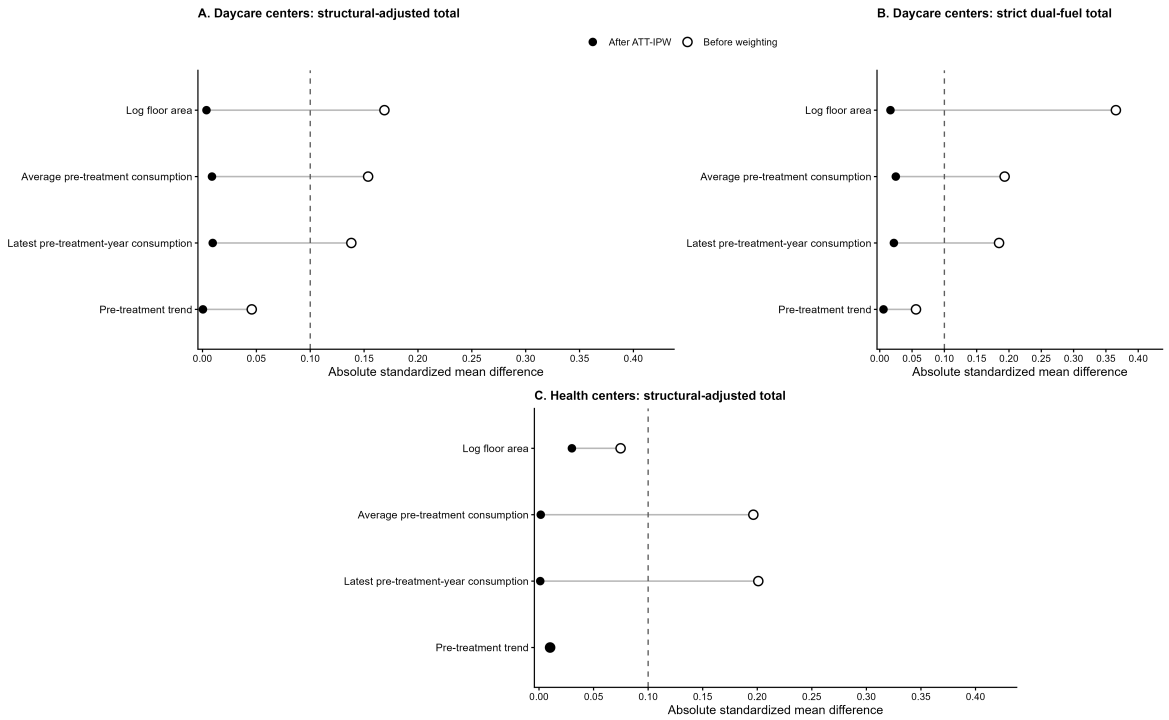


Figure 6: Covariate balance before and after ATT-IPW weighting: total-energy outcomes

Note: Panels A and B report daycare centers using the structural-adjusted and strict dual-fuel total-energy definitions, respectively. Panel C reports health centers using the structural-adjusted definition. Hollow circles indicate absolute standardized mean differences before weighting and filled circles indicate differences after ATT-IPW weighting. The dashed vertical line marks the 0.10 balance threshold. The structural-no-recorded-gas classification is fixed using pre-2021 information.

yet treated at t. Dynamic effects are obtained by aggregating across cohorts at each event time h,

$$ATT(h) = \sum_{g} w_{g,h}\, ATT(g, g+h), \tag{4}$$

where $w_{g,h}$ is the share of cohort g among cohorts observed at event time h. The summary effect reported in the tables is the aggregated ATT over event months 0–12. Estimation uses the outcome-specific ATT-IPW samples with universal base periods, and inference is based on 999 multiplier-bootstrap replications.

Comparison pools. The credibility of the estimates depends on the comparison group, so we report results for four comparison pools. The *not-yet-treated* pool consists of buildings selected in the 2023 and 2024 program years, observed only before their own construction starts. These buildings passed the same screening as treated buildings and are therefore likely to be similar on unobservables, but they are few and contribute only in the early part of the sample period. The second pool adds *cancelled* projects, that is, buildings whose applications were accepted but whose projects were cancelled and for which no construction start is observed. The third pool adds the ATT-IPW-weighted *never-treated* public buildings to the not-yet-treated buildings. The *full pool* combines not-yet-treated, cancelled, and never-treated buildings and is our preferred specification because it provides a comparison group throughout the event window while retaining the screened later-cohort and cancelled applicants.

TWFE and Sun–Abraham event studies. For comparison, we also estimate a conventional event-study specification

$$\log(E_{it}) = \sum_{\substack{h=-12 \\ h\neq -1}}^{12} \beta_h\, R_{it}\, \mathbf{1}\{t - T_i = h\} + \delta_1 HDD_{it} + \delta_2 CDD_{it} + \alpha_i + \gamma_t + \mu_{i,m(t)} + \varepsilon_{it}, \tag{5}$$

where E_{it} denotes energy consumption per square meter for building i in month t, T_i is the retrofit completion date, and $\mathbf{1}\{t - T_i = h\}$ equals one when observation (i, t) occurs h months relative to completion. Building fixed effects α_i absorb time-invariant differences across buildings; year-month fixed effects γ_t absorb common shocks such as aggregate weather, energy prices, and the COVID-19 pandemic; and building-by-calendar-month fixed effects $\mu_{i,m(t)}$ absorb building-specific seasonal profiles. Heating and cooling degree days control for local weather. The coefficient for $h = -1$ is normalized to zero, so all β_h are interpreted relative to the month immediately before treatment. We estimate equation (5) both by TWFE and with the interaction-weighted estimator of Sun and Abraham (2021), which constructs cohort-specific event-time effects and avoids comparisons between newly treated and already-treated buildings.

The coefficients on pre-treatment leads provide the main diagnostic for the design. After weighting and restricting the sample to common support, the absence of systematic differences in pre-treatment trends is consistent with the conditional parallel-trends assumption underlying the DID estimates.

3.4 Average Post-Retrofit Effects

To summarize the dynamic estimates, we report the average effect of the program over event months 0–12. For the Callaway–Sant'Anna estimator, this is the aggregated ATT described above. For the TWFE and Sun–Abraham estimators, it is the equal-weight average of the event-time coefficients $\beta_0, \ldots, \beta_{12}$ from equation (5). We report TWFE estimates both for the full analysis sample and for the outcome-specific ATT-IPW common-support sample. Consequently, the matched estimates are interpreted as average treatment effects for participating buildings that have comparable non-participating buildings within the observed common-support region.

Because the dependent variable is expressed in logs, the treatment coefficient β can be interpreted approximately as the percentage change in energy consumption associated with the retrofit; the exact percentage effect is $100 \times (\exp(\beta) - 1)$. Standard errors for TWFE estimates are clustered at the building level.

We estimate the models separately for structural-adjusted total delivered energy, electricity, and gas, and separately for daycare centers and health centers when examining heterogeneity by building type. Where adequate common support cannot be established for a particular building-type and outcome combination (health-center gas and all hospital outcomes), we do not report a weighted DID estimate, because the available non-participating buildings do not provide a credible empirical counterfactual.

3.5 Heterogeneous Effects by Retrofit Type

To examine heterogeneity in energy savings across retrofit measures, we follow the measure-level approach in Papineau et al. (2025). Instead of using a single post-retrofit indicator, we replace it with a set of indicator variables corresponding to the retrofit measures implemented in each building, such as insulation, windows, HVAC, lighting, and other upgrades.

The specification is

$$\log(E_{it}) = \sum_{j \in \mathcal{J}} \beta_j \, \text{Measure}_{ijt} + \delta_1 HDD_{it} + \delta_2 CDD_{it} + \alpha_i + \gamma_{ym} + \mu_{i,m(t)} + \varepsilon_{it}. \tag{6}$$

where Measure_{ijt} indicates whether retrofit measure $j \in \mathcal{J}$ was installed in building i and month t falls in the post-retrofit window (event months 0–12). Each coefficient β_j therefore measures the average change in log energy consumption associated with retrofit measure j, conditional on the other included measures, weather, building fixed effects α_i, year-month fixed effects γ_{ym}, and building-by-calendar-month fixed effects $\mu_{i,m(t)}$. The measures are grouped into windows and doors, wall and roof insulation, floor insulation, high-efficiency boilers, heat-recovery ventilation, high-efficiency HVAC, high-efficiency lighting, renewable energy, energy monitoring, cool roofs and solar shading, and ancillary works. We also report the implied effect of a *complete*

envelope retrofit, defined as the sum of the coefficients on windows and doors, wall or roof insulation, and floor insulation.

Because the dependent variable is expressed in logarithms, β_j can be interpreted approximately as the percentage change in energy consumption associated with measure j. The exact percentage change is given by

$$100 \times (\exp(\beta_j) - 1)\,.$$

This specification treats the contribution of individual retrofit measures as additive and does not explicitly model interactions among measures. Given the available sample size and substantial overlap in the combination of measures installed across buildings, we focus on these main effects as the primary measure-specific estimates.

3.6 Selection into the Program

Although ATT weighting improves balance on observed pre-treatment characteristics, participation in the Green Remodeling Program is not random. Program participation is voluntary and is also influenced by a program scoring rubric (Section 2.1). Consequently, participating buildings may differ systematically from non-participating buildings along characteristics that are related to both program selection and subsequent energy savings.

To characterize observable selection into the program, we estimate a probit model of Green Remodeling Program participation using the broader universe of public buildings:

$$Pr(GR_i = 1) = \Phi(\beta_0 + \beta_1 Area_i + \beta_2 Vintage_i + \beta_3 Intensity_i + \beta_4 Subsidy50_i), \tag{7}$$

where GR_i indicates participation in the Green Remodeling Program, $Area_i$ is total building floor area, $Vintage_i$ is the building construction-completion year, $Intensity_i$ is pre-retrofit annual electricity consumption per square meter, and $Subsidy50_i$ indicates eligibility for the 50% cost-share rate applicable to Seoul, central government, and other eligible public-sector institutions, relative to the 70% cost-share rate applicable to other local governments.

The selection regression serves two purposes. First, it describes which observable characteristics are associated with participation in the Green Remodeling Program. We use these estimates in Section 4.5 to examine whether program targeting is related to the heterogeneity in realized energy savings documented in Section 4.4. Second, the selection model provides an alternative characterization of participation probabilities that can be compared with the propensity-score specifications used to construct the main ATT-IPW samples.

The selection analysis should not be interpreted as eliminating all potential selection bias. Rather, it complements the common-support ATT-IPW and DID framework by documenting observable determinants of program participation and helping to characterize the population for which the estimated treatment effects are most relevant.

3.7 Machine-Learning Counterfactual Prediction

We complement the ATT-IPW staggered difference-in-differences analysis with a machine-learning (ML) counterfactual exercise following the prediction-based event-study framework of **?** and the ex-ante prediction logic in **?**. The two approaches serve different purposes. The CSDID estimates remain our primary causal estimates and use a comparison group to identify average treatment effects under conditional parallel trends. The ML exercise instead constructs a flexible prediction of each treated building's untreated energy path and therefore provides an independent check on the functional-form assumptions used to describe post-retrofit energy demand. It also allows us to compare data-driven realized savings with the engineering savings projected by the program's ECO2 model.

The ML sample is restricted to public daycare centers and public health centers in the 2021–2022 treatment cohorts, matching the facility types for which the primary structural-total analysis has adequate support. Hospitals are excluded from this exercise. We estimate models in energy-intensity levels (kWh/m^2) for electricity, observed-positive gas consumption, and structural-adjusted total delivered energy. We additionally estimate a strict dual-fuel total-energy specification as a robustness check.

For each outcome, we estimate two state-specific prediction functions. Let $m_0(X_{it})$ denote expected energy use in the untreated state and $m_1(X_{it})$ expected energy use in the post-retrofit state. The untreated model is trained on months -24 through -1 relative to construction start, excluding construction months. The treated-state model is trained on months $+1$ through $+12$ after retrofit completion. The final figures evaluate pre-retrofit fit over months -12 through -1 and post-retrofit outcomes over months $+1$ through $+12$.

The predictor set is intentionally limited to information that is available independently of realized retrofit performance. It includes building age and physical characteristics, floor area and other geometry measures, geographic location, facility type, calendar month, heating and cooling degree days, temperature, precipitation, humidity, wind, cloud cover, snow depth, and historical consumption summaries. The historical features include building-specific pre-period means, dispersion, seasonal means, trends, and calendar-month-specific consumption levels. Realized retrofit-scope variables and realized project costs are excluded from the prediction features to prevent post-treatment information leakage.

We compare five candidate algorithms: ordinary least squares, Lasso, random forest, gradient-boosted trees (XGBoost), and a feed-forward neural network. Model selection uses nested cross-validation grouped at the building level. We first divide buildings into five outer folds. Within each outer training sample, hyperpa-

rameters are selected using three inner building-grouped folds, after which the tuned model is refit and evaluated on the held-out outer-fold buildings. Because all observations from a building remain in the same fold, the procedure prevents the same building from appearing simultaneously in the training and test samples. The same outer fold assignments are reused across outcomes and model classes. The selected algorithm minimizes combined held-out root mean squared error (RMSE) across the untreated-state pre-retrofit predictions and the treated-state post-retrofit predictions.

For the final counterfactual exercise, let Y_{it} denote observed energy use and let $\widehat{Y}_{it}(0)$ and $\widehat{Y}_{it}(1)$ denote the cross-fitted predictions under the untreated and treated states, respectively. The ML-implied realized saving for a treated building-month is

$$\widehat{\tau}_{it}^{\text{realized}} = \widehat{Y}_{it}(0) - Y_{it}, \tag{8}$$

while the model-implied treatment-state saving is

$$\widehat{\tau}_{it}^{\text{ML}} = \widehat{Y}_{it}(0) - \widehat{Y}_{it}(1). \tag{9}$$

Both post-retrofit prediction states are evaluated under the same expected weather scenario so that their difference is not mechanically driven by realized weather shocks. Expected monthly weather is constructed from pre-2021 historical weather for the same building and calendar month, with district-, region-, and calendar-month fallbacks when a building-specific normal is unavailable. We retain predictions based on observed post-retrofit weather as a sensitivity check.

For graphical presentation, monthly outcomes are expressed as deviations from each building's historical same-calendar-month mean, calculated over months -36 through -13. This normalization removes persistent seasonal level differences from the figure but does not alter the underlying level-based model estimation. A credible counterfactual should reproduce the observed pre-retrofit path before treatment; we therefore report event-month pre-period prediction errors as a specification diagnostic.

The ML exercise should not be interpreted as eliminating all possible selection on unobservables. Rather, it complements the ATT-IPW/CSDID design by allowing a highly flexible outcome model and by evaluating predictive performance strictly out of sample. Agreement between the CSDID and ML results is informative because the two approaches construct the untreated path using substantially different empirical strategies. The comparison with ECO2 engineering projections then provides a direct measure of the engineering–realized performance wedge.

4 Results

4.1 Event Study

We begin with graphical evidence from the preferred Callaway and Sant'Anna (2021) event studies. The main event-study outcome is total delivered energy. As described in Section 3.2, the structural-adjusted measure is the primary specification because it retains buildings with strong pre-2021 evidence of electricity use but no recorded gas use, treating missing gas as zero only for buildings classified ex ante as structural no-gas. The strict dual-fuel measure requires both electricity and gas to be observed and positive and is used as a robustness check.

Figure 7 reports the estimates for the structural-adjusted total-energy outcome under the four comparison pools defined in Section 3.3. Two patterns stand out. First, when the comparison group contains never-treated buildings (bottom panels), the pre-treatment coefficients are small and statistically indistinguishable from zero. There is a modest hump between event months -6 and -4, with point estimates of about 0.05–0.06 log points, but the confidence intervals include zero and there is no monotonic trend into treatment. After retrofit, total energy use falls immediately by about 0.17 log points in the first full post-retrofit month and remains 0.10–0.14 log points below the counterfactual through event month 8, with confidence intervals that exclude zero. The estimates attenuate to about -0.06 to -0.07 in months 9–12, when the confidence intervals begin to include zero. Second, when the comparison group is restricted to not-yet-treated buildings, with or without cancelled projects (top panels), the estimates are much less precise. Total energy use falls in event month 0, but the subsequent estimates are close to zero and even positive in months 6–7 before turning negative again at the end of the window. This sensitivity reflects the small size of the not-yet-treated pool: only 2023–2024 projects observed before their own construction start contribute, so the comparison group thins out quickly as event time increases. We therefore regard the full pool, which retains the screened later-cohort and cancelled applicants while adding never-treated buildings, as the preferred specification.

Figure 8 repeats the analysis using the strict dual-fuel outcome. With never-treated buildings in the comparison pool, the post-retrofit estimates are again negative, ranging from about -0.13 in event month 0 to -0.04 to -0.10 thereafter, and they are of similar magnitude to the structural-adjusted estimates. However, they are less precisely estimated, and the pre-treatment coefficients are more variable, with negative leads of about -0.05 to -0.08 in the earliest pre-treatment months that drift toward zero just before treatment. With not-yet-treated controls only, the pre-treatment leads are larger and the post-treatment estimates are noisy. This pattern is consistent with the smaller and more selective sample created by requiring both fuels to be observed and positive in each building-month. We therefore use the structural-adjusted total as the primary combined-energy result and interpret the strict dual-fuel figure as a robustness check on the construction of total delivered energy.

The component-outcome CSDID event studies are reported in Appendix Figures C.1 and C.2. For electricity, the full-pool estimates show flat pre-treatment coefficients and an immediate decline of about 0.24 log points in event month 0, followed by reductions of 0.12–0.17 log points that remain statistically significant in every month through event month 12. For gas, the picture is different. The pre-treatment leads are negative (about −0.05 to −0.10) in the earliest months and rise toward zero before treatment, gas use falls by about 0.14 log points in event month 0, and the subsequent estimates are close to zero or positive and statistically insignificant. Alternative event studies estimated using conventional TWFE and the heterogeneity-robust estimator of Sun and Abraham (2021) are reported in Appendix Figures C.3–C.6.

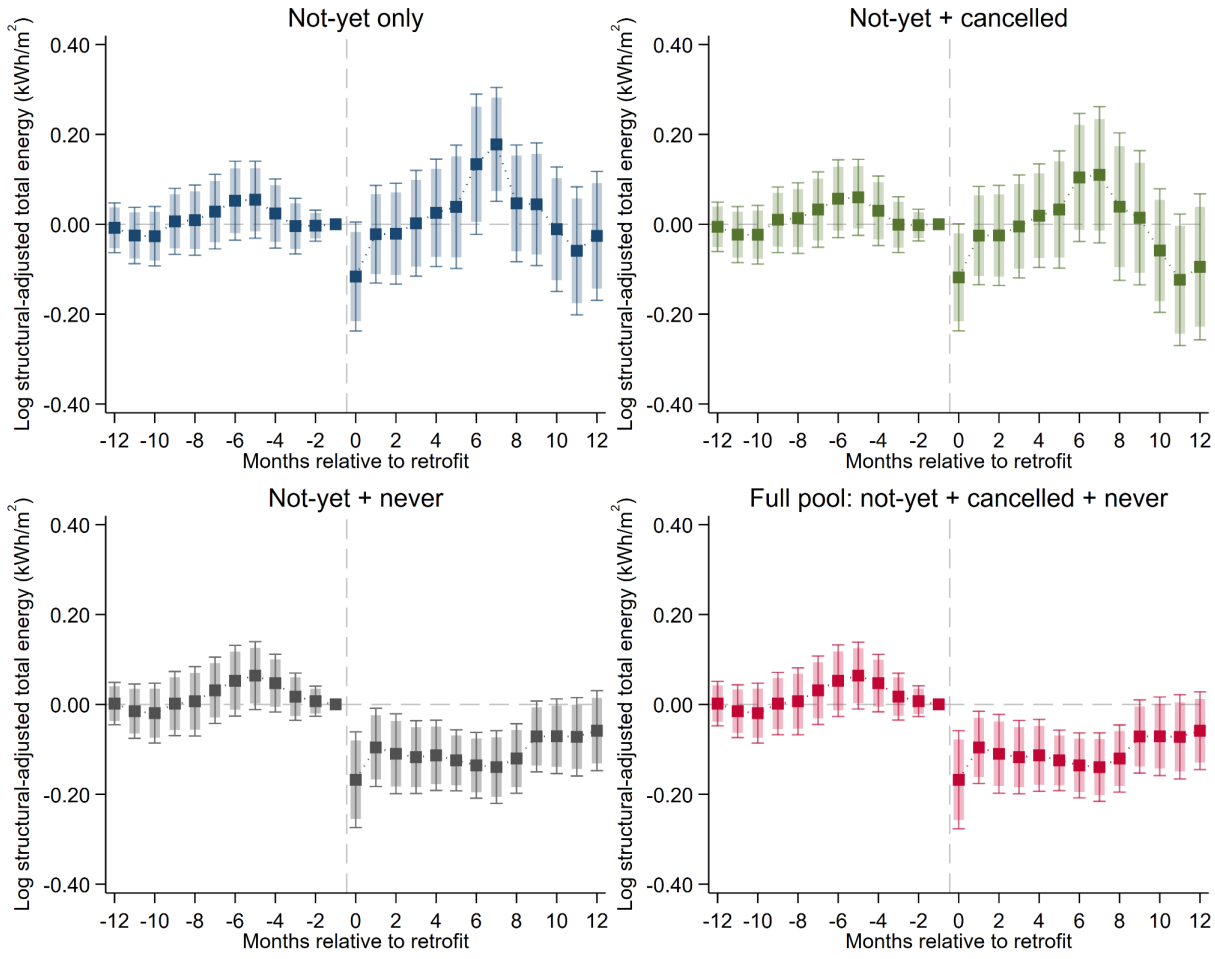


Figure 7: CSDID Event Study: Structural-Adjusted Total Delivered Energy

Note: The figure reports dynamic treatment-effect estimates using the staggered-adoption difference-in-differences estimator of Callaway and Sant'Anna (2021). The outcome is the log of structural-adjusted total delivered energy defined in Section 3.2. The four panels vary the comparison pool across not-yet-treated only (2023–2024 projects observed before their own construction start); not-yet-treated plus cancelled projects with no observed construction start; not-yet-treated plus ATT-IPW-weighted never-treated buildings; and the full pool of not-yet-treated, cancelled, and never-treated buildings. Event month 0 is the first full month after retrofit completion. Vertical bars report 95 percent confidence intervals based on 999 multiplier-bootstrap replications. Construction months are excluded.

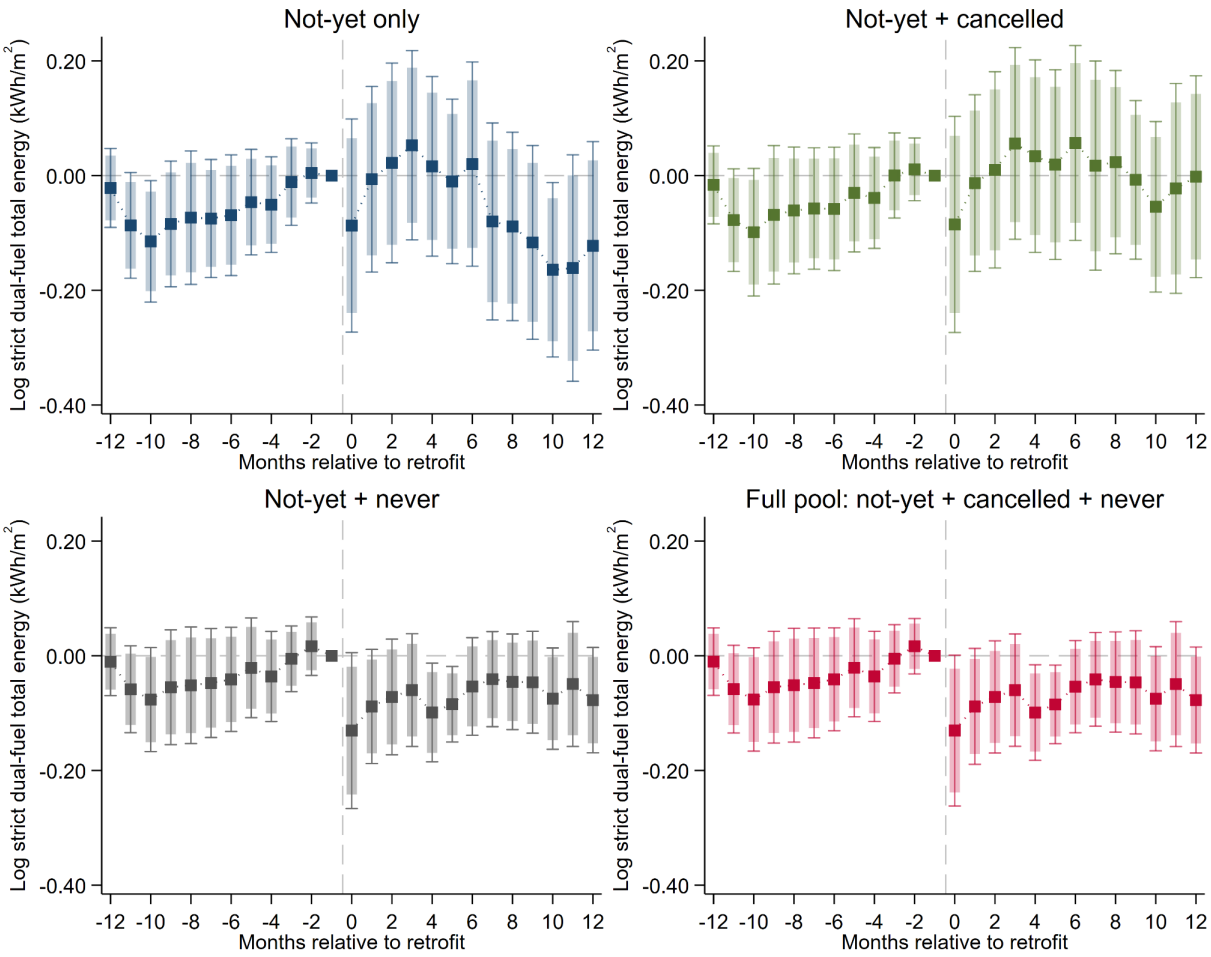


Figure 8: CSDID Event Study: Strict Dual-Fuel Total Delivered Energy

Note: The figure reports the same CSDID event-study specifications as Figure 7, but the outcome is the log of strict dual-fuel total delivered energy. The strict total is observed only when electricity and gas are both observed, finite, and positive. Vertical bars report 95 percent confidence intervals. Construction months are excluded.

4.2 Average Effects of Program Participation

Table 4 summarizes the average post-retrofit effects over event months 0–12. Column (1) reports TWFE estimates for the full analysis sample, column (2) TWFE estimates for the matched ATT-IPW sample, column (3) the Sun–Abraham estimate for the matched sample, and column (4) the preferred Callaway–Sant'Anna estimate using the full comparison pool.

Panel A shows that the program substantially reduces total delivered energy. In the preferred specification, structural-adjusted total energy use falls by 0.107 log points, or 10.2%, and the effect is statistically significant at the 1% level. The TWFE and Sun–Abraham estimates are somewhat larger, at −0.146 to −0.155 log points (13.6–14.4%). Matching barely changes the TWFE estimate (−0.146 in the full sample versus −0.148 in the matched sample), which suggests that observable differences between participants and non-participants do not drive the results.

Panel B shows that electricity savings are remarkably stable across estimators. The estimated effect ranges from −0.148 to −0.154 log points across all four columns, implying reductions of 13.7–14.3%; the preferred Callaway–Sant'Anna estimate is −0.150 (13.9%). The close agreement between the TWFE, Sun–Abraham, and Callaway–Sant'Anna estimates indicates that staggered adoption and treatment-effect heterogeneity do not materially bias the conventional estimates for electricity.

Panel C shows that the gas results are less robust. The TWFE and Sun–Abraham estimates indicate reductions of 0.088–0.098 log points (8.4–9.4%), statistically significant at the 5% or 1% level. However, the preferred Callaway–Sant'Anna estimate is 0.016 with a standard error of 0.079, which is statistically indistinguishable from zero and also from the TWFE estimates. Together with the gas event study in Appendix Figure C.2, which shows a decline only in the first post-retrofit months and upward-drifting pre-treatment leads, this suggests that the gas reductions estimated by the regression-based estimators should be interpreted with caution. The gas sample is also much smaller and consists almost entirely of daycare centers, because health centers do not provide a sufficient sample with usable pre-treatment gas histories.

Taken together, the results indicate that the Green Remodeling Program reduces total delivered energy by roughly 10–14% and electricity by about 14% in the first post-retrofit year, whereas the evidence for gas savings depends on the estimator. That total-energy savings are somewhat smaller than electricity savings in the preferred specification is consistent with the weaker gas response.

Table 4: Program Effects on Energy Consumption

Model	(1)	(2)	(3)	(4)
Panel A: Total Delivered Energy				
Average post-retrofit effect	-0.146***	-0.148***	-0.155***	-0.107***
	(0.025)	(0.027)	(0.026)	(0.034)
Implied change (%)	-13.6	-13.7	-14.4	-10.2
Panel B: Electricity Consumption				
Average post-retrofit effect	-0.154***	-0.148***	-0.154***	-0.150***
	(0.026)	(0.027)	(0.026)	(0.033)
Implied change (%)	-14.2	-13.7	-14.3	-13.9
Panel C: Gas Consumption				
Average post-retrofit effect	-0.088**	-0.090**	-0.098***	0.016
	(0.042)	(0.045)	(0.035)	(0.079)
Implied change (%)	-8.4	-8.6	-9.4	1.6
Estimator	TWFE	TWFE	S&A	C&S
Matched sample	No	Yes	Yes	Yes
Fixed effects				
Building	Yes	Yes	Yes	–
Year-Month	Yes	Yes	Yes	–
Building × Calendar Month	Yes	Yes	Yes	–
Weather controls	Yes	Yes	Yes	–

Note: The table reports average post-retrofit effects of the Green Remodeling Program on the natural logarithm of monthly energy consumption (kWh/m^2). Panel A reports structural-adjusted total delivered energy, Panel B electricity, and Panel C gas. Column (1) reports conventional two-way fixed-effects (TWFE) estimates using the full analysis sample. Column (2) reports TWFE estimates using the matched sample. Column (3) reports the Sun and Abraham (2021) interaction-weighted estimator using the matched sample. Column (4) reports the doubly robust Callaway and Sant'Anna (2021) estimator using the matched sample and the full comparison pool (not-yet-treated, cancelled, and never-treated buildings) and is the preferred specification. For Columns (1)–(3), the reported coefficient is the equal-weight average of event-time effects over event months 0–12. For Column (4), the reported coefficient is the aggregated ATT over event months 0–12. Construction months are excluded. The matched sample refers to the outcome-specific ATT-IPW common-support sample. TWFE and Sun–Abraham specifications include heating and cooling degree days and absorb building, year-month, and building-by-calendar-month fixed effects. Standard errors for Columns (1)–(3) are clustered at the building level. Callaway–Sant'Anna inference uses 999 multiplier-bootstrap repetitions. Implied percentage changes are calculated as $100[\exp(\hat{\beta}) - 1]$. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

4.2.1 Building-Type-Specific Savings Estimates

Table 5 reports estimates separately for public daycare centers (columns 1–4) and public health centers (columns 5–8). Hospitals are not reported because no hospital outcome yields a weighted comparison group that satisfies the common-support and balance requirements.

For health centers, the program produces large and very stable savings. Total delivered energy falls by 0.164–0.178 log points (15.1–16.3%) and electricity by 0.159–0.176 log points (14.7–16.1%) across all four estimators, and every estimate is statistically significant at the 1% level. The preferred Callaway–Sant'Anna estimates imply reductions of 15.1% for total energy and 14.7% for electricity. Gas effects for health centers cannot be estimated because the gas sample does not contain enough treated health centers with usable pre-treatment gas histories.

For daycare centers, savings are smaller and somewhat more sensitive to the estimator. Electricity use falls by 0.105–0.127 log points (9.9–11.9%), and the effect is statistically significant in all four columns, including the preferred Callaway–Sant'Anna specification (9.9%, significant at the 5% level). Total delivered energy falls by 0.113–0.123 log points (10.7–11.5%) in the TWFE and Sun–Abraham specifications, but the Callaway–Sant'Anna estimate is smaller (-0.074, or 7.1%) and not statistically significant at conventional levels. As in the pooled sample, daycare-center gas use falls by 8.4–9.4% in the TWFE and Sun–Abraham specifications, but the Callaway–Sant'Anna estimate (-0.011) is close to zero. Because health-center gas is not estimable, the daycare-center gas estimates essentially coincide with the pooled gas estimates in Table 4.

The larger savings in health centers are consistent with their higher pre-treatment energy intensity and with the engineering literature, which predicts the largest performance gains for public health centers (Woo et al., 2024). They also motivate the seasonal analysis below.

Table 5: Program Effects on Energy Consumption by Building Type

	Public Daycare Center				Public Health Center			
Model	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Panel A: Total Delivered Energy								
Average post-retrofit effect	-0.123***	-0.113***	-0.114***	-0.074	-0.171***	-0.165***	-0.178***	-0.164***
	(0.036)	(0.038)	(0.035)	(0.046)	(0.033)	(0.035)	(0.032)	(0.045)
Implied change (%)	-11.5	-10.7	-10.8	-7.1	-15.7	-15.2	-16.3	-15.1
Panel B: Electricity Consumption								
Average post-retrofit effect	-0.127***	-0.120***	-0.116***	-0.105**	-0.170***	-0.164***	-0.176***	-0.159***
	(0.046)	(0.046)	(0.043)	(0.052)	(0.033)	(0.034)	(0.030)	(0.042)
Implied change (%)	-11.9	-11.3	-10.9	-9.9	-15.6	-15.1	-16.1	-14.7
Panel C: Gas Consumption								
Average post-retrofit effect	-0.087**	-0.090**	-0.098***	-0.011	–	–	–	–
	(0.042)	(0.045)	(0.035)	(0.072)				
Implied change (%)	-8.4	-8.6	-9.4	-1.1	–	–	–	–
Estimator	TWFE	TWFE	S&A	C&S	TWFE	TWFE	S&A	C&S
Matched sample	No	Yes	Yes	Yes	No	Yes	Yes	Yes
Fixed effects								
Building	Yes	Yes	Yes	–	Yes	Yes	Yes	–
Year-Month	Yes	Yes	Yes	–	Yes	Yes	Yes	–
Building × Calendar Month	Yes	Yes	Yes	–	Yes	Yes	Yes	–
Weather controls	Yes	Yes	Yes	–	Yes	Yes	Yes	–

Note: The table reports average post-retrofit effects of the Green Remodeling Program separately for public daycare centers and public health centers. The dependent variables are the natural logarithms of monthly energy consumption (kWh/m^2). Panel A reports structural-adjusted total delivered energy, Panel B electricity, and Panel C gas. Within each building type, the first column reports conventional two-way fixed-effects (TWFE) estimates using the full building-type-specific analysis sample; the second reports TWFE estimates using the matched sample; the third reports the Sun and Abraham (2021) interaction-weighted estimator using the matched sample; and the fourth reports the doubly robust Callaway and Sant'Anna (2021) estimator using the matched sample and the full comparison pool. For TWFE and Sun–Abraham, the reported coefficient is the equal-weight average of event-time effects over event months 0–12. For Callaway–Sant'Anna, the reported coefficient is the balanced aggregated ATT over event months 0–12. Construction months are excluded. TWFE and Sun–Abraham specifications include heating and cooling degree days and absorb building, year-month, and building-by-calendar-month fixed effects. Standard errors for TWFE are clustered at the building level; Sun–Abraham uses two-way clustering by building and calendar month. Callaway–Sant'Anna inference uses 999 multiplier-bootstrap repetitions. The matched sample refers to the outcome-specific ATT-IPW common-support sample. Implied percentage changes are calculated as $100[\exp(\hat{\beta}) - 1]$. Gas effects for public health centers are not reported because the relevant building-type/outcome analysis sample does not contain treated health-center buildings with estimable gas outcomes under the required support restrictions. Dashes indicate that an estimate or regression fixed-effect entry is not applicable or not estimable. S&A denotes Sun–Abraham and C&S denotes Callaway–Sant'Anna. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

4.2.2 Seasonal Patterns by Building Type

To understand when in the year the savings occur, Figure 9 plots calendar-month means of log consumption for participating buildings in the twelve months before construction, for the same buildings in the first thirteen full post-retrofit months, and for the ATT-IPW-weighted comparison group. Because the figure is indexed by calendar month rather than by event time, it shows whether savings are concentrated in the heating or cooling season. The figure is descriptive: unlike the DID estimates, it does not net out common shocks between the pre- and post-retrofit years.

For both building types, post-retrofit consumption lies below pre-retrofit consumption in nearly every calendar month and for every outcome. The seasonal profile of the gap relative to the comparison group, however, differs. In health centers (Panels D and E), post-retrofit total energy and electricity are close to the comparison group during the heating months but fall clearly below it from June to September; in July and August the difference is about 0.15 log points. This suggests that the large health-center savings in Table 5 are concentrated in the cooling season, which is consistent with envelope improvements and high-efficiency cooling equipment reducing air-conditioning loads. In daycare centers (Panels A–C), post-retrofit consumption falls below the comparison group mainly during the heating season (November through February for total energy and gas), while post-retrofit electricity and total consumption are slightly above the comparison group in July through September. Daycare-center gas use, which is concentrated in winter, falls in the heating months, but the post-retrofit and comparison profiles are close to one another for most of the year, consistent with the weak gas results in the Callaway–Sant'Anna specification.

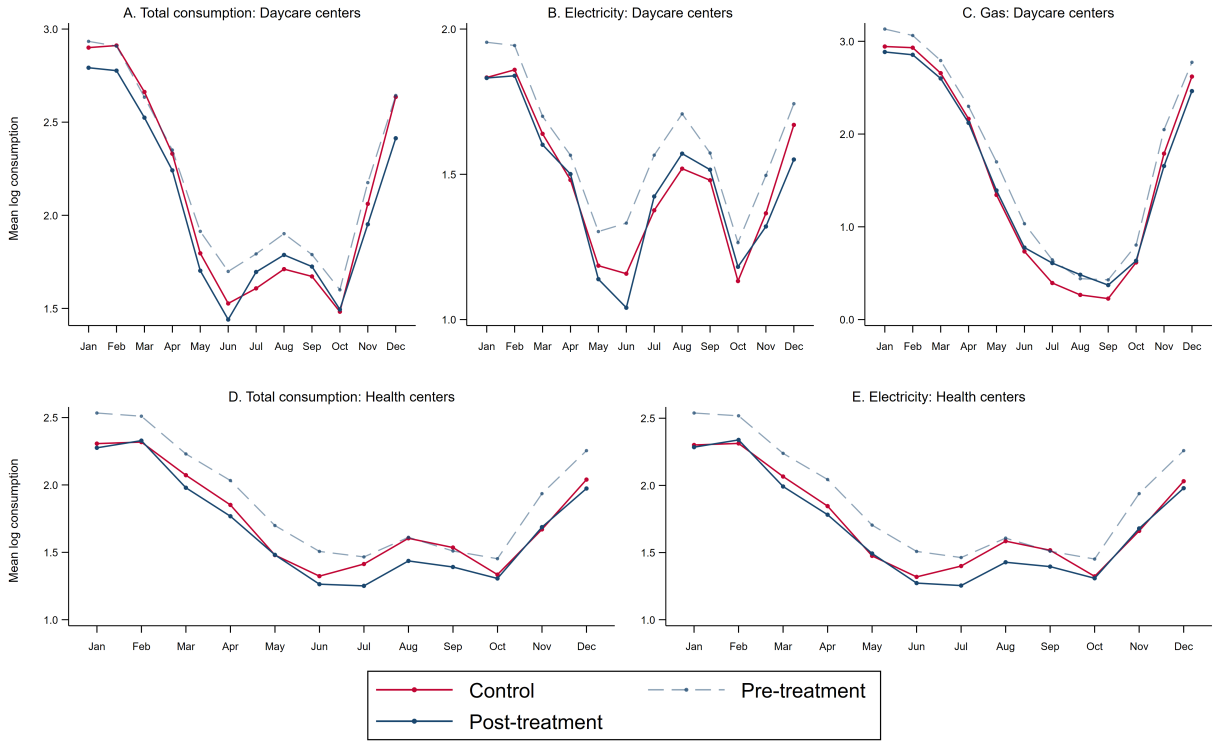


Figure 9: Seasonal Pattern of Energy Consumption by Building Type

Note: The figure reports calendar-month means of log monthly energy consumption for public daycare centers and public health centers. Total consumption refers to the structural-adjusted total delivered-energy measure used as the primary combined-energy outcome. The light-blue dashed line shows participating buildings during the 12 months before construction begins (event months −12 to −1); the dark-blue solid line shows participating buildings during the first 13 full post-retrofit months (event months 0 to 12); and the red solid line shows the ATT-IPW weighted comparison group. Comparison buildings are drawn from the full pool of 2023–2024 not-yet-treated buildings, cancelled projects with no observed construction start, and never-treated buildings; future-treated buildings contribute only while untreated. Construction-period observations are excluded, and each outcome uses its corresponding common-support ATT-IPW sample. Health-center gas is omitted because no estimable matched treated sample is available under the gas-outcome support restrictions.

4.2.3 Robustness to the Construction Window

The average construction duration of 280 days raises the possibility that early post-treatment months are coded while buildings are partially closed, generating a mechanical dip in measured consumption. Our design addresses this concern in two ways. First, all construction months are excluded from the estimation sample, and event month 0 is defined as the first full month after project completion. Second, the event studies show that the savings are not confined to the first post-retrofit month. Although the largest single-month reduction occurs in event month 0, which could partly reflect commissioning or delayed re-occupancy, the structural-adjusted total-energy estimates remain between -0.10 and -0.14 log points through event month 8, and the electricity estimates remain between -0.12 and -0.17 log points and statistically significant through event month 12 (Figure 7 and Appendix Figure C.1). A short-lived vacancy effect cannot explain reductions that persist for a full year after completion.

4.3 Retrofit-Specific Savings Estimates

Figure 10 and Table 6 report measure-by-measure estimates from equation (6). Because the measure indicators enter jointly and many buildings receive similar packages, the estimates should be interpreted as conditional associations rather than as separately identified causal effects of each technology. Overall, the results indicate substantial heterogeneity: a few measures are associated with sizeable reductions in consumption, whereas most estimates are small and imprecise.

For total delivered energy, the largest reductions are associated with renewable energy (-0.127 log points, or 11.9%), windows and doors (-0.113, 10.7%), and wall and roof insulation (-0.107, 10.1%), all statistically significant at the 5% level. For electricity, the largest and most precisely estimated reductions are associated with renewable energy (-0.144, 13.4%) and wall and roof insulation (-0.119, 11.2%), the latter significant at the 1% level. The renewable-energy estimate is consistent with on-site generation offsetting grid electricity purchases. By contrast, several measures that are often emphasized in engineering assessments, including high-efficiency boilers, heat-recovery ventilation, high-efficiency HVAC, energy monitoring, and floor insulation, have estimates close to zero and statistically indistinguishable from no effect for every outcome.

For gas, no individual measure is associated with a reduction that is statistically significant at the 5% level. Wall and roof insulation has a point estimate of -0.108 (10.2%), but it is imprecisely estimated, and the cool roof and solar shading estimate (-0.192, significant at the 10% level) is based on a small number of buildings. This is consistent with the weak gas results in the aggregate estimates.

One measure is associated with higher consumption. High-efficiency lighting is associated with a 9.3% increase in total energy use (significant at the 10% level) and a positive but insignificant effect on electricity. Because lighting retrofits mechanically reduce lighting loads, this pattern is more likely to reflect behavioral or operational responses, such as longer operating hours or brighter lighting, or the types of buildings that

select lighting upgrades, than the technology itself. It underscores that not all retrofit components translate into reductions in measured consumption.

Finally, buildings that undergo a complete envelope retrofit, defined as the combination of windows and doors, wall or roof insulation, and floor insulation, exhibit the largest reductions. The bundle is associated with a 0.160 log-point (14.8%) reduction in total energy use, significant at the 1% level, and a 0.129 log-point (12.1%) reduction in electricity use, significant at the 5% level, although its estimated effect on gas use is close to zero. Taken together, the measure-by-measure results suggest that program savings are driven primarily by envelope measures and on-site renewable generation, and that savings are largest when envelope measures are adopted jointly rather than in isolation.

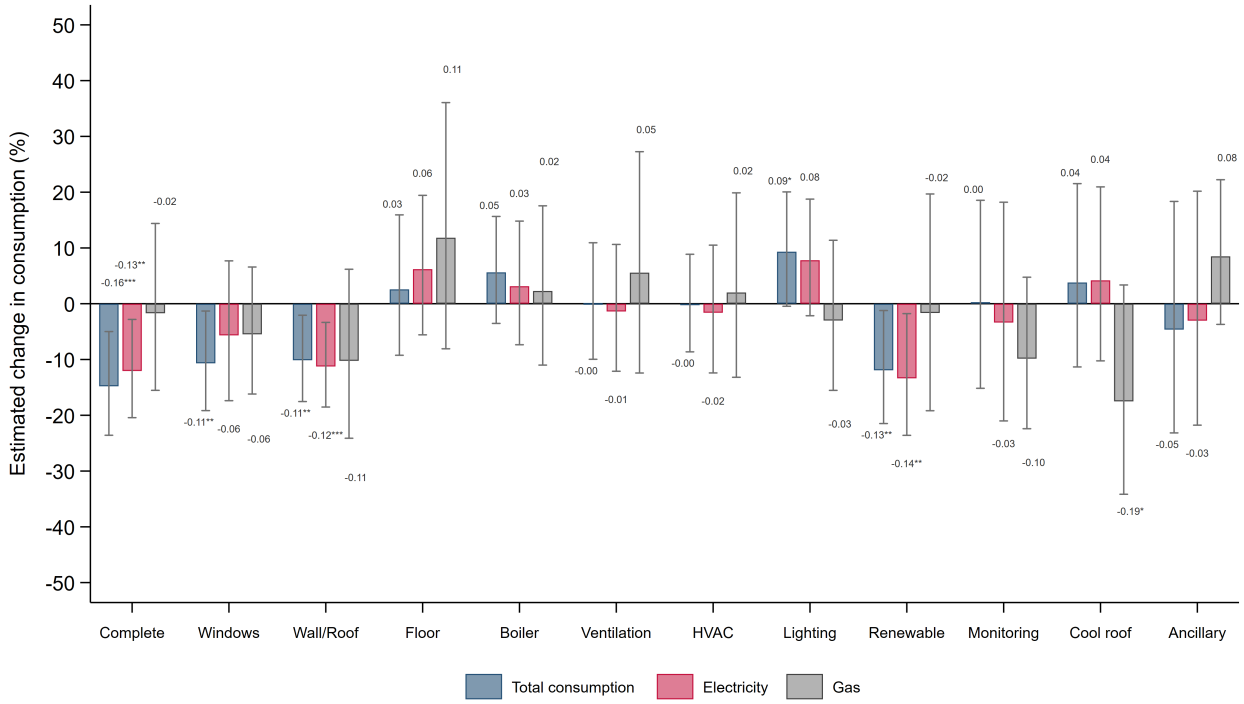


Figure 10: Energy Savings by Retrofit Measure

Note: The figure reports estimated changes in monthly energy consumption associated with individual Green Remodeling retrofit measures. Estimates are based on the outcome-specific ATT-IPW common-support samples. Total consumption denotes the structural-adjusted total delivered-energy measure used as the primary combined-energy outcome. Bars transform the estimated log-point coefficients into percentage changes as $100[\exp(\hat{\beta}) - 1]$, and vertical whiskers report the corresponding transformed 95% confidence intervals. Numerical labels report the underlying log-point coefficients, with significance indicated by stars. The retrofit-measure indicators are included jointly in each regression. The post-retrofit period covers event months 0–12 beginning with the first full month after project completion, and construction-period observations are excluded. All specifications control for heating and cooling degree days and absorb building, year-month, and building-by-calendar-month fixed effects; standard errors are clustered at the building level. "Complete envelope retrofit" is the imputed broad measure-by-measure bundle combining windows/doors, wall-or-roof insulation, and floor insulation. Ancillary works combine non-core measures including asbestos survey, demolition/waste disposal, other Green Remodeling works, landscaping, structural reinforcement, EPD finish materials, water-saving devices, instantaneous water heaters, and electrical capacity upgrades. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 6: Energy Savings by Retrofit Measure

Energy Type	Total consumption	Electricity	Gas
Retrofit measures			
Windows & doors	-0.113**	-0.058	-0.056
	(0.051)	(0.068)	(0.061)
Wall & roof insulation	-0.107**	-0.119***	-0.108
	(0.044)	(0.044)	(0.086)
Floor insulation	0.025	0.060	0.112
	(0.062)	(0.060)	(0.100)
High-efficiency boiler	0.055	0.031	0.023
	(0.046)	(0.055)	(0.071)
Heat recovery ventilation	-0.001	-0.014	0.054
	(0.053)	(0.059)	(0.095)
High-efficiency HVAC	-0.003	-0.016	0.020
	(0.045)	(0.059)	(0.082)
High-efficiency lighting	0.089*	0.075	-0.031
	(0.048)	(0.049)	(0.071)
Renewable energy	-0.127**	-0.144**	-0.017
	(0.059)	(0.064)	(0.100)
Energy monitoring	0.003	-0.034	-0.104
	(0.085)	(0.103)	(0.077)
Cool roof & solar shading	0.037	0.041	-0.192*
	(0.080)	(0.076)	(0.115)
Ancillary works	-0.048	-0.031	0.082
	(0.110)	(0.110)	(0.061)
Bundle			
Complete envelope retrofit	-0.160***	-0.129**	-0.017
	(0.056)	(0.051)	(0.077)
Fixed effects			
Building	Yes	Yes	Yes
Year-Month	Yes	Yes	Yes
Building × Calendar Month	Yes	Yes	Yes
Weather controls	Yes	Yes	Yes

Note: The table reports measure-specific estimates from the outcome-specific ATT-IPW common-support samples. The dependent variables are the natural logarithms of monthly structural-adjusted total delivered energy, electricity consumption, and gas consumption, all measured in kWh/m^2. Each coefficient is the interaction between the post-retrofit period and the indicated retrofit measure. The post-retrofit period covers event months 0–12 beginning with the first full month after project completion, and construction-period observations are excluded. Models control for heating and cooling degree days and absorb building, year-month, and building-by-calendar-month fixed effects. Standard errors clustered at the building level are reported in parentheses. The complete envelope retrofit estimate is the imputed broad measure-by-measure bundle combining windows/doors, wall-or-roof insulation, and floor insulation. Ancillary works combine non-core measures including asbestos survey, demolition/waste disposal, other Green Remodeling works, landscaping, structural reinforcement, EPD finish materials, water-saving devices, instantaneous water heaters, and electrical capacity upgrades. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

4.4 Heterogeneity by Vintage, Cohort, and Baseline Intensity

Beyond building-type and retrofit-measure heterogeneity, realized savings may vary with building age, with the program cohort in which the building was retrofitted, and with the building's baseline energy intensity. Figure 11 reports estimated effects on log total energy (Panels A–C), log electricity (Panels D–F), and log gas (Panels G–I) consumption separately for daycare centers and health centers. Effects are estimated from the matched-sample TWFE specification interacted with terciles of building vintage, the program cohort, and terciles of pre-2021 energy intensity.

Three patterns emerge. First, health centers show larger savings than daycare centers in almost every subgroup, so the building-type difference in Table 5 is not driven by a particular vintage or cohort. Among health centers, the oldest buildings show the largest reductions in total energy and electricity (about 0.33 log points), but reductions are also large and statistically significant for the middle and youngest terciles. Among daycare centers, savings are largest and statistically significant in the middle vintage tercile (about 0.21 log points for total energy and 0.23 for electricity), while the estimates for the oldest and youngest terciles are smaller and imprecise. Second, savings are similar across the 2021 and 2022 cohorts, with overlapping confidence intervals. The 2022 daycare estimates are imprecise, reflecting the smaller number of 2022 projects with a full year of post-retrofit data. Third, electricity savings increase with baseline intensity: buildings in the lowest pre-treatment-intensity tercile show small and statistically insignificant electricity effects, whereas buildings in the middle and high terciles show significant reductions of about 0.15–0.28 log points. For total energy, the largest reductions are in the middle tercile for both building types. For daycare-center gas, the only significant reduction is in the lowest-intensity tercile, and none of the vintage or cohort estimates are statistically significant.

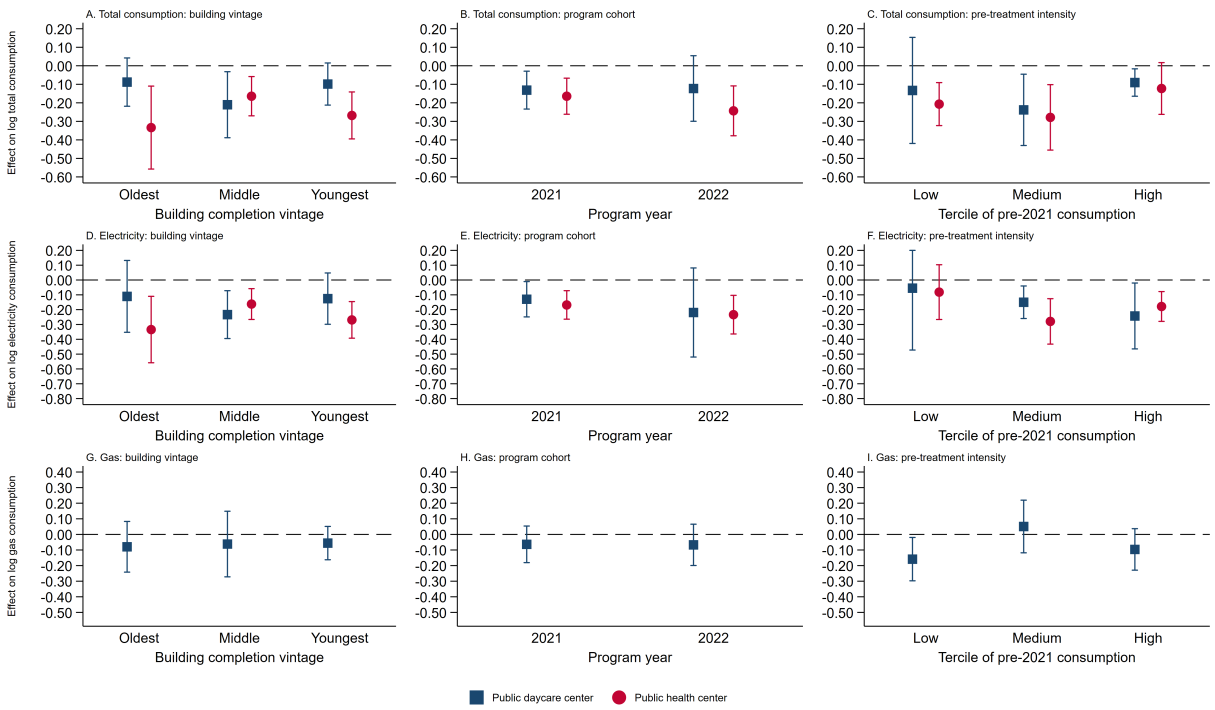


Figure 11: Heterogeneity by Vintage, Cohort, and Pre-treatment Intensity

Note: The figure reports heterogeneity in the estimated effects of the Green Remodeling Program across building vintage, program cohort, and pre-treatment energy intensity. Estimates are shown separately for total consumption, electricity, and gas, and by public daycare centers and public health centers; hospitals are excluded. Total consumption refers to the structural-adjusted total delivered-energy measure used as the primary combined-energy outcome. Each point reports the equal-weight average treatment effect over event months 0–12, beginning with the first full month after project completion, and vertical bars denote 95% confidence intervals. Construction-period observations are excluded. Estimates are based on outcome-specific ATT-IPW common-support samples and use the matched comparison group of never-treated buildings and cancelled projects with no observed construction start. All specifications control for heating and cooling degree days and absorb building, year-month, and building-by-calendar-month fixed effects; standard errors are clustered at the building level. Building vintage and pre-treatment energy intensity are divided into terciles. Pre-treatment intensity is calculated using pre-2021 consumption. Program-cohort estimates are reported for the 2021 and 2022 treatment cohorts only. Cells that do not satisfy the minimum support requirements are not plotted.

4.5 Selection into the Program

Table 7 reports average marginal effects from the probit selection regression specified in Section 3.6. Four findings emerge.

Table 7: Probit Selection Regression — Average Marginal Effects

Variable	**dy/dx**	**Std. err.**	**z**	**P > |z|**
Total floor area (m^2)	4.38e−07	2.33e−06	0.19	0.851
Building completion year	0.000333	0.000567	0.59	0.557
Pre-retrofit energy intensity (kWh/m^2)	−0.00115	0.000620	−1.86	0.063
Subsidy rate = 50% (vs. 70%)	−0.0719	0.0402	−1.79	0.074

Note: The dependent variable is an indicator for acceptance into the Green Remodeling Program. Marginal effects are computed from a probit model estimated on the universe of public buildings ($N = 645$, 635 treated buildings and 11 cancelled). Standard errors are computed using the observed information matrix. The Subsidy50 indicator equals one for buildings eligible for the 50% cost-share rate (Seoul, central government, and public-sector institutions) and zero for buildings eligible for the 70% rate (other local governments).

First, selection is not random. The model jointly rejects the null of no selection ($\chi^2(4) = 18.62$, $p = 0.001$), indicating that the weighting and comparison-group choices in Section 3.1 carry identification work beyond what the unweighted summary statistics in Table 3 suggest. The weighted estimates are therefore best interpreted as conditional on the observable selection process documented here.

Second, the cost-share rate is the strongest predictor of acceptance. Buildings under the 50% subsidy rate (Seoul, central government, and public institutions) are approximately 7.2 percentage points less likely to be accepted than buildings under the 70% rate (other local governments), consistent with the view that the residual cost burden faced by applicant institutions is a binding constraint on participation. This pattern suggests that participation is shaped by the financial capacity of public institutions to co-fund retrofits as well as by the engineering characteristics of the buildings themselves.

Third, higher pre-retrofit energy intensity reduces the probability of acceptance. A one-kWh/m^2 increase in baseline energy intensity lowers the probability of acceptance by approximately 0.12 percentage points (significant at the 10% level). Combined with the heterogeneity results in Section 4.4, which show that electricity savings are small and insignificant in the lowest-intensity tercile and larger in the middle and high terciles, this finding points to a potential mis-targeting pattern: the selection process may channel subsidies away from the higher-intensity buildings where electricity savings tend to be larger.

Fourth, floor area and building vintage have small and statistically insignificant effects on acceptance.

4.6 Threats to Identification and Robustness

The validity of the DID estimates rests on the assumption that, absent the retrofit, participating buildings would have followed the same consumption path as the weighted comparison buildings. We examine five threats to this assumption: unobserved differences between buildings selected in different years, free-riding, spatial spillovers, behind-the-meter solar generation, and common shocks such as COVID-19.

Placebo test of the not-yet-treated counterfactual. If the timing of selection were correlated with pre-existing consumption trends, later cohorts would be poor counterfactuals for earlier ones. We test this with an early-cohort placebo. We restrict the sample to the 2023 and 2024 cohorts, assign the 2023 cohort a placebo treatment date 24 months before its actual completion month, and use the 2024 cohort as the untreated comparison group. The estimation window ends before the earliest actual construction start, so no building has yet been exposed to retrofit construction. Figure 12 shows that the placebo event-study coefficients fluctuate around zero with wide confidence intervals and no systematic post-placebo decline. The aggregated placebo ATT is -0.036 (SE 0.132, $p = 0.784$). Thus, buildings selected in different program years do not display differential consumption trends before construction.

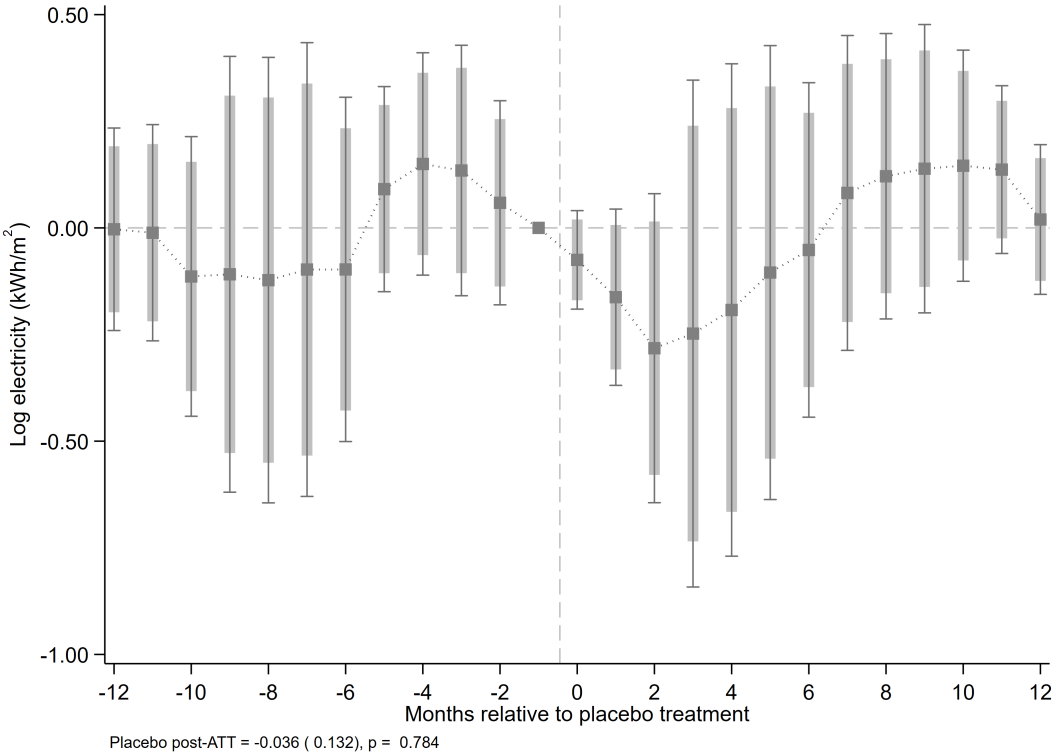


Figure 12: Placebo Test of the Not-Yet-Treated Counterfactual: Electricity

Note: The figure reports an early-cohort placebo test using the Callaway and Sant'Anna (2021) estimator. The outcome is log electricity consumption per square meter. The sample is restricted to the 2023 and 2024 retrofit cohorts. For the 2023 cohort, placebo treatment is assigned 24 months before the actual completion month, while the 2024 cohort serves as the untreated comparison group. The estimation window ends before the earliest actual construction start, so no building has yet been exposed to retrofit construction. Vertical bars represent 95 percent confidence intervals.

Free-riding. A concern specific to subsidy programs is that some participants would have improved their buildings' energy performance even without the program. If so, part of the estimated savings would reflect changes that would have occurred anyway. We examine this using accepted applicants whose projects were cancelled. These buildings expressed the same interest in retrofitting and passed the same initial screening as participants but did not receive the subsidized retrofit. If free-riding were important, cancelled applicants would be expected to reduce consumption by a similar amount through other means. Figure 13 and Table 8 compare annual consumption changes relative to the year before the project year for cancelled applicants and for matched approved participants. Over years +1 to +3, matched approved buildings reduce electricity use by 7.3%, whereas cancelled applicants increase it by 5.3%, a difference of 12.5 percentage points. For gas, both groups reduce consumption, but approved buildings reduce it by 8.1 percentage points more (25.7% versus 17.6%). Because the number of cancelled projects with usable data is very small (four for electricity and five for gas), this comparison is only suggestive, but the pattern is inconsistent with substantial free-riding.

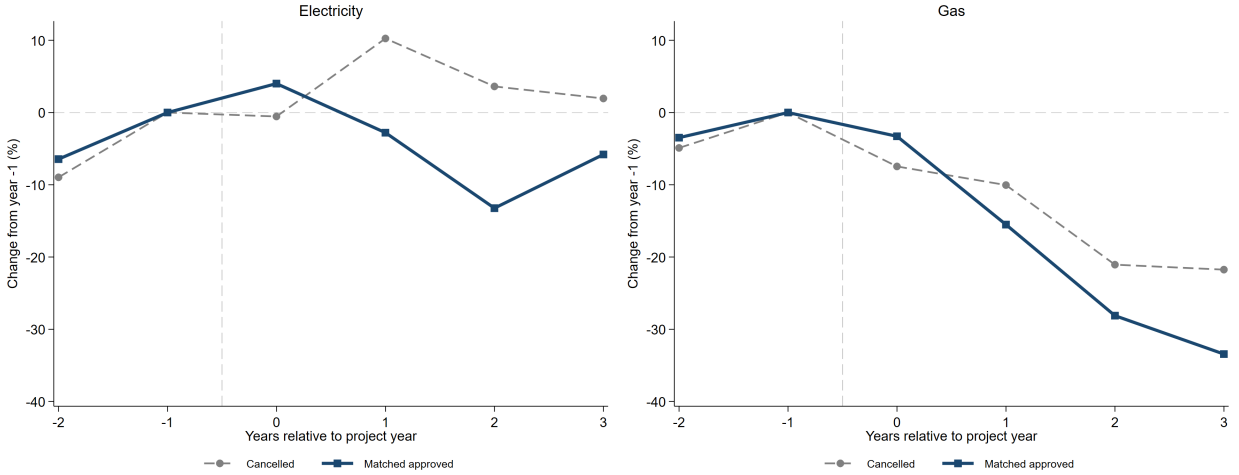


Figure 13: Free-Riding Diagnostic: Approved versus Cancelled Applicants

Note: The figure plots the mean percentage change in annual electricity (left) and gas (right) consumption relative to the year before the project year (year −1) for accepted applicants whose projects were cancelled and for matched approved participants. Year 0 is the project year.

Table 8: Free-Riding Diagnostic: Mean Post-Project Changes in Consumption

	Cancelled	Matched approved	Additional decline	N (cancelled)
Electricity	+5.3%	−7.3%	12.5 pp	4
Gas	−17.6%	−25.7%	8.1 pp	5

Note: Entries report mean percentage changes in annual consumption over years +1 to +3 relative to year −1, excluding the project year (year 0) as a transition year. Additional decline is the difference between the change for matched approved buildings and the change for cancelled applicants, in percentage points. N reports the number of cancelled applicants with usable consumption data.

Spatial spillovers. Retrofits may generate spillovers to nearby non-participating buildings, for example through information sharing among facility managers, local contractor learning, or municipal energy-saving

campaigns that coincide with the program. Such spillovers would reduce consumption among nearby comparison buildings and bias the estimates toward zero. We conduct two tests. First, we re-estimate the electricity and gas effects after excluding comparison buildings located within a minimum radius of any Green Remodeling project, for radii of 1, 3, 5, and 10 km. Figure 14 shows that the electricity ATT is essentially unchanged at about -20% across all radii, and the gas estimates remain statistically imprecise and close to zero, with no systematic movement toward larger savings as nearby controls are removed. Because this exercise uses a different comparison pool and specification from the preferred estimate, the level of the electricity estimate differs from Table 4; the object of interest is its stability across radii. Second, holding the treated buildings fixed, we compare estimates that draw never-treated comparison buildings from the same municipalities as treated buildings with estimates that draw them only from municipalities with no treated building. If municipality-level spillovers reduced consumption among nearby controls, the estimate using same-municipality controls would be attenuated toward zero. Table 9 shows that the electricity estimates are nearly identical (-13.7% versus -14.1%) and close to the preferred estimate in Table 4, and the gas estimates are imprecise in both cases. Together, these results provide little evidence that local spillovers materially bias the main estimates.

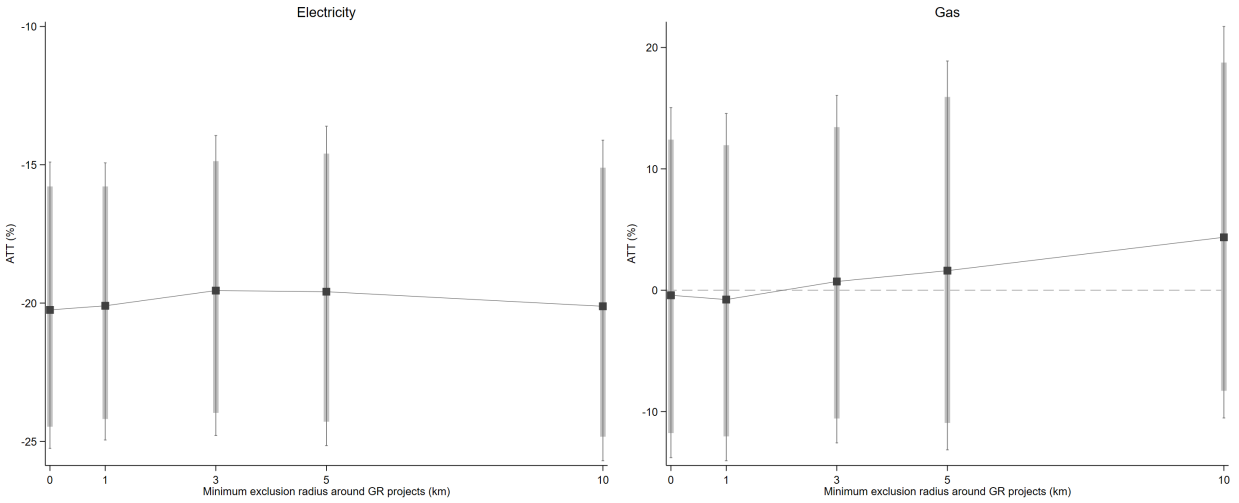


Figure 14: Spatial Spillover Robustness: Excluding Nearby Comparison Buildings

Note: The figure reports ATT estimates, in percent, for electricity (left) and gas (right) after excluding never-treated comparison buildings located within the indicated minimum distance of any Green Remodeling project. A radius of 0 km corresponds to no exclusion. Vertical bars report 95 percent confidence intervals.

Table 9: Municipality-Level Spillover Robustness

Outcome	Never-treated control pool	ATT	SE	Approx. effect
Electricity	Same treated municipality	−0.147	0.035	−13.7%
Electricity	Outside treated municipalities	−0.152	0.034	−14.1%
Gas	Same treated municipality	0.039	0.080	+4.0%
Gas	Outside treated municipalities	0.064	0.084	+6.6%

Note: The table reports ATT estimates on log consumption holding the treated buildings fixed and varying only the geographic source of the never-treated comparison buildings. "Same treated municipality" draws never-treated controls from municipalities that contain at least one treated building; "Outside treated municipalities" draws them only from municipalities with no treated building. Approximate effects are calculated as $100[\exp(\hat{\beta}) - 1]$.

Behind-the-meter generation. Renewable energy, typically rooftop solar photovoltaics, is part of the program's eligible scope. If electricity meters record net grid consumption, installing solar panels mechanically lowers measured electricity use without any change in energy efficiency, which could overstate the program's efficiency effect. Appendix Table B.2 re-estimates the main specifications after excluding treated buildings whose retrofit scope includes renewable energy. The electricity estimates fall from about −0.15 to between −0.100 and −0.116 log points (9.5–11.0%), but they remain statistically significant at the 1% level for all four estimators, including the Callaway–Sant'Anna estimator (−0.100). Thus, roughly two-thirds of the electricity effect remains after excluding renewable-energy projects, which is consistent with the measure-specific estimates in Section 4.3: on-site generation contributes to measured electricity savings, but envelope improvements generate savings on their own. The gas estimates are similar to the full-sample estimates: negative in the TWFE and Sun–Abraham specifications and insignificant in the Callaway–Sant'Anna specification.

COVID-19 and other common shocks. The sample period includes the COVID-19 pandemic, which may have affected the occupancy and energy demand of public facilities. Appendix Figure B.1 plots mean monthly electricity and gas consumption by building type from 2018 to 2025. The series show strong seasonal variation, particularly for gas, but no clear common decline in energy use during the pandemic period, and the patterns differ across building types. In addition, year-month fixed effects absorb any shock common to all buildings, and the treated cohorts completed their retrofits after the initial pandemic period of 2020. Finally, we find no statistically significant relationship between annual active user counts and energy consumption (Appendix Table B.1), which provides little support for the hypothesis that changes in occupancy drive our results.

4.7 Machine-Learning Counterfactual Estimates

We next use the cross-fitted ML framework in Section 3.7 as a complementary counterfactual exercise that does not rely on a comparison group. Random forest achieves the lowest combined held-out RMSE for the three primary outcomes: 4.77 kWh/m^2 for electricity, 3.68 kWh/m^2 for gas, and 5.10 kWh/m^2 for structural-

adjusted total energy. XGBoost is selected for the strict dual-fuel total-energy robustness outcome, with a combined held-out RMSE of 3.86 kWh/m^2. Across the primary outcomes, the selected models also exhibit small average held-out bias: -0.08 kWh/m^2 for electricity, essentially zero for gas, and -0.10 kWh/m^2 for structural-adjusted total energy. Appendix Table F.1 reports the full comparison across all five candidate algorithms.

Pre-retrofit prediction errors provide an additional diagnostic of the counterfactual. For structural-adjusted total energy, the mean event-month prediction error over months -12 through -1 is -0.08 kWh/m^2, and none of the twelve monthly 95% confidence intervals excludes zero. The same is true for gas, where the average error is -0.01 kWh/m^2. Electricity has a mean pre-period error of -0.07 kWh/m^2, with 11 of 12 monthly confidence intervals containing zero. These diagnostics indicate that the selected models reproduce the pre-retrofit path reasonably well before they are used to extrapolate the untreated post-retrofit counterfactual.

Table 10: Machine-Learning Counterfactual Savings over the First 12 Post-Retrofit Months

Outcome	Selected model	Buildings	Realized saving	ML-predicted saving
Structural-adjusted total	Random forest	414	22.61	24.39
Electricity	Random forest	425	16.86	19.27
Gas	Random forest	102	25.59	23.09
Strict dual-fuel total	XGBoost	95	28.62	26.81

Note: Savings are measured in kWh/m^2 and summed over the first 12 complete post-retrofit months for buildings with all 12 months available. "Realized saving" is the cross-fitted no-retrofit counterfactual minus observed post-retrofit consumption, $\sum_{m=1}^{12}[\widehat{Y}_{im}(0) - Y_{im}]$. "ML-predicted saving" is the difference between the cross-fitted untreated- and treated-state predictions, $\sum_{m=1}^{12}[\widehat{Y}_{im}(0) - \widehat{Y}_{im}(1)]$. The structural-adjusted total is the primary combined-energy outcome; strict dual-fuel total is reported as a robustness check.

The 12-month savings estimates in Table 10 show that the two ML-based savings measures are similar in magnitude. For structural-adjusted total energy, the mean counterfactual-implied realized saving is 22.6 kWh/m^2 over the first twelve complete post-retrofit months, compared with a model-implied treatment-state saving of 24.4 kWh/m^2. For electricity, the corresponding values are 16.9 and 19.3 kWh/m^2, while for gas they are 25.6 and 23.1 kWh/m^2. The balanced 12-month samples contain 414 buildings for structural-adjusted total energy, 425 for electricity, and 102 with sufficient positive gas observations. The gas results are based on a much smaller and more selected sample of buildings with positive gas use and are therefore not directly comparable with the DID gas estimates.

The structural-total comparison also sharpens the engineering–realized comparison developed in Section 5. In the event-study plotting sample, the ECO2 engineering benchmark implies mean monthly savings of 3.76 kWh/m^2. The ML no-retrofit counterfactual implies realized savings of 2.05 kWh/m^2 per month, and the difference between the ML untreated- and treated-state predictions is 2.01 kWh/m^2 per month. Thus, both empirical ML quantities are only about 53–55% of the engineering benchmark. Because the ML counterfactual does not use a comparison group, this close agreement suggests that the engineering–realized wedge is not

driven by the particular DID comparison-group construction.

The detailed event-time plots are reported in Appendix Figures F.1–F.4. They show the observed path, the cross-fitted treated-state prediction, and the no-retrofit counterfactual. The structural-adjusted and strict total-energy figures also display the ECO2 engineering benchmark. Appendix Figure F.4 repeats the exercise for the strict dual-fuel total-energy definition. Because that definition requires both fuels to be observed and positive in the same building-month, it is based on a substantially smaller and more selected sample and is therefore interpreted as a robustness check rather than as an alternative primary total-energy estimate. We keep the detailed figures in the appendix because the CSDID estimates remain the primary causal results; the ML analysis is intended as a complementary prediction-based validation and a bridge to the engineering–realized comparison.

5 Discussion

This paper empirically estimates the energy savings from retrofits of public buildings in South Korea. In the preferred specification, retrofits reduce total delivered energy by about 10% and electricity by about 14%, while the evidence for gas savings depends on the estimator. We find clear evidence that South Korea's public-building Green Remodeling Program reduces realized energy consumption after retrofits, and that these reductions are not uniform across facilities or measures. The patterns in our results closely mirror the main conclusions in the broader retrofit-evaluation literature summarized by Giandomenico et al. (2022), who emphasized that measured savings from retrofit programs are typically modest on average, vary widely across contexts and measures, and depend strongly on evaluation design. Across 39 credible evaluations (covering 23 programs), they reported an average measured reduction of roughly 7.2% in electricity and/or fuel use, and emphasized that none of the studies in their sample achieved "deep" savings of 50% or more. Our estimates lie slightly above this average, which is plausible given that the Green Remodeling Program bundles several envelope and equipment measures, but they are far from deep savings.

However, while South Korea's public-building Green Remodeling Program reduces realized energy consumption in participating buildings, realized reductions fall well short of the program's engineering-based projections. Our results connect to two strands in the literature: the engineering–realized gap in energy-efficiency programs and the heterogeneity of realized savings across building types and retrofit measures. We discuss each in turn, leading with the engineering–realized comparison, since it provides the most consequential takeaway for program design.

5.1 The Engineering-Realized Gap and Data-Driven Counterfactuals

Predicted energy savings are computed by the Korea Authority of Land and Infrastructure Safety (KALIS) for each treated building using the ECO2 simulation tool, conditional on the planned retrofit scope. We compare these engineering predictions with realized savings calculated from utility data. Because the paper's

primary combined-energy outcome is the structural-adjusted total-energy measure, the main engineering–realized comparison is restricted to total energy and to the current matched common-support sample of public daycare centers and public health centers; hospitals are excluded.

Figure 15 plots engineering-predicted total energy savings against realized structural-adjusted total-energy savings for the matched sample. The 45-degree line represents one-for-one realization of engineering projections, while the fitted line summarizes the empirical relationship between projected and realized savings. The figure is intended as a building-level descriptive diagnostic of the performance wedge; the paper's causal program-effect estimates continue to come from the matched DID, Sun–Abraham, and CSDID specifications.

Most buildings lie below the 45-degree line. Across the 301 buildings in the matched sample (96 daycare centers and 205 health centers), the engineering model predicts average total-energy savings of 37.9%, whereas realized savings average 11.7%. Engineering projections therefore overstate realized savings by a factor of about 3.2, a ratio very close to the factor of three reported by Fowlie et al. (2018) for the U.S. Weatherization Assistance Program. The building-level average realized saving is also close to the causal estimates in Section 4.2 (10–14% for total energy), which indicates that the descriptive and causal measures tell a consistent story. The fitted relationship between predicted and realized savings is positive but flat: the OLS slope is 0.28 (SE 0.11), so a building predicted to save 10 percentage points more than another building saves, on average, only about 3 percentage points more in practice. Engineering predictions thus contain some information about which buildings will save more, but they substantially overstate both the level and the dispersion of predicted differences across buildings.

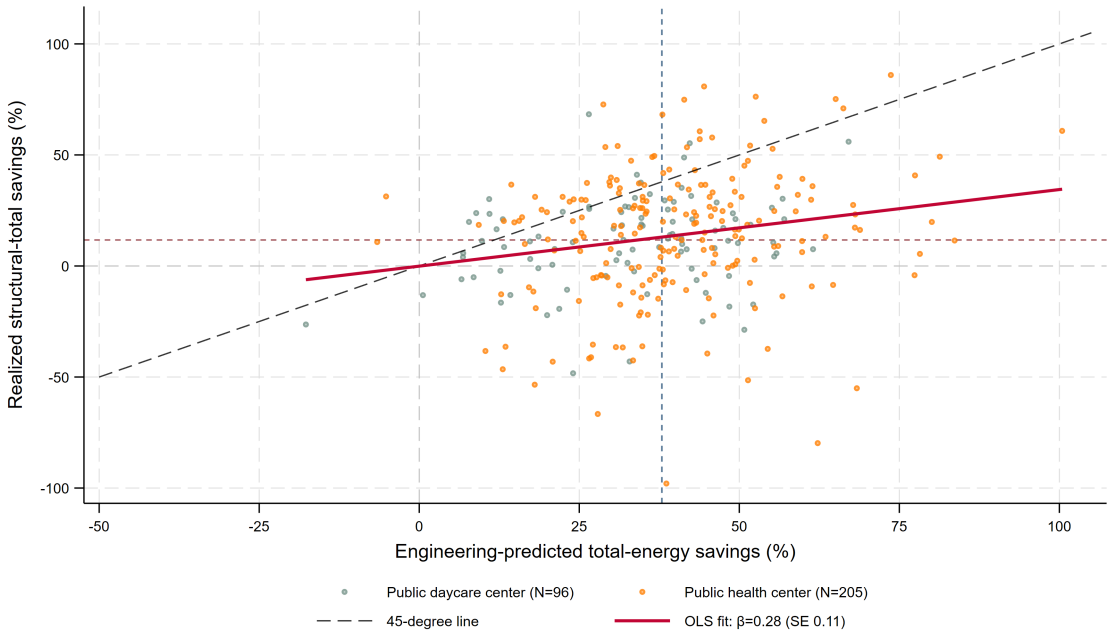


Figure 15: Engineering-Predicted vs. Realized Total-Energy Savings: Matched Sample

Note: Each point is a treated public daycare center or public health center in the current structural-adjusted total-energy matched sample for which both the ECO2 engineering prediction and realized utility-based saving are observed. Hospitals are excluded. The dashed 45-degree line represents equality between engineering-predicted and realized savings; the fitted line reports the empirical relationship between the two measures. Vertical and horizontal dashed reference lines report their sample means. The realized building-level saving is a descriptive pre/post quantity under the current retrofit-timing definition and should not be interpreted as a replacement for the paper's causal ATT estimates.

Figure 16 compares the distributions of engineering-predicted and realized total-energy savings under two sample definitions. Panel A uses the current matched sample, while Panel B uses the broader treated full sample. This comparison separates two features of the performance wedge: differences in average savings and differences in the dispersion of realized outcomes. The figure also shows whether the wedge is specific to the common-support restriction or remains visible in the broader treated population. In the matched sample, engineering-predicted savings average 37.9% with a standard deviation of 16.4 percentage points, whereas realized savings average 11.7% with a standard deviation of 31.6 percentage points. In the full sample, the corresponding values are 38.9% (SD 16.8) and 14.1% (SD 35.1). Realized savings are therefore not only lower on average but also roughly twice as dispersed as predicted savings, and a substantial share of buildings show no realized savings at all.

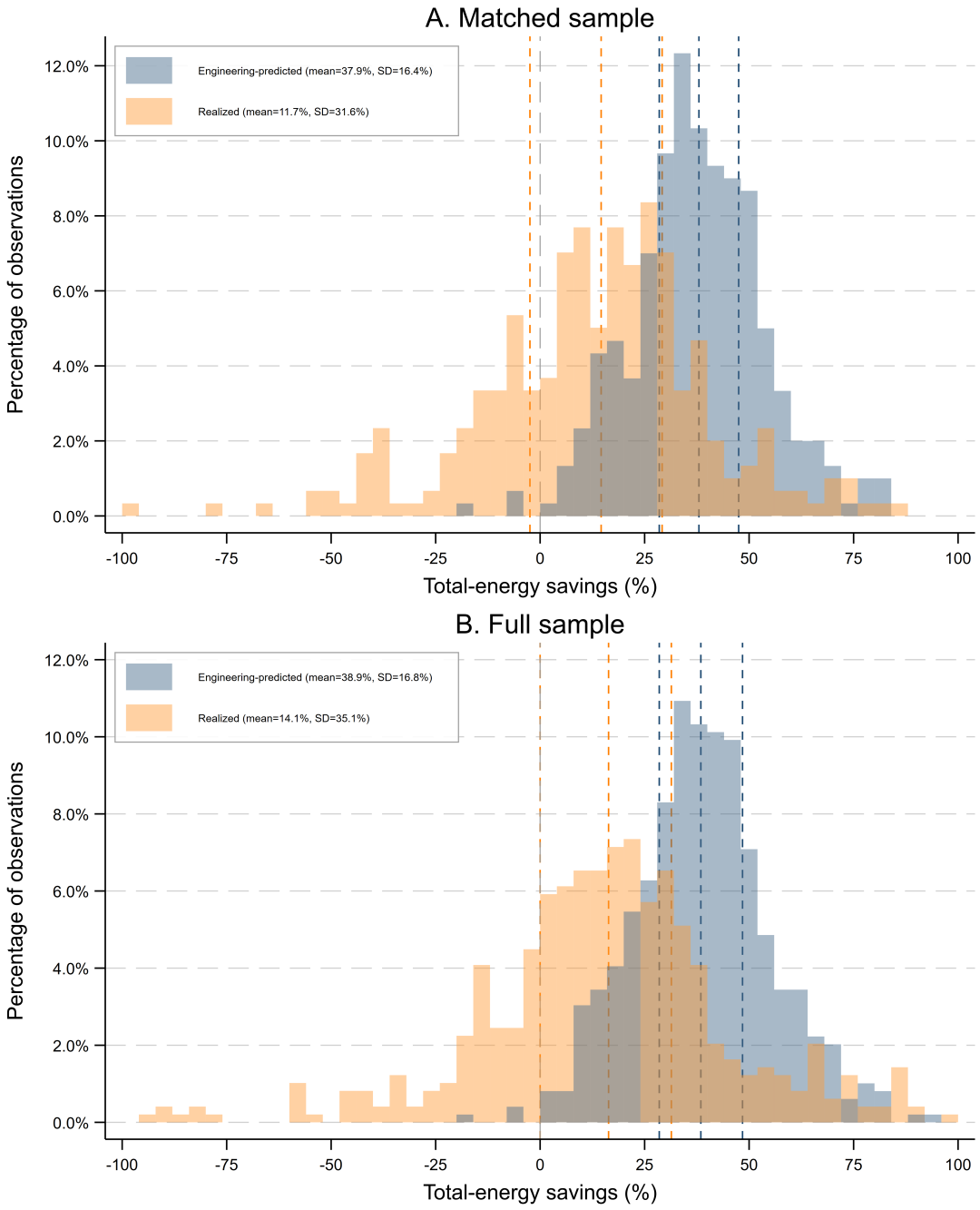


Figure 16: Distributions of Engineering-Predicted and Realized Total-Energy Savings

Note: Panel A reports the current matched structural-adjusted total-energy sample; Panel B reports the broader treated full sample. Histograms show the percentage of buildings in each savings interval. Engineering-predicted savings and realized savings are plotted separately, and the figure legends report the corresponding means and standard deviations. Dashed vertical lines denote the first quartile, median, and third quartile of each distribution. Hospitals are excluded from both panels. The full-sample panel is descriptive and does not impose the outcome-specific ATT-IPW common-support restriction used in the main matched analysis.

The engineering–realized gap also differs across facility types. Figure 17 reports mean engineering-predicted and realized total-energy savings separately for public daycare centers and public health centers. The connecting segment directly displays the difference between projected and realized savings within each building type. Panel A uses the matched sample and Panel B uses the broader full sample. Presenting the two sample definitions side by side makes clear whether the qualitative building-type pattern depends on the matching and support restrictions. In the matched sample, daycare centers realize 8.3% savings against a predicted 32.9%, and health centers realize 13.3% against a predicted 40.2%. The full-sample values are similar (11.4% versus 33.8% for daycare centers and 15.6% versus 41.9% for health centers). Health centers have both higher predicted and higher realized savings, consistent with the building-type estimates in Table 5, but the gap of roughly 25–27 percentage points is similar for both building types. The engineering model thus ranks the two building types correctly while overstating the savings of both.

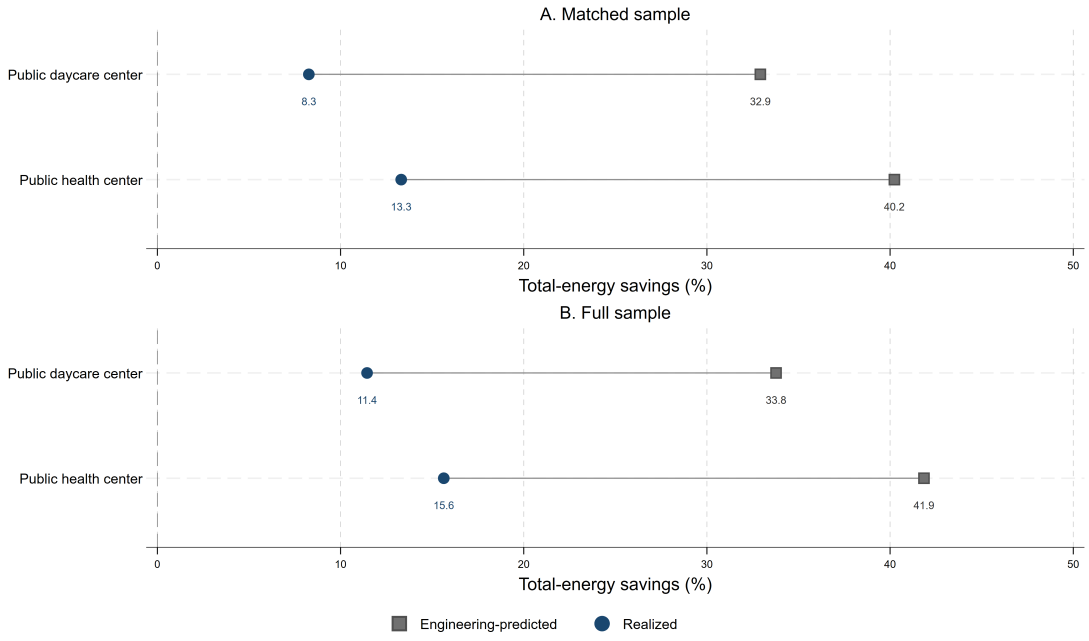


Figure 17: Engineering-Predicted and Realized Total-Energy Savings by Building Type

Note: The figure reports mean engineering-predicted and realized total-energy savings for public daycare centers and public health centers. Panel A uses the current matched structural-adjusted total-energy sample and Panel B uses the broader treated full sample. Squares denote engineering predictions, circles denote realized savings, and the connecting segment shows the engineering–realized gap within each building type. Numeric labels report the corresponding mean percentage savings. Hospitals are excluded.

Across these comparisons, engineering projections are systematically more optimistic than the utility-based realized outcomes, while realized savings are considerably more dispersed across buildings. The fact that the same qualitative wedge is visible in both the matched and broader full sample suggests that it is not solely an artifact of the common-support restriction. We nevertheless treat the matched comparison as the main descriptive benchmark because it aligns with the sample used for the paper's primary structural-adjusted total-energy analysis. Appendix Figure E.1 reports the corresponding building-level scatter for the broader

full sample.

To examine whether the engineering–realized performance wedge is systematically associated with the retrofit technologies installed, we next regress the building-level overprediction gap on the set of retrofit-technology indicators. The dependent variable is the ECO2 engineering-predicted total-energy saving minus the realized total-energy saving, measured in percentage points. Figure 18 reports the conditional technology coefficients for the preferred matched structural-adjusted total-energy sample and, as a sensitivity comparison, the previous full-sample total-energy construction.

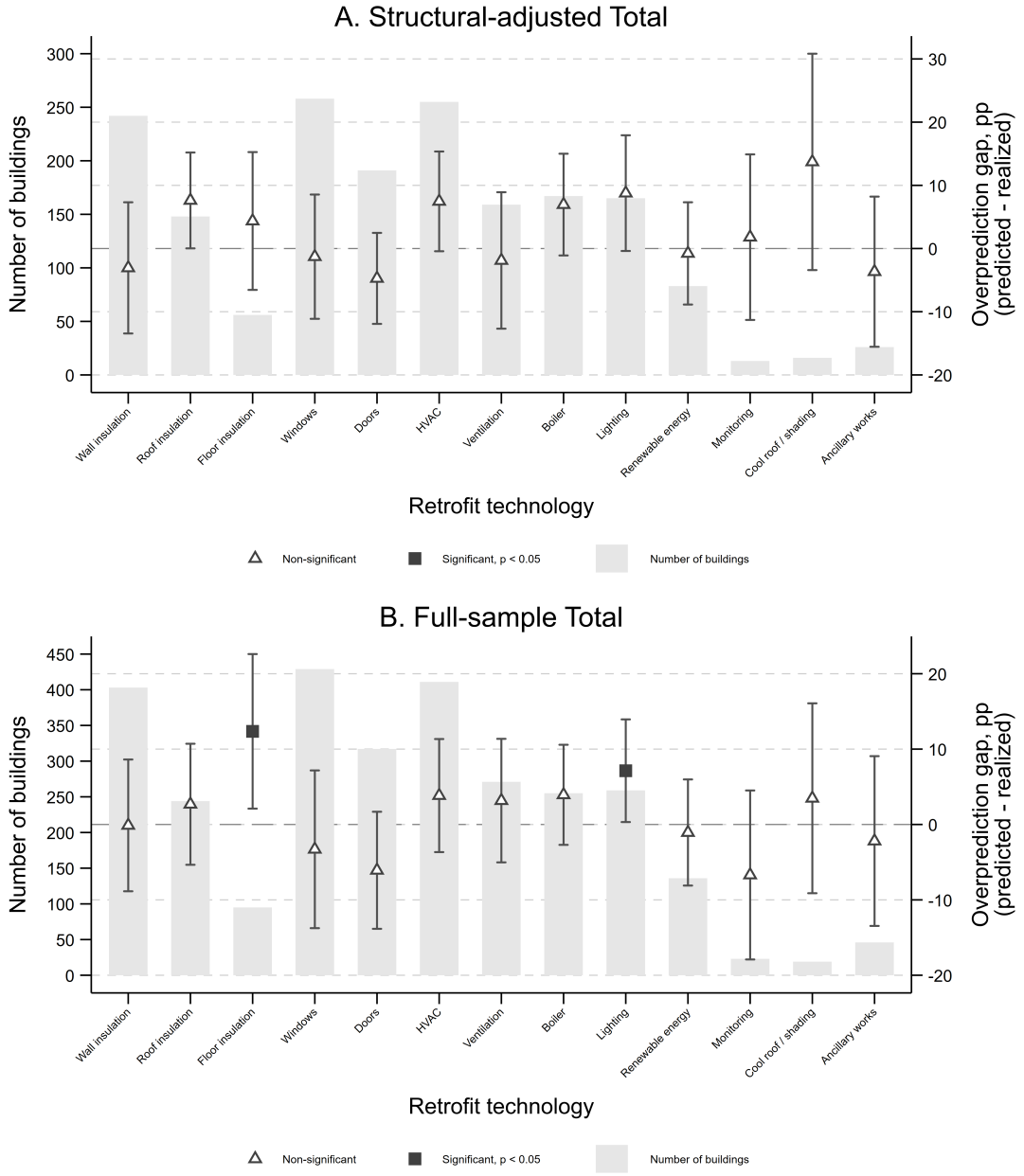


Figure 18: Technology-Specific Heterogeneity in the Engineering Overprediction Gap

Note: The outcome is the engineering overprediction gap, defined as ECO2 engineering-predicted total-energy savings minus realized total-energy savings, in percentage points. Panel A uses the current matched structural-adjusted total-energy sample; Panel B uses the previous full-sample total-energy construction. Both panels include public daycare centers and public health centers only; hospitals are excluded. Technology indicators enter jointly. Regressions control for log baseline realized total-energy use, building type, project year, and log project cost when observed. Points report conditional technology coefficients and vertical whiskers report 95% confidence intervals. Filled squares indicate $p < 0.05$ and open triangles indicate estimates that are not significant at the 5% level. Gray bars report the number of buildings receiving each technology. Positive coefficients indicate a larger engineering overprediction gap. The estimates are descriptive technology-gap associations rather than causal technology-specific treatment effects.

The average engineering–realized wedge is highly similar under the two total-energy constructions: 26.2 percentage points in the preferred structural-adjusted matched sample ($N = 301$) and 25.4 percentage points under the previous full-sample method ($N = 494$). By contrast, the technology-specific estimates are less stable and generally imprecise. In the preferred matched sample, no technology coefficient is significant at the 5% level, but four technologies are associated with larger overprediction gaps at the 10% level: lighting (8.8 percentage points), roof insulation (7.6), HVAC (7.5), and boilers (7.0). Because HVAC is installed in 85% of the buildings, it accounts for the largest implied contribution to the mean gap (6.3 percentage points, or 24%). Doors, wall insulation, and windows are associated with smaller gaps, although not significantly so. In the full sample, floor insulation (12.4 percentage points) and lighting (7.1) are significant at the 5% level. Lighting is thus the only technology associated with a larger gap in both samples, which matches the measure-specific consumption estimates in Section 4.3, where lighting upgrades were associated with higher rather than lower energy use. Mechanical systems (HVAC and boilers) are also associated with larger gaps in the preferred sample and showed no measurable savings in the consumption regressions, whereas the envelope measures that generated measurable savings (wall insulation, windows, and doors) are associated with smaller gaps. This distinction is important: the aggregate engineering overprediction gap is robust across the two constructions, whereas attribution of that gap to any single retrofit technology is substantially more uncertain. Detailed technology prevalence and decomposition shares are reported in Appendix Table E.1.

These findings are consistent with prior evidence that engineering models can overstate realized savings in energy-efficiency programs. Liang et al. (2018), for example, report realized savings substantially below engineering predictions, while the broader evaluation literature similarly documents that measured savings are typically more modest than engineering projections (Giandomenico et al., 2022). In this setting, the comparison between ECO2 projections and utility-based outcomes therefore provides a direct measure of the implementation and performance wedge that motivates the data-driven counterfactual exercise below.

The machine-learning counterfactual provides a second perspective on this performance wedge. Following the logic of **?**, we use realized utility data to learn flexible untreated and treated energy relationships rather than relying on the physical equations embedded in the engineering model. For the primary structural-adjusted total-energy outcome, the ECO2 benchmark implies average monthly savings of 3.76 kWh/m^2, whereas the ML counterfactual implies realized savings of 2.05 kWh/m^2 and the ML treatment-state comparison predicts 2.01 kWh/m^2. In other words, the two data-driven estimates recover only about 53–55% of the engineering benchmark. This result is consistent with the broader evidence that engineering projections can systematically exceed realized retrofit performance.

The ML exercise also has a program-design interpretation. The selected models are chosen on held-out predictive performance rather than on how well they reproduce the estimated treatment effect, and the primary outcomes select random forests while the strict dual-fuel robustness outcome selects XGBoost. This is

consistent with **?**, who argue that realized program data can improve prediction and ultimately the allocation of efficiency investments relative to engineering projections alone. In the Green Remodeling Program, our results suggest that historical utility data, building characteristics, and weather can be used to recalibrate expected savings before future subsidy rounds. We view this as a complement to, rather than a substitute for, engineering audits: engineering models describe the physical potential of a retrofit package, while cross-fitted ML predictions incorporate empirical information about how similar buildings have actually performed.

5.2 Heterogeneity by Building Type and Retrofit Measure

A second contribution of our analysis is the documentation of heterogeneity, which is itself a first-order policy result rather than a robustness exercise. Davis et al. (2014) and Ito (2015) both reported heterogeneous treatment effects across subgroups, and these heterogeneity findings drove their policy implications. In our setting, two heterogeneity findings carry comparable weight.

First, the energy-saving effects differ by building type and season. Health centers deliver larger and more precisely estimated savings, about 15% for both total energy and electricity, than daycare centers, where electricity falls by about 10–12% and the preferred estimate for total energy is smaller and imprecise. The calendar-month profiles suggest that health-center savings are concentrated in the cooling season, when post-retrofit consumption falls clearly below the comparison group, whereas daycare-center savings appear mainly in the heating months. The heterogeneity analysis also shows that electricity savings are small in the lowest-baseline-intensity tercile and larger for buildings with medium and high baseline intensity. These patterns are consistent with Giandomenico et al. (2022), who emphasize that measured savings vary widely across contexts. They also suggest that savings are largest where there is more energy demand to reduce, which is relevant for targeting because the selection model in Section 4.5 shows that higher-intensity buildings are, if anything, less likely to be accepted into the program.

Second, envelope measures and on-site renewable generation are associated with the largest and most precisely estimated savings, and bundled envelope retrofits deliver the largest reductions in total energy use. Insulation measures are among the most promising retrofits across methods in the literature (Giandomenico et al., 2022). Liang et al. (2018) also found a significant impact from insulation retrofits, and Adan and Fuerst (2016) provided evidence that the single most effective energy-saving measure is insulation, which reduces total annual energy consumption by 8%. In our data, wall and roof insulation is associated with an 11.2% reduction in electricity and a 10.1% reduction in total energy use, renewable-energy adoption with a 13.4% reduction in electricity, and complete envelope retrofits with a 14.8% reduction in total energy and a 12.1% reduction in electricity. These patterns echo the engineering and review literature emphasizing insulation as a reliable driver of measured savings (Adan and Fuerst, 2016; Liang et al., 2018; Giandomenico et al., 2022), although the realized magnitudes are smaller than what engineering models projected.

Our findings also document that not all retrofit components reduce measured energy consumption. High-efficiency boilers, heat-recovery ventilation, high-efficiency HVAC, and energy monitoring show no measurable savings, and high-efficiency lighting is associated with higher total energy use. These patterns are plausible operationally: mechanical systems depend heavily on commissioning and on how they are operated, and lighting upgrades may be accompanied by longer operating hours. They reinforce the broader point that engineering models do not always anticipate the operational changes that accompany retrofits, and they line up with the technology-gap results above, in which lighting and mechanical systems are associated with larger engineering overprediction.

Our bundle results differ from those of Papineau et al. (2025), who found no evidence of electricity reductions from bundled residential retrofits but a 35% reduction in gas use. In our setting, complete envelope retrofits reduce electricity use by about 12% but have no measurable effect on gas. A plausible explanation is that the buildings in our sample, especially health centers, rely heavily on electricity for both cooling and heating, and that the gas sample is small and limited to daycare centers. The comparison nonetheless supports the interpretation that a subset of components drives savings and that savings are largest when envelope measures are implemented jointly rather than in isolation.

The practical implication for the Green Remodeling Program is that the most reliable path to larger realized savings likely combines prioritizing high-return envelope packages and on-site renewable generation with stronger post-retrofit verification and operational support for mechanical systems and lighting, where installed equipment does not automatically translate into lower consumption.

5.3 Cost-Effectiveness

Table 11: Program Cost per Estimated Reductions in Energy Consumption and Carbon Dioxide

	Total project cost	National subsidy	Local share
Program cost (billion KRW)	147.8	100.9	47.0
Program cost (million USD)	98.1	66.9	31.2
Estimated annual electricity reduction (kWh)	3,724,431	3,724,431	3,724,431
Estimated annual gas reduction (kWh)	2,410,900	2,410,900	2,410,900
Estimated annual energy reduction (kWh)	6,135,330	6,135,330	6,135,330
Estimated lifetime energy reduction (kWh)	122,706,620	122,706,620	122,706,620
Estimated annual CO_2 reduction (tCO_2)	2,064	2,064	2,064
Estimated lifetime CO_2 reduction (tCO_2)	41,275	41,275	41,275
Program cost per kWh saved (USD/kWh)	0.799	0.545	0.254
Program cost per CO_2 reduced (USD/tCO_2)	2,376	1,621	755

Notes: This table follows the cost-effectiveness accounting approach in Ito (2015), which divides program cost by estimated reductions in energy consumption and carbon dioxide emissions. The sample includes the 2021–2022 treated cohort of 509 buildings. Annual energy savings are based on estimated reductions of 3,724,431 kWh of electricity and 2,410,900 kWh of gas. CO_2 reductions are calculated using emissions factors of 0.424 $kgCO_2$/kWh for electricity and 0.201 $kgCO_2$/kWh for natural gas. Lifetime reductions assume that annual savings persist for 20 years. Dollar values use an exchange rate of 1,507 KRW/USD. Program cost per kWh saved divides program cost by lifetime energy savings.

Following Ito (2015), Table 11 evaluates cost-effectiveness by dividing program costs by estimated reductions in energy consumption and CO_2 emissions. Because green remodeling is a capital investment, we focus on lifetime savings over an assumed 20-year persistence period. The estimated post-retrofit reductions in electricity and gas use imply annual energy savings of approximately 6.14 million kWh for the 2021–2022 treated cohort, consisting of 3.72 million kWh of electricity savings and 2.41 million kWh of gas savings. Using emissions factors of 0.424 $kgCO_2$/kWh for electricity and 0.201 $kgCO_2$/kWh for natural gas, these savings correspond to approximately 2,064 tCO_2 of avoided emissions per year, or about 4.1 tCO_2 per treated building annually.[2]

Program cost data show that the 2021–2022 cohort received total retrofit investment of 147.8 billion KRW, of which 100.9 billion KRW came from the national subsidy and 47.0 billion KRW from local government funding. Assuming that retrofit-induced emissions reductions persist for 20 years, the program abates approximately 41,275 tCO_2 over the lifetime of the investments. This implies a realized abatement cost of 3.58 million KRW/tCO_2, or approximately \$2,376/tCO_2, when total project costs are used. Using only the national subsidy yields a lower but still substantial abatement cost of 2.44 million KRW/tCO_2, or approximately \$1,621/tCO_2, and the local share alone corresponds to about \$755/tCO_2.[3] Expressed per unit of en-

[2]The cost-effectiveness analysis is restricted to the 2021–2022 program cohort because these are the years for which project-level program cost data are available. The cohort includes 509 treated buildings, with total floor area of 250,885 m^2.

[3]Dollar values use an exchange rate of 1,507 KRW/USD.

ergy, the program costs about \$0.80 per lifetime kWh saved based on total project costs, or \$0.55 per kWh based on the national subsidy alone, which far exceeds the retail price of electricity or gas in Korea. Energy-bill savings alone therefore cannot recover the investment.

Figure 19 compares these realized cost-effectiveness estimates with the engineering-implied benchmark. The engineering prediction implies a cost of approximately \$711/tCO$_2$, whereas the realized total project cost is approximately \$2,376/tCO$_2$. Thus, the realized cost per ton is about 3.3 times larger than the engineering-implied benchmark, mirroring the ratio of predicted to realized savings documented above. The national subsidy alone also exceeds the engineering benchmark, at approximately \$1,621/tCO$_2$. These values are well above standard social-cost-of-carbon benchmarks, suggesting that the program is not cost-effective when evaluated only as a carbon-abatement policy. However, the program also targets public-use buildings such as childcare centers, health centers, and other facilities serving vulnerable populations. Therefore, a carbon-only metric likely understates the broader social benefits of the program, including improvements in indoor environmental quality, thermal comfort, and public-service infrastructure.

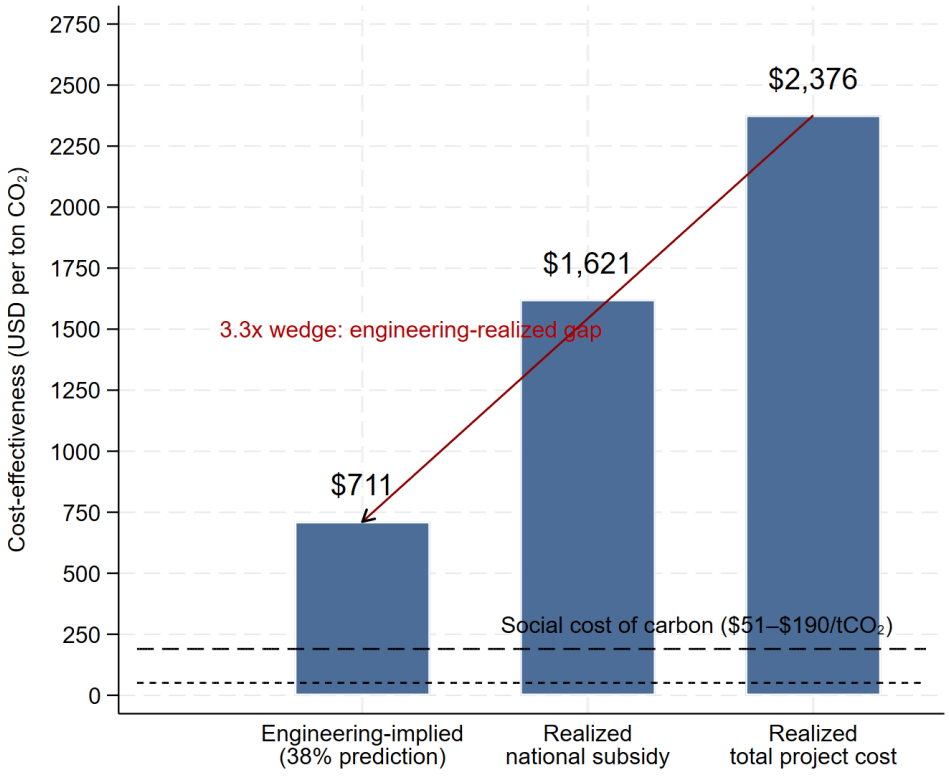


Figure 19: Realized cost per ton of CO$_2$ and Engineering-implied Benchmark

Note: Estimates are computed using actual 2021–2022 program costs and estimated annual reductions in electricity and gas consumption. The sample includes 509 treated buildings. Lifetime abatement assumes that annual emissions reductions persist for 20 years. Dollar values use an exchange rate of 1,507 KRW/USD. The dashed horizontal lines show social-cost-of-carbon benchmarks of \$51–\$190/tCO$_2$.

Table 12 breaks down the cost-effectiveness calculation by building type. Although health centers deliver larger percentage savings, daycare centers are considerably more cost-effective per ton of CO_2. Daycare centers have a larger total floor area (124,983 m^2 versus 73,355 m^2) and a much lower cost per square meter (KRW 538,034 versus KRW 959,930), so they generate about twice the annual emissions reductions (1,196 versus 585 tCO_2) at a lower total cost. The implied lifetime abatement cost is \$1,865/tCO_2 for daycare centers and \$3,994/tCO_2 for health centers. The hospital estimate (\$2,554/tCO_2) is based on only eight buildings for which we could not construct a balanced comparison group, so we do not emphasize it. Because the building-type calculation sums building-type-specific savings, its total (1,913 tCO_2 per year and \$2,564/tCO_2) differs slightly from the pooled calculation in Table 11. The comparison shows that the facility type with the largest percentage savings is not necessarily the most cost-effective target, because project costs per square meter vary substantially across facility types.

Table 12: Cost-Effectiveness by Building Type

	Daycare	Health center	Hospital	Total project cost
Number of buildings	186	315	8	509
Total floor area (m^2)	124,983	73,355	52,547	250,885
Total cost (billion KRW)	67.2	70.4	10.2	147.8
Cost per m^2 (KRW)	538,034	959,930	193,341	589,205
Annual CO_2 reduction (tCO_2/yr)	1,196	585	132	1,913
Lifetime CO_2 reduction (tCO_2)	23,920	11,700	2,640	38,260
Annual cost per tCO_2 reduced (USD/tCO_2)	37,309	79,873	51,072	51,275
Lifetime cost per tCO_2 reduced (USD/tCO_2)	1,865	3,994	2,554	2,564

Notes: The sample includes the 2021–2022 treated cohort of 509 buildings. Annual CO_2 reductions are calculated from building-type-specific estimated reductions in electricity and gas use using emissions factors of 0.424 $kgCO_2$/kWh for electricity and 0.201 $kgCO_2$/kWh for natural gas. Lifetime reductions assume that annual savings persist for 20 years. Annual cost per tCO_2 divides total cost by annual reductions; lifetime cost per tCO_2 divides total cost by lifetime reductions. Dollar values use an exchange rate of 1,507 KRW/USD. The last column reports totals across building types. Hospital estimates are based on eight buildings and should be interpreted with caution.

Table 13 places these estimates in the context of prior evaluations of energy-efficiency programs. Our ratio of projected to realized savings is similar to that found by Fowlie et al. (2018) for the Weatherization Assistance Program and somewhat smaller than the ratio found by Davis et al. (2014) for appliance replacement in Mexico. The realized abatement cost, however, is several times larger than in these studies. This reflects the high cost of whole-building retrofits per unit of energy saved and the fact that the program is designed as a capital renewal program for public facilities rather than as a least-cost abatement instrument. These estimates are not directly comparable because programs differ in setting, cost definitions, discounting, emissions factors, and in whether non-energy benefits are included.

Table 13: Comparison with Prior Evaluations of Energy-Efficiency Programs

Study and setting	Projected vs. realized savings	Realized cost-effectiveness
This paper: Korean public-building Green Remodeling	Engineering prediction 37.9% vs. realized 11.7% (matched sample)	\$2,376/tCO$_2$ (total project cost); \$1,621/tCO$_2$ (national subsidy only)
Davis et al. (2014): Mexican appliance replacement	Savings about one-quarter of ex ante prediction	More than \$450–\$500/tCO$_2$; \$0.29/kWh
Fowlie et al. (2018): U.S. Weatherization Assistance Program	Projected savings more than three times realized savings	More than \$200/tCO$_2$; negative social rate of return
Ito (2015): California electricity rebate	–	\$381/tCO$_2$ short run; \$160/tCO$_2$ long run

Note: Prior-study estimates are taken from the cited papers and are expressed in the units reported there. They are not directly comparable to our estimates because programs differ in setting, cost definitions, discounting, emissions factors, and whether non-energy benefits are included. Our estimates exclude local air-pollution benefits and non-energy benefits such as thermal comfort, health, building quality, and public-service improvements.

5.4 Implications for Program Design

Four implications follow from our results. First, subsidies sized to engineering projections systematically overstate expected benefits. In this setting, realized savings are only about 30% of the ECO2 projection at the building level and 53–55% of the engineering benchmark in the ML counterfactual, so engineering projections should be discounted substantially when used for ex ante cost-effectiveness analysis. Because the program collects building-level utility data, historical realized performance can be used to recalibrate expected savings before future subsidy rounds (**?**).

Second, program effectiveness can be improved by targeting. At the measure level, envelope improvements (wall and roof insulation, windows and doors, and complete envelope packages) and on-site renewable generation are associated with measurable savings, whereas high-efficiency boilers, HVAC, ventilation, and lighting are not and are associated with larger engineering overprediction. At the building level, electricity savings are larger for buildings with higher baseline energy intensity, but the selection model indicates that such buildings are not favored in the current selection process. Because cost per square meter differs substantially across facility types, targeting should also consider cost-effectiveness rather than percentage savings alone: daycare centers deliver smaller percentage savings than health centers but at less than half the abatement cost per ton.

Third, the cost-share rule affects which institutions participate. Institutions eligible for the 50% subsidy rate are less likely to be accepted than those eligible for the 70% rate, which suggests that the residual cost burden matters for participation. Adjusting cost-share rules could change the composition of participants, but our estimates do not identify how such changes would affect realized savings.

Fourth, the gap between projected and realized outcomes points to the importance of post-retrofit verifica-

tion. Subsidy disbursements that depend on installed measures rather than on realized performance leave operational margins of adjustment unconstrained. Strengthening post-retrofit commissioning, especially for mechanical systems and lighting, and multi-year measurement and verification would help align realized savings with the projections that motivate program funding. Finally, even with these improvements, the program is unlikely to be cost-effective as a carbon-abatement policy alone; its justification rests also on non-energy benefits for the vulnerable populations that use these facilities, which future evaluations should quantify.

6 Conclusion

This study provides large-scale causal evidence that South Korea's public-building Green Remodeling Program reduces realized energy consumption in participating facilities. Using propensity-score-weighted comparison groups and heterogeneity-robust staggered difference-in-differences estimators, we find that retrofits reduce total delivered energy by about 10% in the preferred specification (11–14% across alternative estimators) and electricity use by about 14%. The electricity results are remarkably stable across estimators and comparison groups, whereas the gas reductions estimated by two-way fixed-effects and Sun–Abraham specifications are not robust to the Callaway–Sant'Anna estimator. The results survive an early-cohort placebo test, a comparison with cancelled applicants that is inconsistent with free-riding, tests for spatial spillovers, and the exclusion of buildings that installed on-site renewable energy.

A key substantive insight is that savings are not uniform across building types and retrofit measures. Health centers deliver larger savings than daycare centers, concentrated in the cooling season, and electricity savings are larger for buildings with higher baseline energy intensity. At the measure level, savings are concentrated in envelope measures and on-site renewable generation, and complete envelope retrofits deliver the largest reductions in total energy use, whereas boilers, HVAC, ventilation, and lighting are not associated with measurable savings.

A further implication of our analysis is that realized energy savings are far smaller than engineering-model predictions. At the building level, realized total-energy savings average 11.7%, compared with 37.9% predicted by the program's ECO2 simulation, and a machine-learning counterfactual that does not rely on a comparison group recovers only 53–55% of the engineering benchmark. The wedge is of similar magnitude to that documented for residential weatherization in the United States, even though public buildings largely lack the split-incentive problem, which suggests that occupant behavior, operational practice, and measure selection are important sources of the gap. Translated into emissions, the realized abatement cost is about \$2,376/tCO_2, roughly 3.3 times the engineering-implied cost and well above standard estimates of the social cost of carbon. Simulation-based assessments may therefore substantially overstate the real-world effects of retrofit programs, which reinforces the importance of validating projected savings with observed consumption data.

These findings imply that program effectiveness can be improved by prioritizing envelope-focused retrofit

bundles, recalibrating ex ante savings predictions with realized utility data, considering cost-effectiveness across facility types when selecting projects, and strengthening post-retrofit commissioning and multi-year measurement and verification. Our results can help policymakers, engineers, and building owners assess the effectiveness of the Green Remodeling Program and contribute to the broader policy debate on whether publicly supported building retrofit programs are an effective instrument for reducing energy demand and achieving climate mitigation goals. This is particularly important given the scale of government policies and incentives that support energy efficiency in buildings, as well as initiatives to extend these efforts to private buildings.

Several limitations qualify the interpretation of our findings and suggest avenues for future research. First, our estimates cover the first twelve months after retrofit completion, a horizon constrained by the availability of post-retrofit utility records and the timing of project completion for later projects in the 2022 cohort. As more post-retrofit months become available, researchers can improve precision and examine longer-run persistence, potential rebound, and technology degradation; our cost-effectiveness calculations assume that first-year savings persist for 20 years. Second, not all participating buildings could be included because of missing energy data, failed address matches, or a lack of comparable non-participants. In particular, hospitals and health-center gas consumption could not be evaluated with a balanced comparison group, so our estimates should be generalized to these facilities with caution. Third, our cost data are available only at the project level, so we cannot attribute costs to individual technologies; the survey of participating institutions described in Section 2, which collects technology-specific costs and information on operating hours, setpoints, and budget rules, will allow future work to examine the behavioral and operational mechanisms behind the engineering–realized gap and to construct component-level cost-effectiveness estimates. Finally, complementary data on procurement processes and contractor selection could clarify whether the measure-specific overprediction we document reflects installation quality or subsidy-driven measure choice.

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Appendix A Propensity-Score Weighting and Balance Diagnostics

A.1 Detailed Covariate Balance

This appendix reports the detailed building-level covariate statistics underlying the balance plots in Section 3.1. Before-weighting columns use the outcome-specific treated and control samples prior to the final common-support restriction. After-weighting columns use the final common-support population and the corresponding ATT-IPW weights. The continuous covariates are reported on the same scale on which they enter the propensity-score specification.

Table A.1: Building attributes before and after ATT-IPW weighting: Daycare centers.

	Before weighting		After weighting	
Covariate	Treated	Control	Treated	Control
Panel A: Electricity				
Log building floor area	6.269 (0.579)	6.148 (0.496)	6.275 (0.579)	6.268 (0.488)
Log average pre-treatment electricity consumption	1.839 (0.599)	1.741 (0.655)	1.851 (0.588)	1.859 (0.629)
Log latest pre-treatment-year electricity consumption	1.788 (0.604)	1.697 (0.658)	1.799 (0.594)	1.805 (0.636)
Pre-treatment electricity trend	-0.040 (0.131)	-0.038 (0.113)	-0.041 (0.131)	-0.041 (0.128)
Panel B: Gas				
Log building floor area	6.443 (0.608)	6.167 (0.506)	6.428 (0.555)	6.427 (0.497)
Log average pre-treatment gas consumption	2.281 (0.539)	2.184 (0.481)	2.306 (0.478)	2.310 (0.420)
Log latest pre-treatment-year gas consumption	2.239 (0.544)	2.142 (0.490)	2.266 (0.481)	2.265 (0.411)
Pre-treatment gas trend	-0.039 (0.116)	-0.030 (0.139)	-0.038 (0.114)	-0.041 (0.151)
Panel C: Structural-adjusted total energy				
Log building floor area	6.273 (0.585)	6.149 (0.495)	6.256 (0.560)	6.254 (0.494)
Log average pre-treatment structural-adjusted total energy	2.485 (0.530)	2.371 (0.550)	2.480 (0.528)	2.476 (0.459)
Log latest pre-treatment-year structural-adjusted total energy	2.438 (0.541)	2.328 (0.559)	2.433 (0.537)	2.428 (0.475)
Pre-treatment structural-adjusted total-energy trend	-0.041 (0.124)	-0.037 (0.112)	-0.042 (0.123)	-0.042 (0.108)
Panel D: Strict dual-fuel total energy				
Log building floor area	6.443 (0.608)	6.166 (0.506)	6.439 (0.596)	6.429 (0.502)
Log average pre-treatment strict dual-fuel total energy	2.679 (0.373)	2.568 (0.346)	2.672 (0.365)	2.681 (0.355)
Log latest pre-treatment-year strict dual-fuel total energy	2.635 (0.383)	2.524 (0.365)	2.628 (0.372)	2.636 (0.369)
Pre-treatment strict dual-fuel total-energy trend	-0.042 (0.085)	-0.037 (0.116)	-0.042 (0.083)	-0.043 (0.110)

Note: Entries report building-level means with standard deviations in parentheses. Before-weighting columns are unweighted and use the outcome-specific sample eligible before final common-support restriction. After-weighting columns use the final common-support sample and the corresponding main ATT-IPW weights. Panel A reports electricity, Panel B gas, Panel C the structural-adjusted total-energy definition, and Panel D the strict dual-fuel total-energy definition. The structural-no-recorded-gas classification is fixed using pre-2021 information. Regional indicators remain included in the propensity-score model where applicable. A double dash indicates that the corresponding building-type–outcome model was not estimable.

Table A.2: Building attributes before and after ATT-IPW weighting: Health centers.

	Before weighting		After weighting	
Covariate	Treated	Control	Treated	Control
Panel A: Electricity				
Log building floor area	5.286 (0.499)	5.418 (0.827)	5.291 (0.501)	5.307 (0.736)
Log average pre-treatment electricity consumption	2.191 (0.685)	2.026 (0.690)	2.177 (0.674)	2.186 (0.818)
Log latest pre-treatment-year electricity consumption	2.161 (0.696)	1.989 (0.706)	2.146 (0.682)	2.156 (0.826)
Pre-treatment electricity trend	-0.029 (0.137)	-0.034 (0.139)	-0.030 (0.137)	-0.031 (0.124)
Panel B: Gas				
Log building floor area	–	–	–	–
Log average pre-treatment gas consumption	–	–	–	–
Log latest pre-treatment-year gas consumption	–	–	–	–
Pre-treatment gas trend	–	–	–	–
Panel C: Structural-adjusted total energy				
Log building floor area	5.291 (0.503)	5.413 (0.816)	5.295 (0.506)	5.311 (0.761)
Log average pre-treatment structural-adjusted total energy	2.200 (0.681)	2.055 (0.680)	2.189 (0.674)	2.190 (0.822)
Log latest pre-treatment-year structural-adjusted total energy	2.170 (0.692)	2.019 (0.696)	2.160 (0.683)	2.161 (0.831)
Pre-treatment structural-adjusted total-energy trend	-0.031 (0.136)	-0.033 (0.136)	-0.030 (0.136)	-0.031 (0.118)
Panel D: Strict dual-fuel total energy				
Log building floor area	–	–	–	–
Log average pre-treatment strict dual-fuel total energy	–	–	–	–
Log latest pre-treatment-year strict dual-fuel total energy	–	–	–	–
Pre-treatment strict dual-fuel total-energy trend	–	–	–	–

Note: Entries report building-level means with standard deviations in parentheses. Before-weighting columns are unweighted and use the outcome-specific sample eligible before final common-support restriction. After-weighting columns use the final common-support sample and the corresponding main ATT-IPW weights. Panel A reports electricity, Panel B gas, Panel C the structural-adjusted total-energy definition, and Panel D the strict dual-fuel total-energy definition. The structural-no-recorded-gas classification is fixed using pre-2021 information. Regional indicators remain included in the propensity-score model where applicable. A double dash indicates that the corresponding building-type–outcome model was not estimable.

We do not report a separate hospital before/after weighting table because no hospital building-type–outcome combination produced a final ATT-IPW sample satisfying the common-support and balance criteria used for the weighted DID analysis. In the health-center table, gas and strict dual-fuel total are marked as not estimable for the same reason. This presentation keeps the appendix focused on samples that enter the weighted treatment-effect analysis while making the excluded combinations explicit.

A.2 Pre-Treatment Monthly Energy Profiles

The standardized-mean-difference diagnostics are complemented by monthly pre-treatment energy profiles. These figures hold the final common-support population fixed and show participants, the same comparison

buildings before weighting, and those comparison buildings after ATT-IPW reweighting. They are descriptive balance diagnostics rather than treatment-effect estimates.

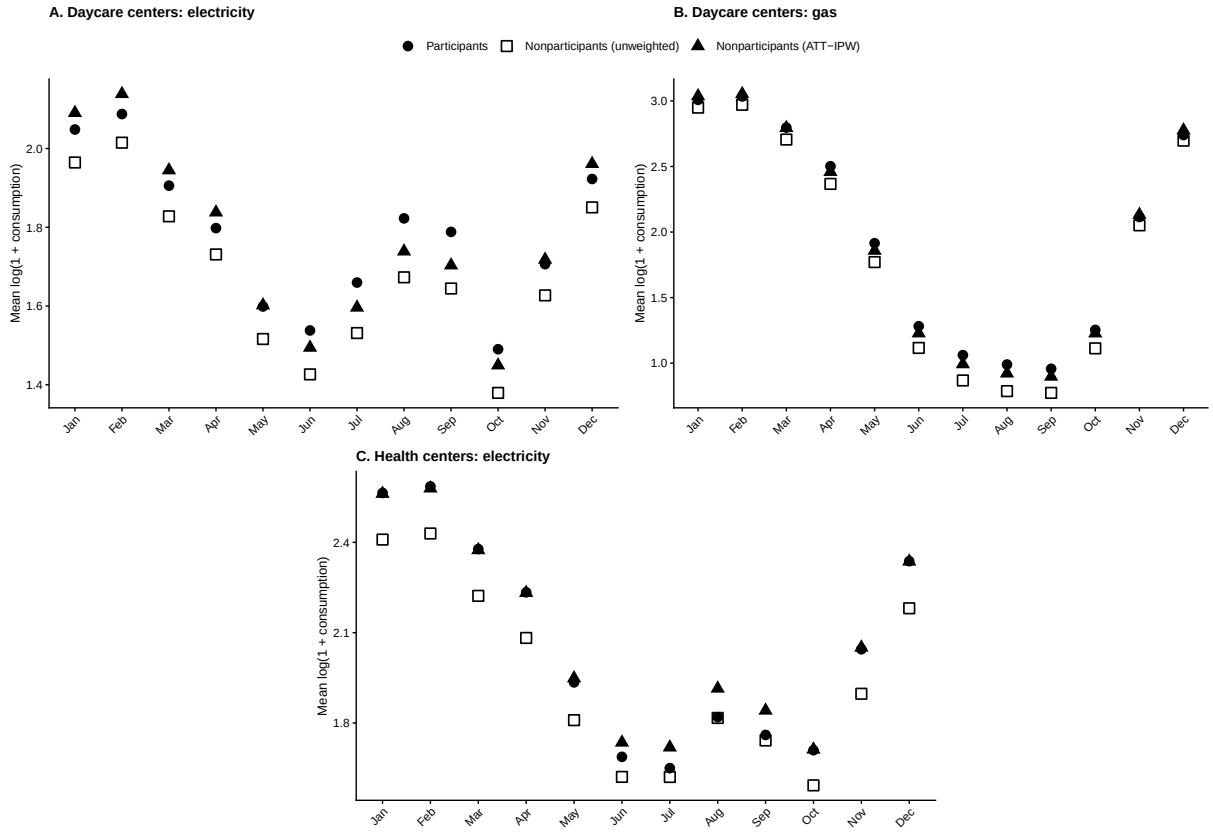


Figure A.1: Pre-treatment monthly energy profiles: electricity and gas components

Note: Panels A and B report daycare-center electricity and gas; Panel C reports health-center electricity. Each panel reports mean $\log(1 + \text{consumption})$ by calendar month during the pre-treatment period. Filled circles denote participating buildings, open squares denote the unweighted common-support comparison group, and triangles denote the same comparison group after ATT-IPW weighting. Monthly values are first averaged within buildings across pre-treatment years and then aggregated across buildings.

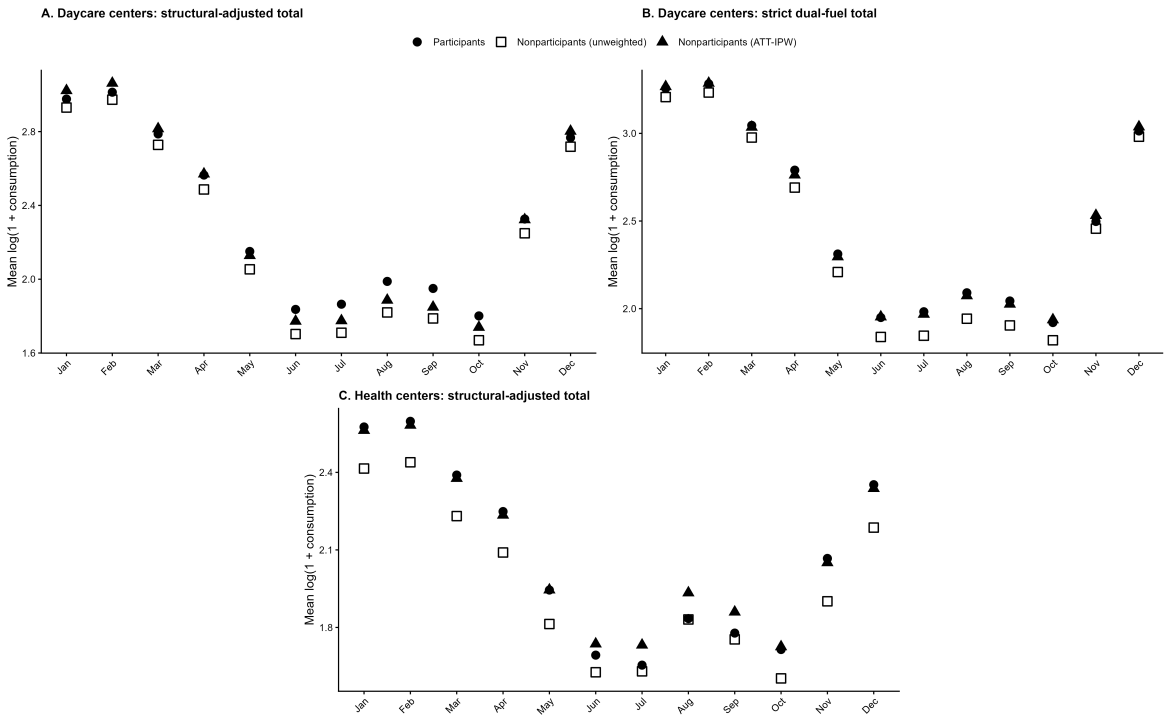


Figure A.2: Pre-treatment monthly energy profiles: total-energy definitions

Note: Panels A and B report daycare centers under the structural-adjusted and strict dual-fuel total-energy definitions, respectively. Panel C reports health centers under the structural-adjusted definition. Symbols have the same interpretation as in Appendix Figure A.1. Each outcome uses its own final common-support population and corresponding ATT-IPW weights.

A.3 Additional Weighting Diagnostics

The balance tables and figures summarize the principal continuous-covariate diagnostics. We additionally examine regional balance indicators, treated and control propensity-score distributions before and after the common-support restriction, the distribution of ATT weights, and the effective sample size of the weighted control group. These diagnostics are used together rather than relying on a single balance statistic.

The unavailable building-type–outcome combinations are driven by data and support limitations rather than by omitting an otherwise satisfactory weighted estimate. In particular, health-center gas has too few treated buildings with usable pre-treatment gas histories, while the hospital samples do not provide a sufficiently balanced common-support comparison under the weighting specifications used in the paper.

Appendix B Additional Robustness Checks

This section reports additional checks referred to in Section 4.6: (1) whether changes in building occupancy could generate the estimated declines in energy use; (2) whether common shocks such as the COVID-19 pandemic affected energy use during the sample period; and (3) whether the estimated electricity savings are driven by behind-the-meter renewable generation.

B.1 Occupancy

First, we estimate equation 10, using the number of daycare, health center, and hospital users as the independent variable. We focus on active user count because it provides a consistent, facility-specific proxy for utilization intensity over time, whereas other measures, such as the duration of energy use, are not readily available at the required temporal or spatial resolution.

For daycare centers, we use annual statistics from the Korea Childcare Promotion Institute (KCPI). The KCPI reports the number of active users by province and daycare center type. We aggregate annual active users across daycare centers that are eligible to apply for the Green Remodeling Program. For health centers and hospitals, we follow the same approach using data from the Korean Statistical Information Service (KOSIS), which provides annual transaction counts by province. We aggregate these counts across eligible health centers and hospitals only.

Using these two groups, we estimate the following specification to examine whether changes in active user volume are associated with changes in energy consumption over time:

$$E_{it} = \beta_1 K_{it} + \alpha_i + \gamma_t + \varepsilon_{it}, \tag{10}$$

where E_{it} denotes energy consumption for unit i at time t. The term K_{it} is the number of active users for

unit i at time t. Unit (province) fixed effects α_i control for time-invariant differences across units, and year fixed effects γ_t control for common shocks affecting all units in a given period. The error term ε_{it} captures unobserved determinants of energy consumption.

Table B.1 reports the estimates separately for daycare centers and for the combined sample of health centers and hospitals, which we combine because of small sample sizes. In both groups and for both fuels, the estimated relationship between active user counts and energy consumption is statistically insignificant. These results provide little empirical support for the alternative hypothesis that changes in utilization or occupancy systematically drive energy consumption trends in a way that could mechanically generate our main estimates.

Table B.1: Energy Consumption by Yearly Active User Count

	Panel A: Electricity Consumption	
Model	Daycare Center	Health Center & Hospital
Variables		
Active Users	20.38	0.27
	(41.88)	(0.29)
Fit statistics		
Observations	112	119
R^2	0.99	0.99
	Panel B: Gas Consumption	
Model	Daycare Center	Health Center & Hospital
Variables		
Active Users	-44.11	0.03
	(85.65)	(0.18)
Fit statistics		
Observations	112	119
R^2	0.99	0.98
Fixed-effects		
Province	Yes	Yes
Year	Yes	Yes

Note: The table reports the relationship between yearly energy consumption and the total active user count, aggregated by province. Standard errors are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

B.2 COVID-19 and Common Time Shocks

Figure B.1 plots mean monthly electricity and gas consumption per square meter by building type from January 2018 to December 2025. Energy use displays strong seasonal variation, particularly for gas, but there is no clear common decline in consumption during the COVID-19 period, and the patterns differ across building types. Combined with the year-month fixed effects in all TWFE and Sun–Abraham specifications and the use of contemporaneous comparison buildings in the Callaway–Sant'Anna estimates, this suggests

that pandemic-related changes in facility use are unlikely to explain our results.

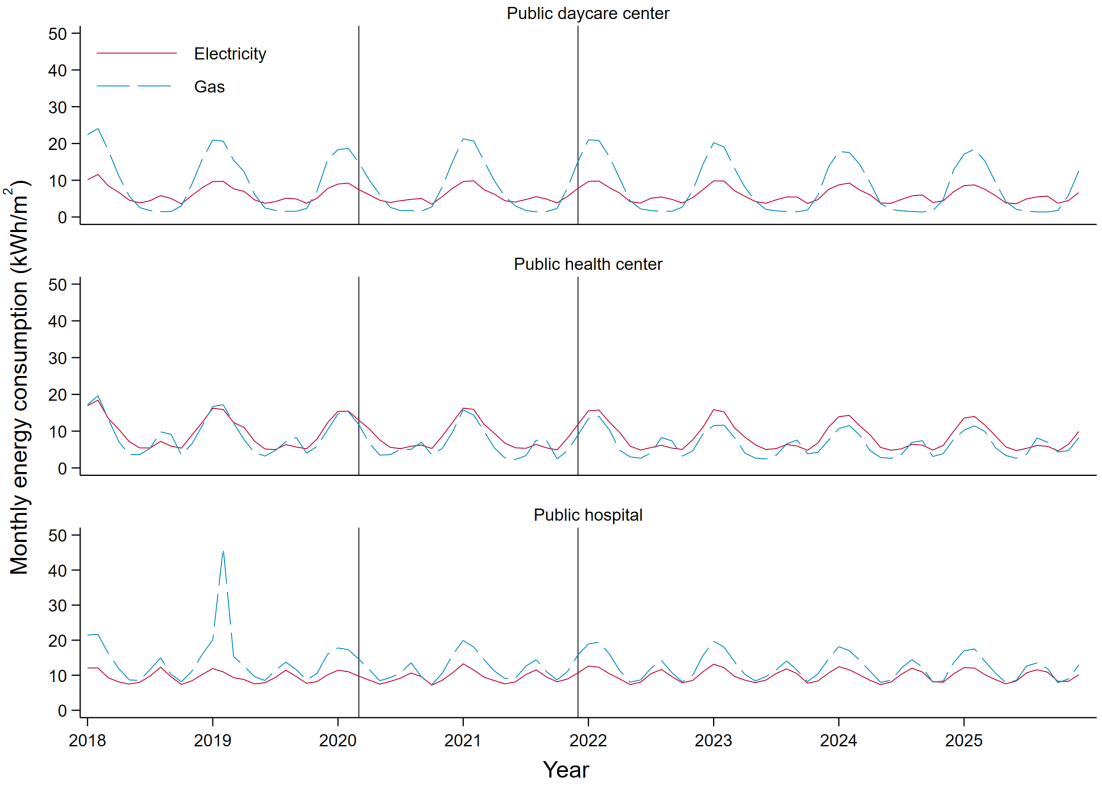


Figure B.1: Monthly Energy Consumption by Building Type, 2018–2025

Note: Mean monthly electricity and gas consumption per square meter by building type, January 2018–December 2025. Vertical lines indicate March 2020 and December 2021.

B.3 Excluding Renewable-Energy Retrofits

Table B.2 re-estimates the main specifications after excluding treated buildings whose retrofit scope includes renewable-energy measures. The results are discussed in Section 4.6.

Table B.2: Program Effects Excluding Renewable-Energy Retrofits

	(1)	(2)	(3)	(4)
Panel A: Electricity Consumption				
Green Remodeling × Post	-0.116^{***}	-0.103^{***}	-0.107^{***}	-0.100^{***}
	(0.024)	(0.024)	(0.024)	(0.038)
Observations	521,899	521,899	521,899	510,762
R^2	0.877	0.836	0.839	–
Panel B: Gas Consumption				
Green Remodeling × Post	-0.091^{*}	-0.095^{**}	-0.106^{**}	0.079
	(0.046)	(0.046)	(0.042)	(0.082)
Observations	235,613	229,719	229,719	225,703
R^2	0.929	0.919	0.924	–
Estimator	TWFE	TWFE	S&A	C&S
Matched sample	No	Yes	Yes	Yes
Building FE	Yes	Yes	Yes	–
Year-Month FE	Yes	Yes	Yes	–

Note: The table reports estimates of the effect of Green Remodeling participation on the natural logarithm of electricity and gas consumption after excluding treated buildings whose retrofit scope includes renewable-energy measures. Column (1) reports TWFE estimates using the full sample. Column (2) reports TWFE estimates using the matched sample with inverse-probability weights. Column (3) reports Sun–Abraham estimates using the matched sample. Column (4) reports Callaway–Sant'Anna estimates using the matched sample. Construction-period observations are excluded. Standard errors are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Appendix C Event-Study Robustness and Component Outcomes

This section reports the event-study results that complement the main CSDID analysis of total delivered energy. We first show the CSDID estimates for the electricity and gas components separately. We then report conventional two-way fixed effects (TWFE) and Sun–Abraham event studies for the two total-energy constructions and for the electricity and gas components. The structural- adjusted and strict dual-fuel total-energy outcomes are defined in Section 3.2.

C.1 CSDID Component Outcomes

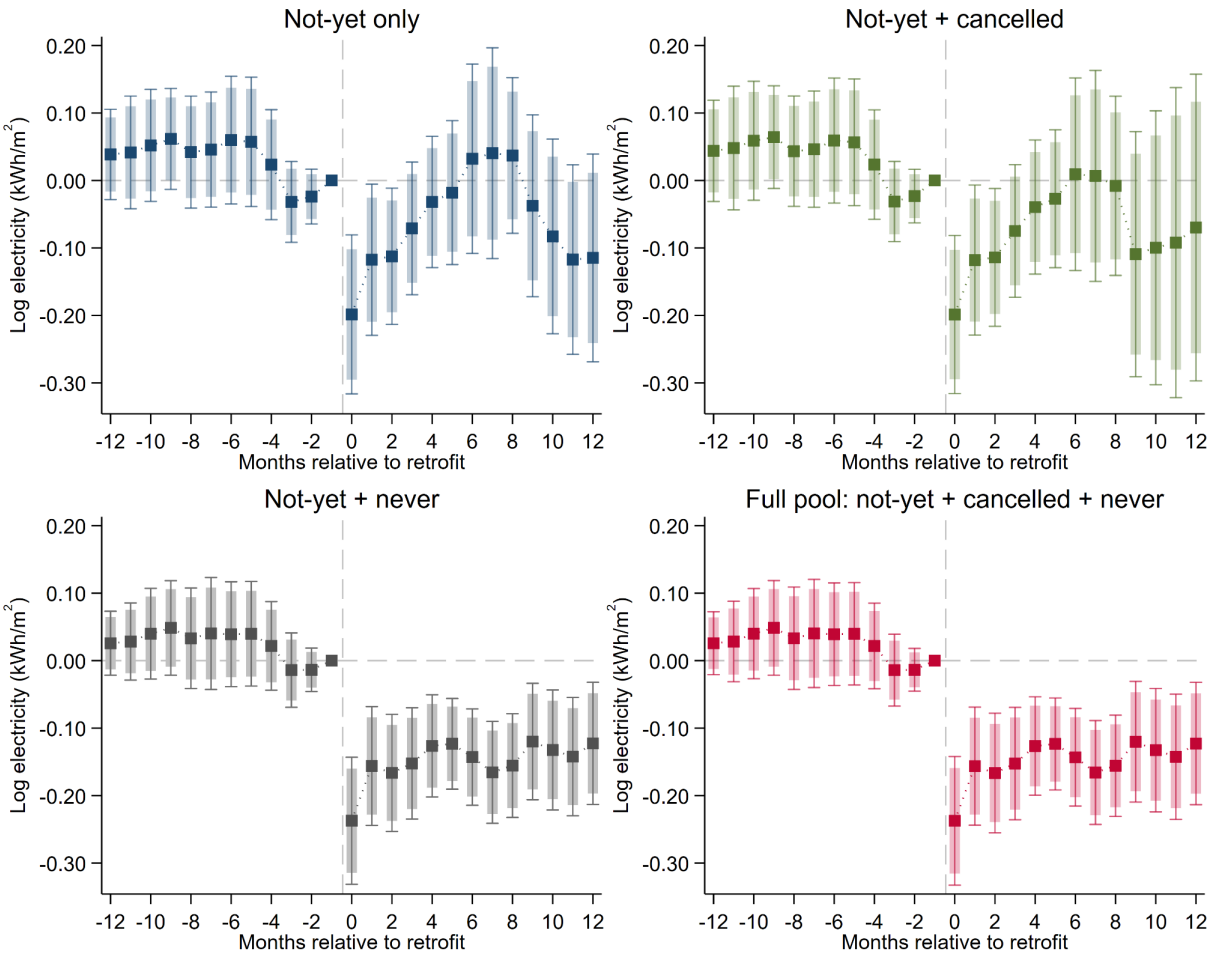


Figure C.1: CSDID Event Study: Electricity

Note: The figure reports CSDID dynamic treatment-effect estimates for log electricity consumption. Event month 0 is the first full month after retrofit completion. The four panels vary the comparison pool across not-yet-treated only; not-yet-treated plus cancelled; not-yet-treated plus never-treated; and the full comparison pool. Vertical bars report 95 percent confidence intervals. Construction months are excluded.

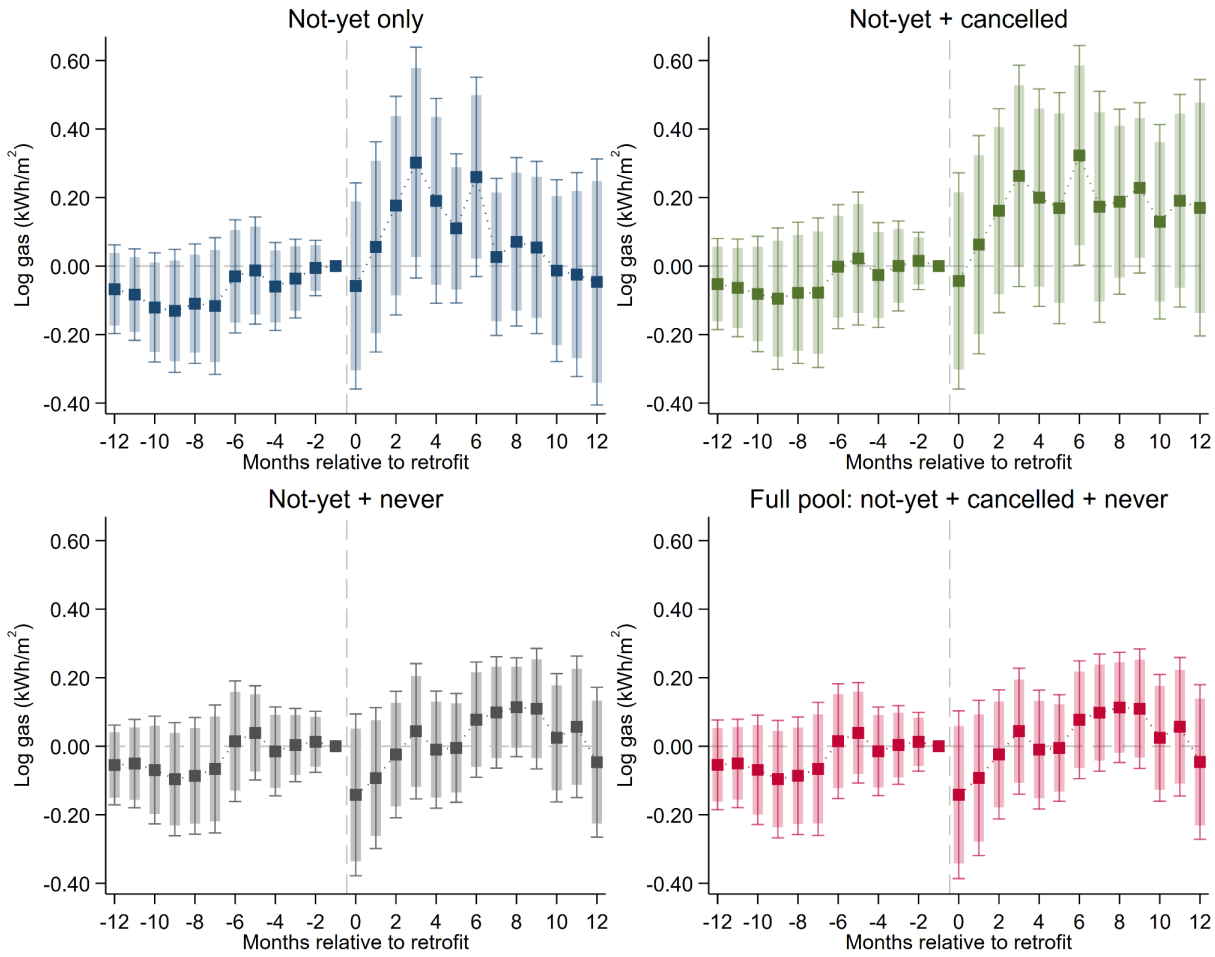


Figure C.2: CSDID Event Study: Gas

Note: The figure reports CSDID dynamic treatment-effect estimates for log gas consumption using the same four comparison-pool definitions as Appendix Figure C.1. Vertical bars report 95 percent confidence intervals. Construction months are excluded.

C.2 TWFE and Sun–Abraham Event Studies

Because conventional TWFE event studies can use already-treated units as controls when treatment timing is staggered, we treat these estimates as robustness checks rather than as the primary dynamic specification. The Sun–Abraham estimator provides a heterogeneity-robust comparison that avoids such already-treated comparisons. Each figure reports both estimators under two comparison-group definitions: never-treated buildings only and never-treated plus cancelled buildings.

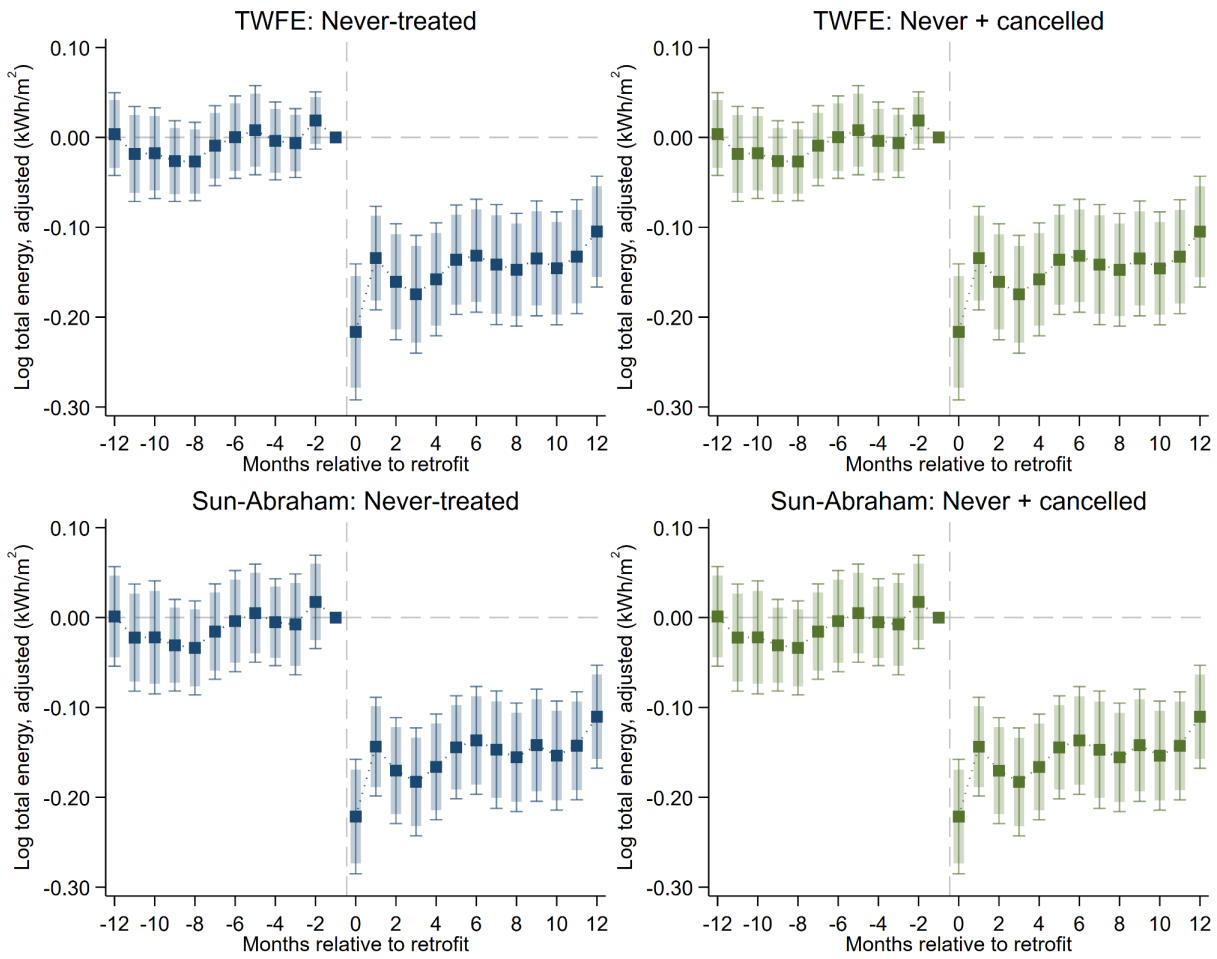


Figure C.3: TWFE and Sun–Abraham Event Studies: Structural-Adjusted Total Delivered Energy

Note: The outcome is the log of structural-adjusted total delivered energy defined in Section 3.2. The top row reports TWFE estimates and the bottom row reports Sun–Abraham estimates. The left column uses never-treated controls and the right column adds eligible cancelled controls. Vertical bars report 95 percent confidence intervals. Construction months are excluded.

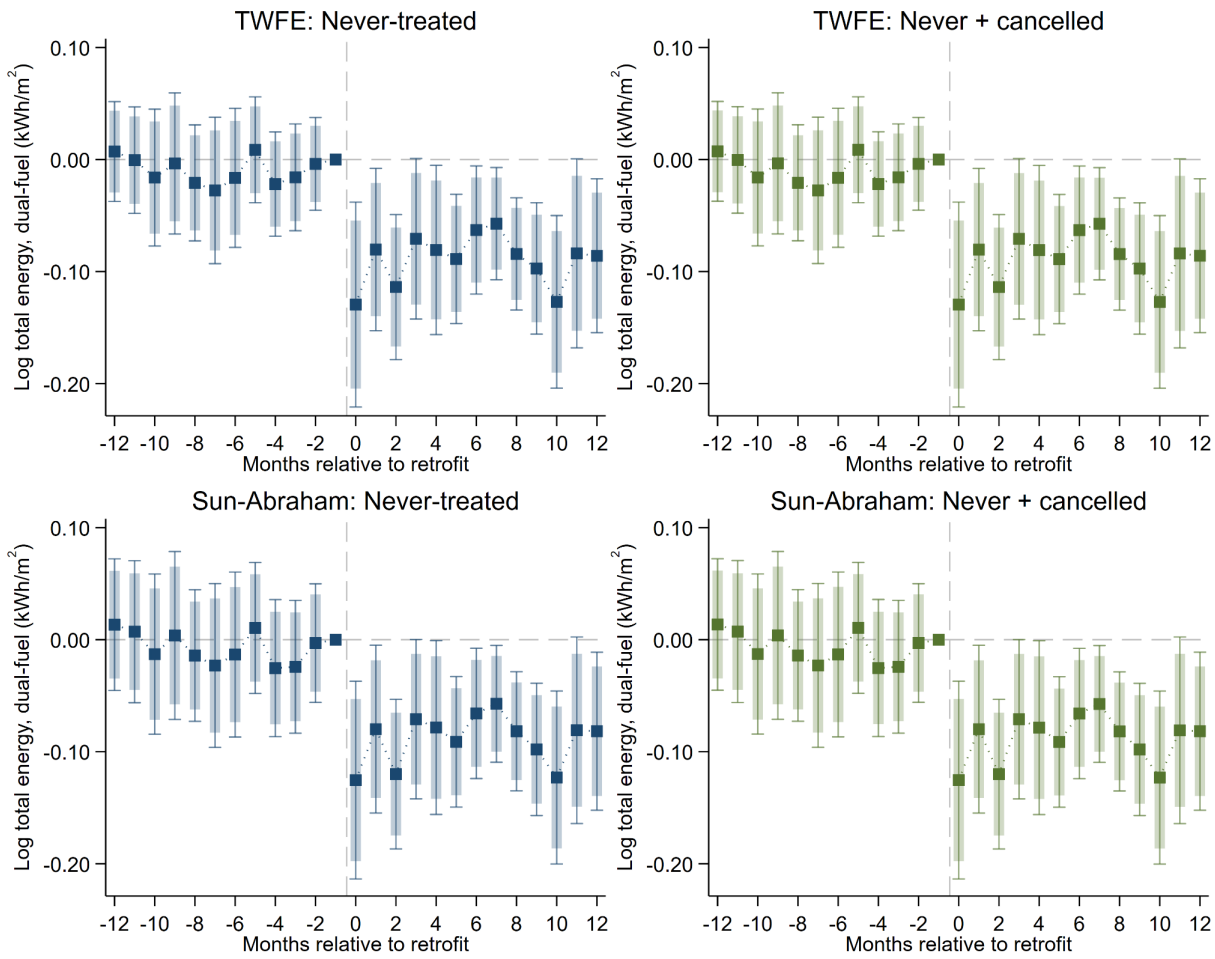


Figure C.4: TWFE and Sun–Abraham Event Studies: Strict Dual-Fuel Total Delivered Energy

Note: The outcome is the log of strict dual-fuel total delivered energy defined in Section 3.2. Panel organization and confidence intervals follow Appendix Figure C.3.

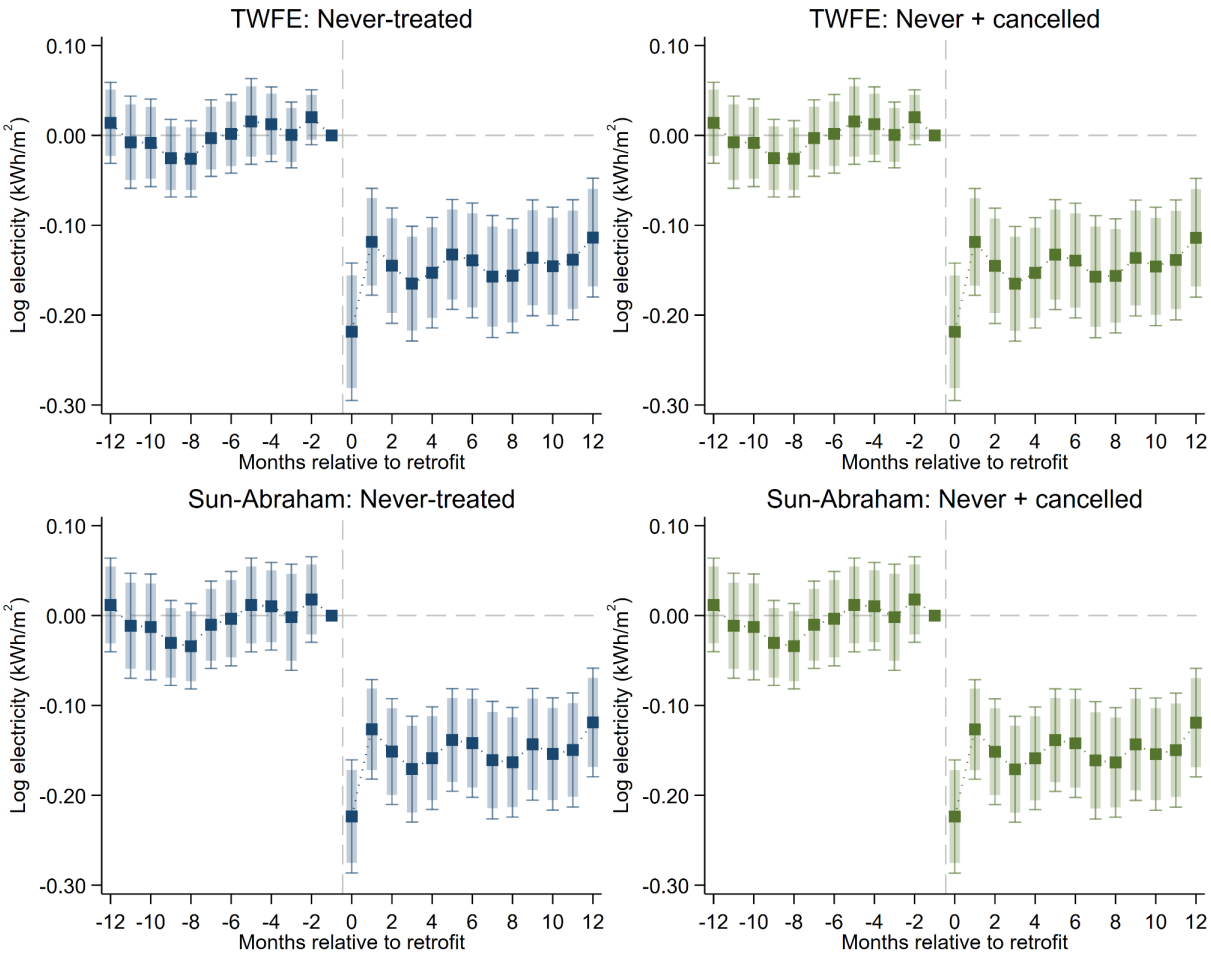


Figure C.5: TWFE and Sun–Abraham Event Studies: Electricity

Note: The outcome is log electricity consumption. The top row reports TWFE estimates and the bottom row reports Sun–Abraham estimates. The left column uses never-treated controls and the right column adds eligible cancelled controls. Vertical bars report 95 percent confidence intervals. Construction months are excluded.

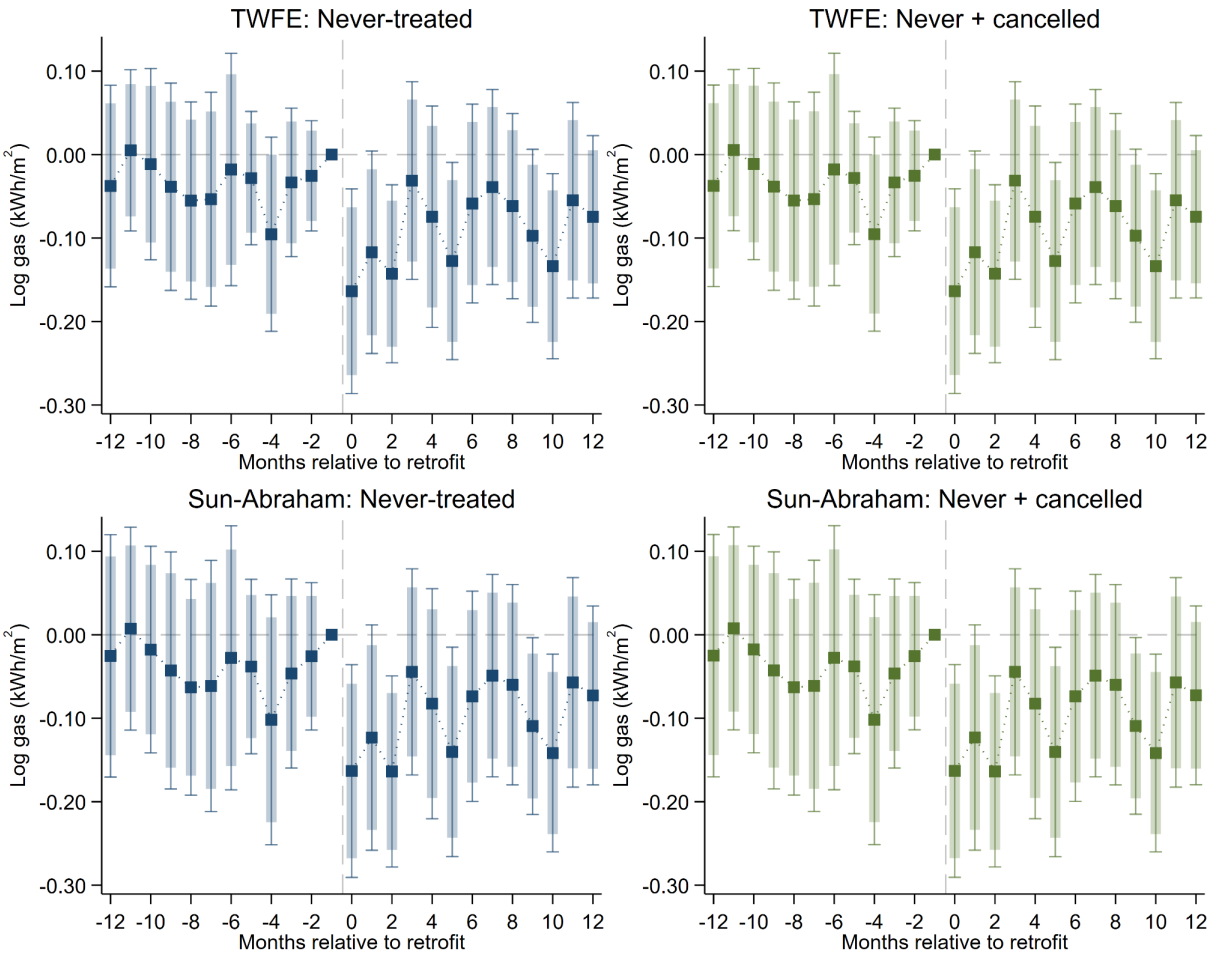


Figure C.6: TWFE and Sun–Abraham Event Studies: Gas

Note: The outcome is log gas consumption. Panel organization and confidence intervals follow Appendix Figure C.5.

Appendix D Additional Figures

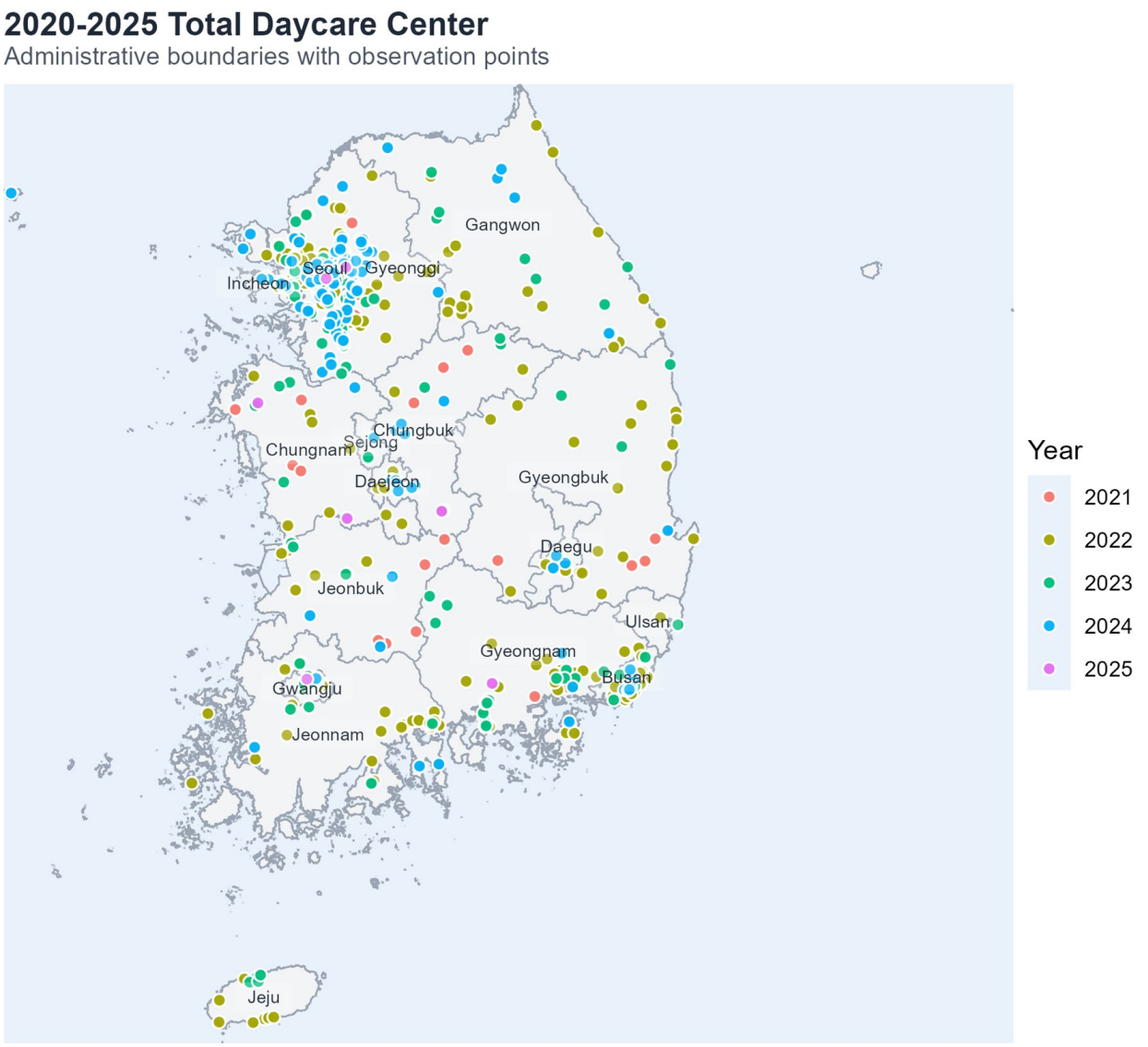


Figure D.1: Green Remodeled Daycare Centers: 2021-2025

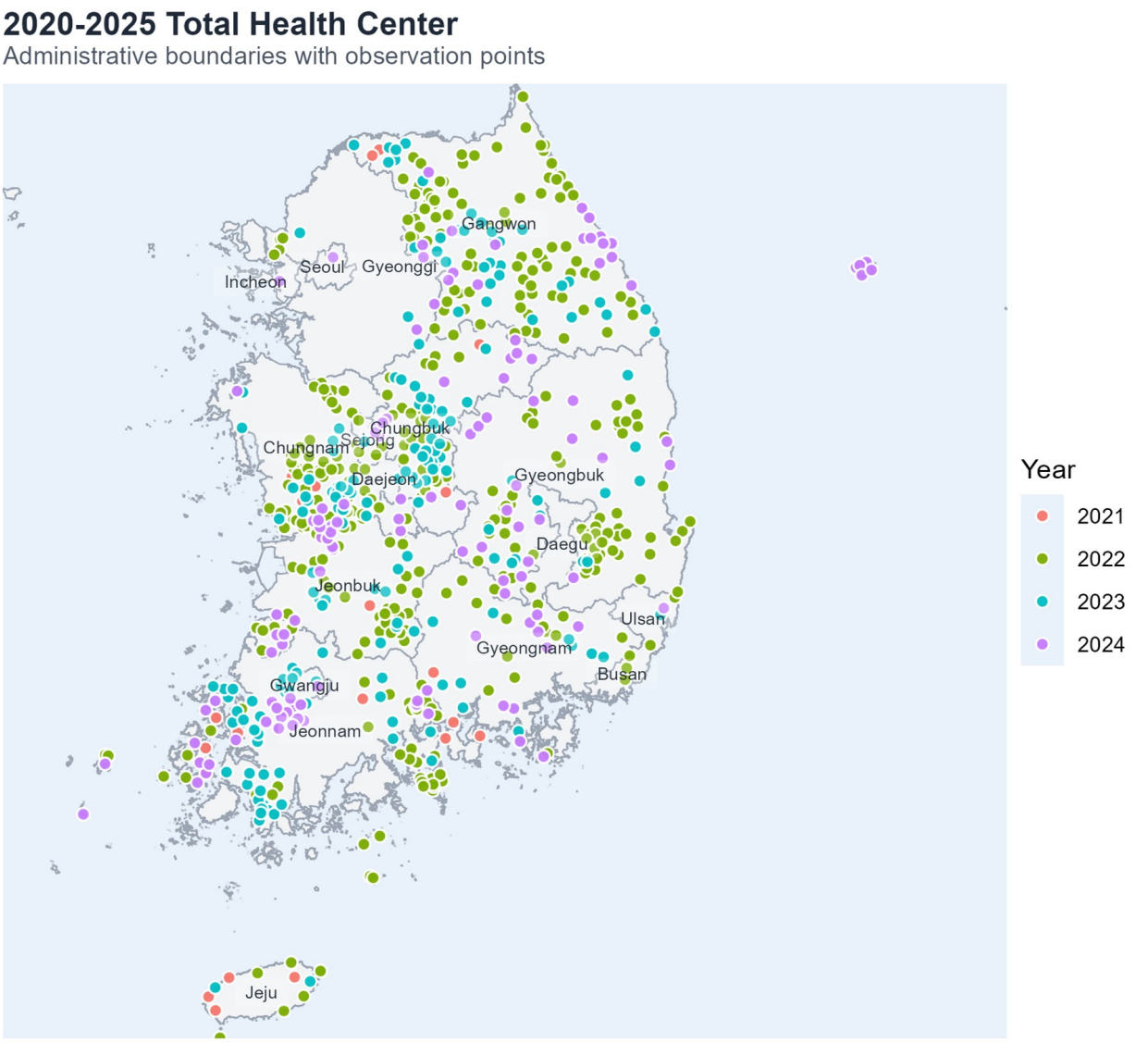


Figure D.2: Green Remodeled Health Care Centers: 2021-2024

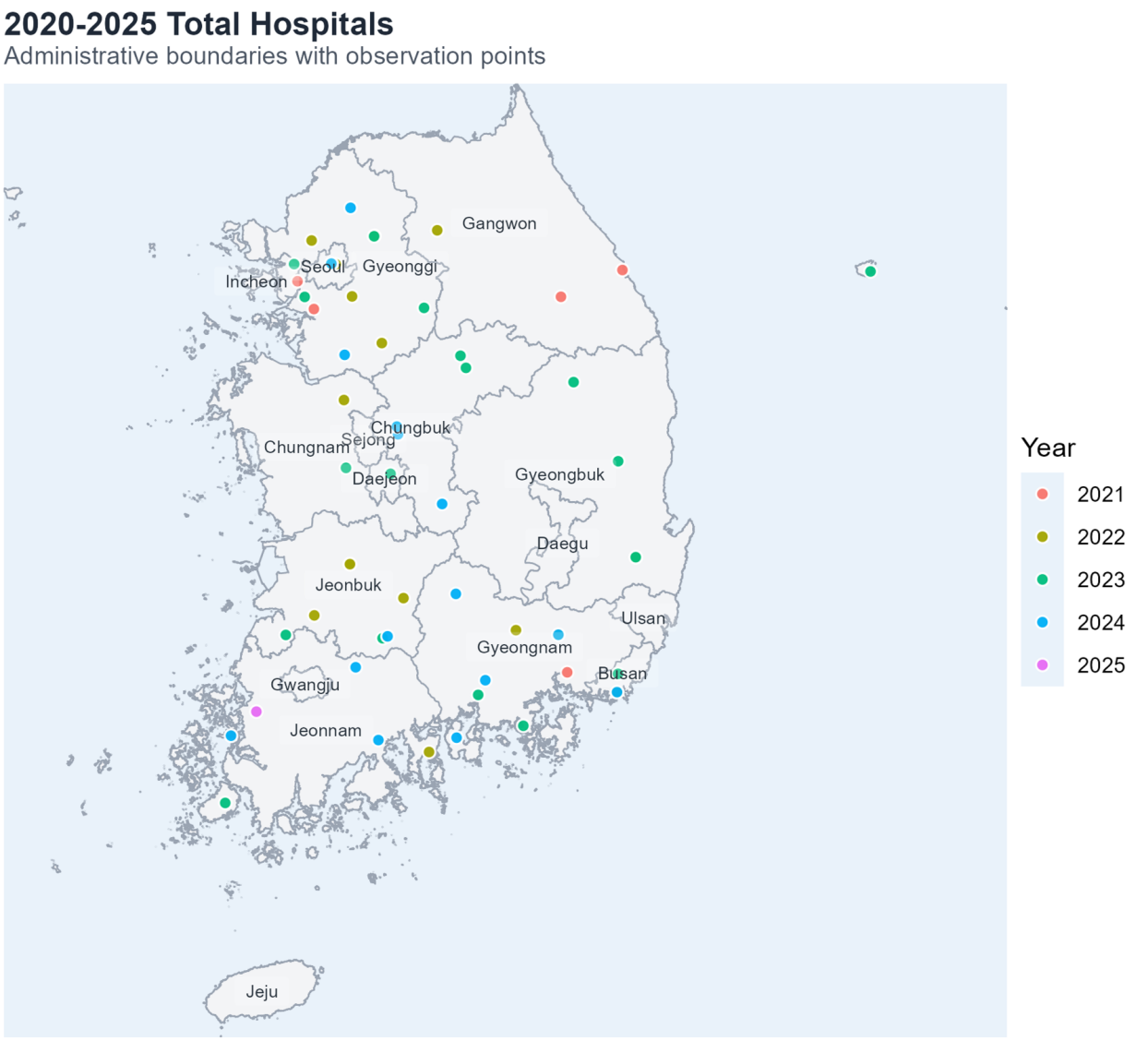


Figure D.3: Green Remodeled Hospitals: 2021-2025

Appendix E Full-Sample Engineering–Realized Comparison

The main text reports the engineering–realized comparison using the current matched structural-adjusted total-energy sample. As a descriptive robustness check, this appendix repeats the building-level scatter using the broader 2021–2022 treated sample of public daycare centers and public health centers. The full-sample specification does not impose the outcome-specific ATT-IPW common-support restriction used in the main analysis and therefore should be interpreted as a broader descriptive comparison rather than as an alternative causal estimate.

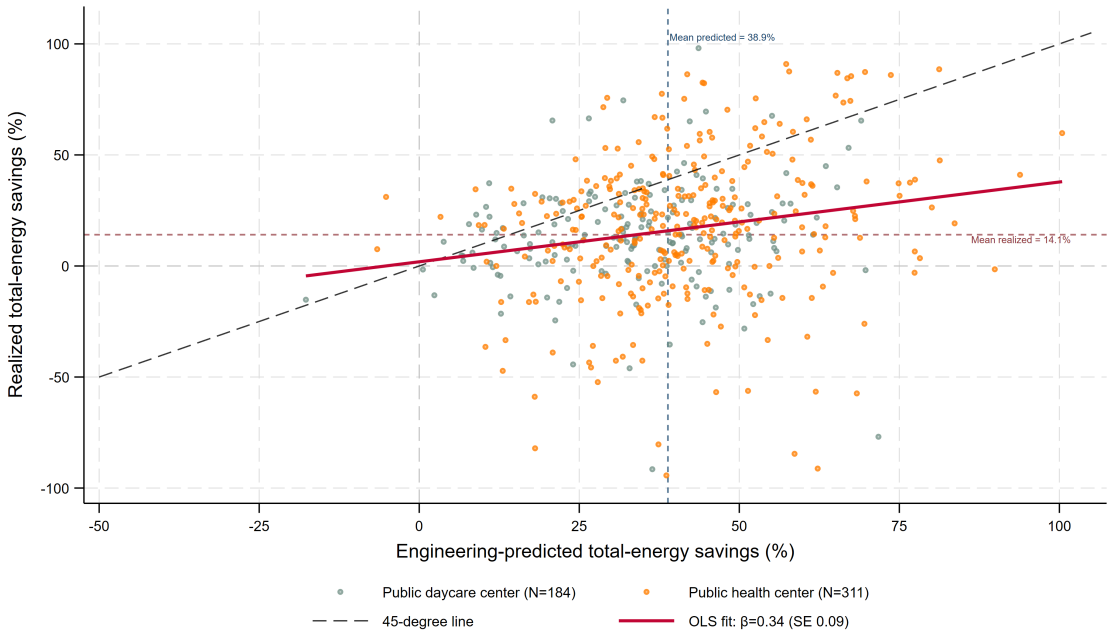


Figure E.1: Engineering-Predicted vs. Realized Total-Energy Savings: Full Sample

Note: Each point is a treated public daycare center or public health center in the broader 2021–2022 full sample for which both the ECO2 engineering prediction and realized total-energy saving are observed. Hospitals are excluded. The dashed 45-degree line represents equality between predicted and realized savings, while the fitted line summarizes their empirical relationship. Vertical and horizontal dashed reference lines report sample means. The full-sample calculation follows the original descriptive pre/post construction and does not impose the outcome-specific ATT-IPW common-support restriction used for the main matched analysis. Accordingly, this figure is interpreted as a robustness and descriptive comparison rather than as a causal ATT estimate.

E.1 Technology Decomposition of the Engineering Overprediction Gap

Figure 18 in the main text shows the technology-specific regression coefficients graphically. Table E.1 provides the corresponding decomposition. For each retrofit technology, the table reports the conditional coefficient, its prevalence in the estimation sample, the implied contribution to the mean engineering overprediction gap, and that contribution as a share of the observed mean gap. Panel A uses the current matched structural-adjusted total-energy sample, while Panel B uses the previous full-sample total-energy construction.

Table E.1: Technology Decomposition of the Engineering Overprediction Gap

Technology	$\hat{\beta}_k$ pp	Prevalence %	Contribution pp	Share of gap %	Sig.
Panel A. Structural-adjusted total (matched)			($N = 301$; mean gap = 26.2 pp)		
Wall insulation	-3.06	80.4	-2.46	-9.4	
Roof insulation	7.61	49.2	3.74	14.3	*
Floor insulation	4.36	18.6	0.81	3.1	
Windows	-1.29	85.7	-1.11	-4.2	
Doors	-4.73	63.5	-3.00	-11.4	
HVAC	7.46	84.7	6.32	24.1	*
Ventilation	-1.89	52.8	-1.00	-3.8	
Boiler	6.95	55.5	3.86	14.7	*
Lighting	8.77	54.8	4.81	18.3	*
Renewable energy	-0.77	27.6	-0.21	-0.8	
Monitoring	1.80	4.3	0.08	0.3	
Cool roof / shading	13.71	5.3	0.73	2.8	
Ancillary works	-3.67	8.6	-0.32	-1.2	
All technologies, net			**12.26**	**46.8**	
Positive effects only			**20.35**	**77.7**	
Panel B. Previous method (full sample)			($N = 494$; mean gap = 25.4 pp)		
Wall insulation	-0.12	81.6	-0.10	-0.4	
Roof insulation	2.68	49.4	1.33	5.2	
Floor insulation	12.35	19.2	2.38	9.3	**
Windows	-3.29	86.8	-2.86	-11.2	
Doors	-6.08	64.2	-3.90	-15.3	
HVAC	3.84	83.2	3.19	12.6	
Ventilation	3.17	54.9	1.74	6.8	
Boiler	3.94	51.6	2.03	8.0	
Lighting	7.13	52.4	3.74	14.7	**
Renewable energy	-1.06	27.5	-0.29	-1.2	
Monitoring	-6.70	4.7	-0.31	-1.2	
Cool roof / shading	3.48	3.8	0.13	0.5	
Ancillary works	-2.20	9.3	-0.20	-0.8	
All technologies, net			**6.87**	**27.0**	
Positive effects only			**14.54**	**57.2**	

Note: The dependent variable is the building-level engineering overprediction gap, defined as ECO2 engineering-predicted total-energy savings minus realized total-energy savings, in percentage points. Panel A uses the current matched structural-adjusted total-energy sample. Panel B uses the previous full-sample construction from the original technology-gap analysis. Both panels are restricted to public daycare centers and public health centers; hospitals are excluded. Each regression jointly includes all listed retrofit-technology indicators and controls for log baseline realized total energy, building type, project year, and log project cost when observed. Contribution is calculated as $\hat{\beta}_k \times Pr(Technology_k = 1)$. Share of gap divides each contribution by the corresponding sample mean overprediction gap. Because the regression includes an intercept and controls and technologies can overlap, contribution shares need not sum to 100%. Robust standard errors are used to calculate significance. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

The decomposition should be interpreted descriptively. Retrofit technologies are frequently installed jointly, and the regressions include both overlapping technology indicators and additional controls. Consequently, the reported shares are not causal attribution parameters and need not sum to 100%. Their primary purpose is to show which installed technologies are most strongly associated with variation in the engineering–realized wedge and whether those patterns are robust to the alternative total-energy construction.

Appendix F Machine-Learning Counterfactual Analysis

This appendix provides the model-selection and predictive diagnostics underlying the machine-learning counterfactual analysis in Sections 3.7 and 4.7. The exercise uses the same 2021–2022 treated daycare-center and health-center population across model classes, with building-grouped nested cross-validation to ensure that all months from a held-out building remain outside the corresponding training sample.

F.1 Model Selection and Predictive Performance

Five candidate algorithms are compared: OLS, Lasso, random forest, XGBoost, and a feed-forward neural network. Each candidate is evaluated with five outer building folds and three inner tuning folds. The final algorithm is selected separately for each outcome by the lowest combined held-out RMSE across the untreated-state pre-retrofit predictions and the treated-state post-retrofit predictions.

Table F.1: Machine-Learning Model Selection by Held-Out RMSE

Outcome	OLS	Lasso	Random forest	XGBoost	Neural net
Structural-adjusted total	5.935	5.760	**5.095**	5.242	5.169
Electricity	5.799	5.678	**4.772**	5.064	5.221
Gas	7.186	4.323	**3.684**	3.698	4.388
Strict dual-fuel total	8.804	4.338	3.890	**3.858**	4.841

Note: Entries are combined held-out RMSE in kWh/m^2 from 5 outer × 3 inner building-grouped nested cross-validation. Combined RMSE pools untreated-state pre-retrofit predictions and treated-state post-retrofit predictions. Bold entries indicate the selected algorithm for each outcome.

Random forest is selected for structural electricity, structural gas, and structural-adjusted total delivered energy. XGBoost is selected for the strict dual-fuel total robustness definition. The ranking is not imposed in advance: for example, random forest lowers combined RMSE relative to OLS by about 18% for electricity, 49% for gas, and 14% for structural-adjusted total energy. For strict dual-fuel total, XGBoost lowers combined RMSE by about 56% relative to OLS.

For the random-forest candidates, hyperparameters are tuned within each outer fold over the fraction of predictors considered at each split (0.30 or 0.60) and minimum terminal-node sizes of 5 or 12, using an 0.80 sampling fraction. The selected model is refit with 500 trees. XGBoost is tuned over learning rates of 0.03–0.08, maximum tree depths of 2–3, minimum child weights of 10–20, and row/column subsampling rates

of 0.80–0.90, with early stopping used to determine the number of boosting rounds. The neural-network candidate uses three hidden layers with 64, 32, and 16 units. These tuning choices are determined only within the inner folds.

F.2 Counterfactual Diagnostics

Table F.2: Predictive Diagnostics for Selected Machine-Learning Models

Outcome	Model	Held-out bias	Mean pre-error	Pre CIs incl. 0	Weather diff.
Structural-adjusted total	Random forest	−0.105	−0.084	12/12	−0.011
Electricity	Random forest	−0.079	−0.073	11/12	0.014
Gas	Random forest	−0.003	−0.013	12/12	−0.072
Strict dual-fuel total	XGBoost	−0.189	−0.204	12/12	−0.046

Note: All values except the CI count are in kWh/m^2. Held-out bias is the combined mean prediction error over the outer-fold test observations. Mean pre-error averages the event-month mean prediction error over months −12 through −1. "Pre CIs incl. 0" reports the number of pre-retrofit event-month 95% confidence intervals that contain zero. "Weather diff." is the mean difference between predicted savings under expected historical weather and predicted savings under observed post-retrofit weather.

Table F.2 reports diagnostics for the selected models. For the primary structural-total model, all twelve pre-retrofit event-month prediction-error confidence intervals contain zero. The same is true for gas and strict dual-fuel total; for electricity, eleven of twelve intervals contain zero. We also compare treatment-effect predictions evaluated under expected historical weather with predictions evaluated under realized post-retrofit weather. The mean difference is small relative to the estimated savings: approximately −0.01 kWh/m^2 per month for structural total, 0.01 for electricity, and −0.07 for gas.

The full ML population contains 500 treated daycare and health-center buildings. The untreated-state training samples contain 478 buildings for structural total, 491 for electricity, 110 with sufficient positive gas observations, and 109 for strict total. The corresponding treated-state training samples contain 472, 487, 114, and 113 buildings, respectively. The balanced annual savings summaries require all twelve post-retrofit months and therefore use somewhat smaller samples.

F.3 Event-Time Counterfactual Figures

Figures F.1–F.4 plot deviations from each building's historical same-calendar-month mean. The blue series is observed energy use. The orange series is the cross-fitted prediction for the relevant observed state: untreated before retrofit and treated after retrofit. The green post-retrofit series is the predicted no-retrofit counterfactual. Shaded bands summarize uncertainty around the monthly means. The gray vertical band separates the construction/transition period from the first complete post-retrofit month.

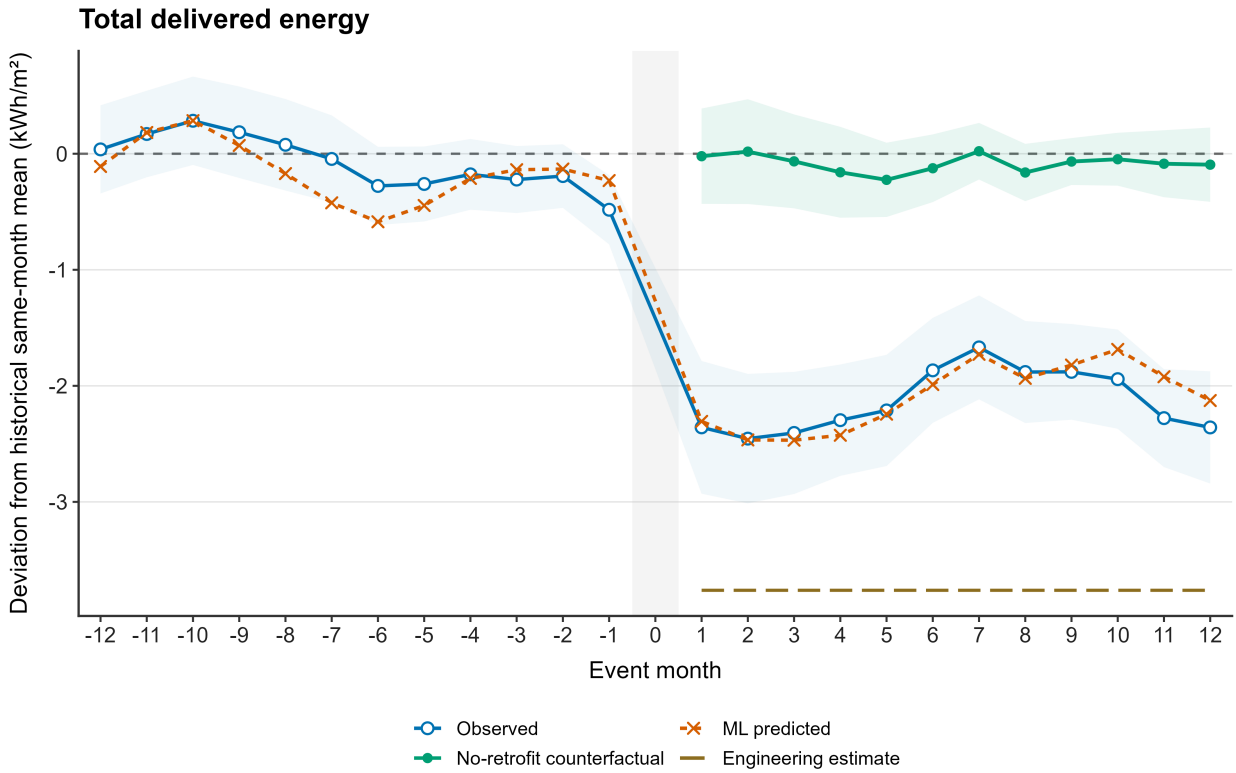


Figure F.1: Machine-Learning Counterfactual: Structural-Adjusted Total Delivered Energy

Note: Outcomes are monthly kWh/m^2 deviations from the same-building, same-calendar-month historical mean. The selected model is random forest, chosen by 5×3 building-grouped nested cross-validation. The orange series reports cross-fitted state predictions and the green series reports the no-retrofit post-treatment counterfactual. The ECO2 engineering line is the mean monthly engineering saving translated to the same absolute kWh/m^2 scale.

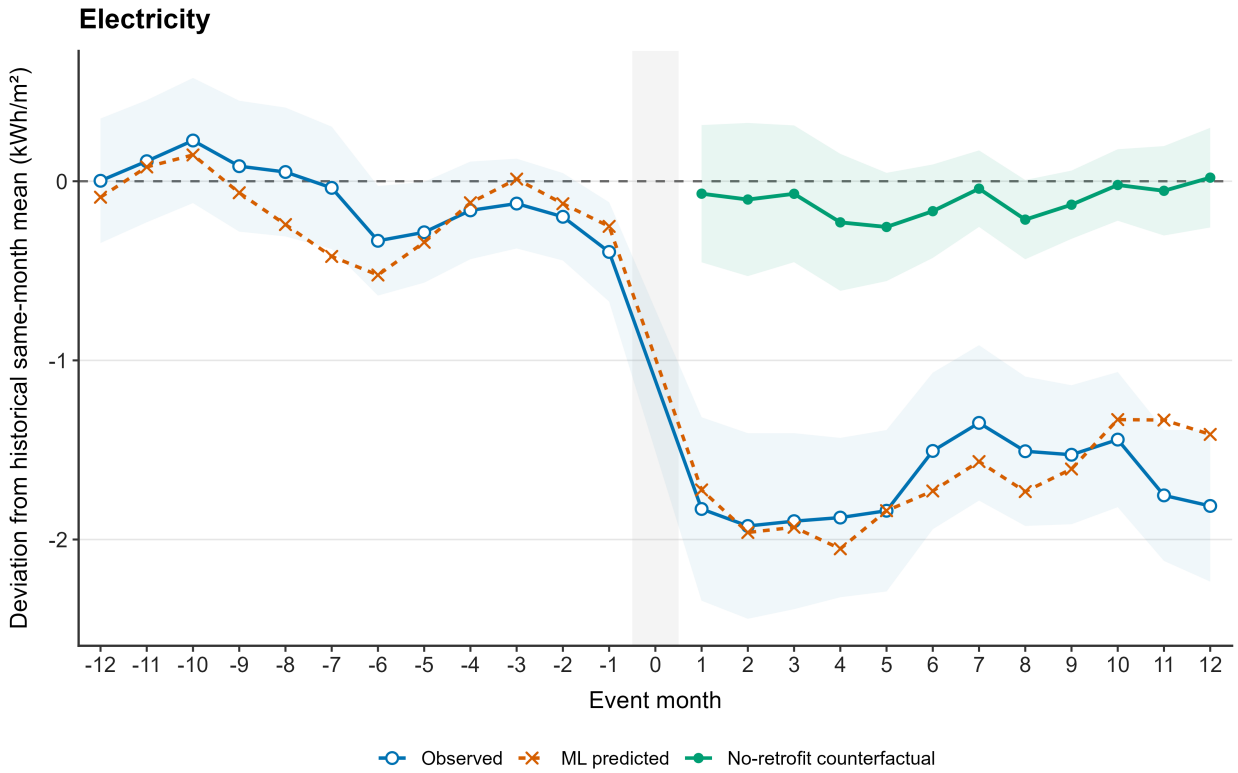


Figure F.2: Machine-Learning Counterfactual: Electricity

Note: The selected model is random forest. Definitions and normalization follow Appendix Figure F.1.

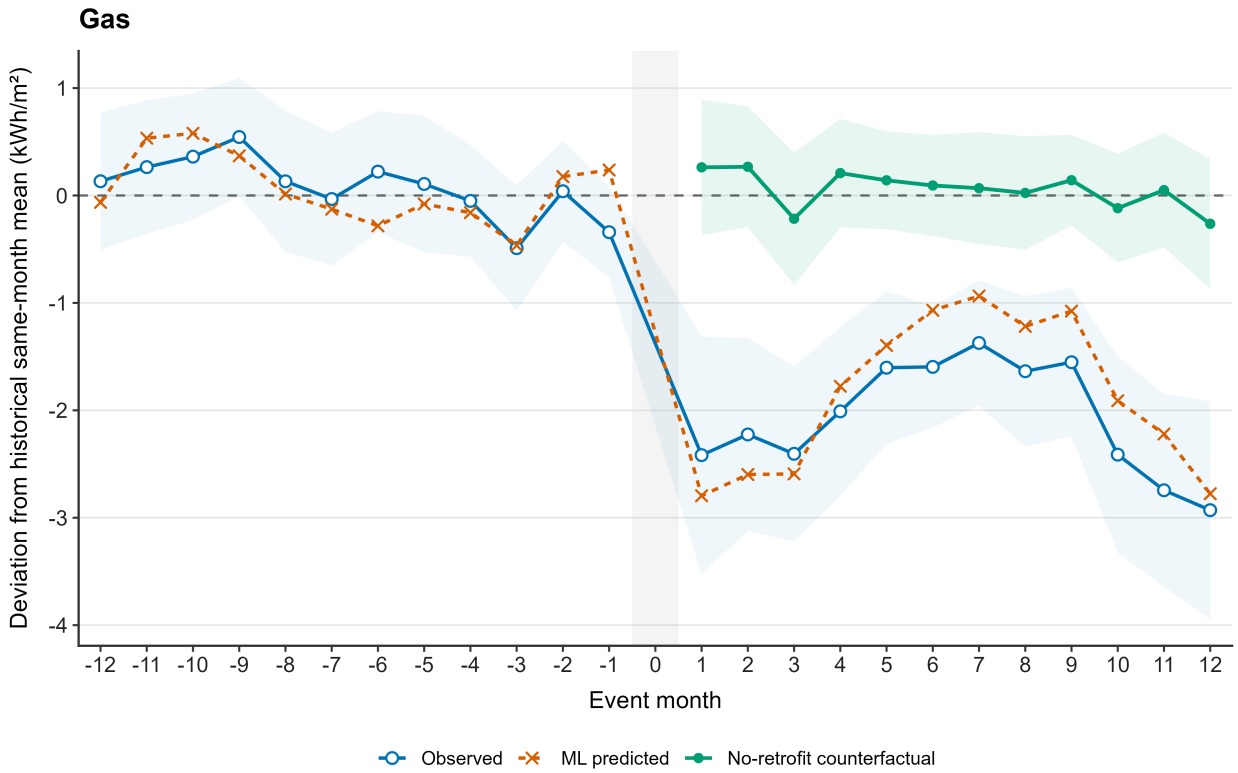


Figure F.3: Machine-Learning Counterfactual: Gas

Note: The gas outcome uses observed positive gas consumption only. The selected model is random forest. Definitions and normalization follow Appendix Figure F.1.

F.4 Strict Dual-Fuel Total Robustness

The strict dual-fuel total outcome is available only when both electricity and gas are observed and positive in the same building-month. It therefore uses a substantially smaller and more selected sample than the structural-adjusted total-energy outcome. For this reason, we treat the strict-total specification as a robustness analysis rather than as an alternative primary combined-energy measure.

Its smaller common observation set selects XGBoost rather than random forest. The selected strict-total model has a combined held-out RMSE of 3.86 kWh/m^2 and a mean held-out bias of -0.19 kWh/m^2. Among the 95 buildings with all twelve post-retrofit months, the mean counterfactual-implied realized saving is 28.6 kWh/m^2 and the model-implied treatment-state saving is 26.8 kWh/m^2. The ECO2 benchmark in the corresponding plotting sample is 2.66 kWh/m^2 per month, compared with 2.38 kWh/m^2 of counterfactual-implied realized saving.

Figure F.4 repeats the event-time counterfactual exercise using this strict dual-fuel definition. The post-retrofit pattern is qualitatively similar to the structural-adjusted total-energy result: observed and ML-predicted consumption remain below the no-retrofit counterfactual throughout most of the post-treatment window. However, the narrower engineering–realized wedge in the strict sample should not be interpreted as evidence that the strict definition is intrinsically more accurate. It reflects a different, more selective population that requires contemporaneously observed positive use of both fuels.

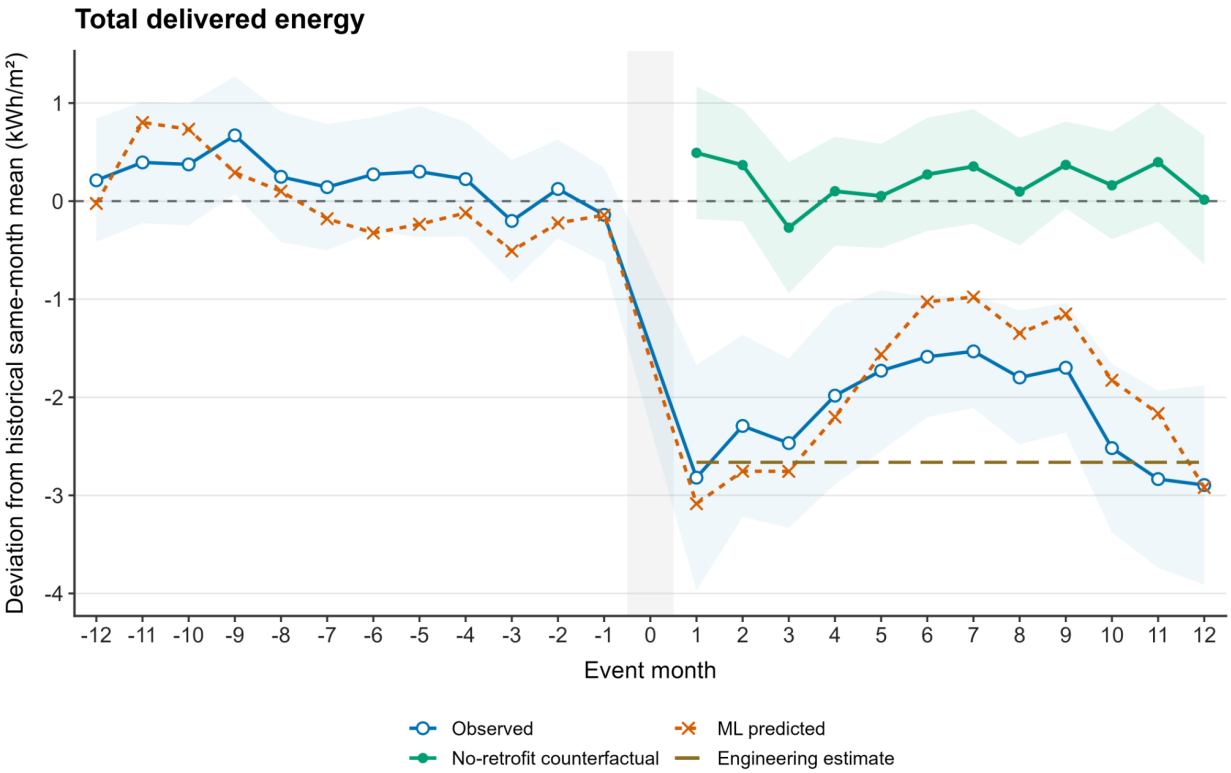


Figure F.4: Machine-Learning Counterfactual: Strict Dual-Fuel Total Delivered Energy

Note: The strict dual-fuel total is defined only for building-months in which both electricity and gas are observed and positive. The selected model is XGBoost, chosen by 5×3 building-grouped nested cross-validation. The blue series reports observed consumption, the orange series reports the cross-fitted prediction for the relevant observed state, and the green post-retrofit series reports the no-retrofit counterfactual. The ECO2 engineering line is translated to the same absolute kWh/m^2 scale. Because the strict definition produces a smaller and more selected dual-fuel sample, this figure is interpreted as a robustness check rather than as the primary total-energy result.